

**EVIDENCE-BASED PATHWAYS TO RESILIENCE
OF SUBNATIONAL ECONOMIC UNITS**

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ABSTRACT

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This study examined subnational economic resilience in the Philippines using 2018 to 2024 Provincial Product Accounts data. It described sectoral growth patterns, tested multivariate differences across six temporal intervals, and classified 117 subnational economic units using growth velocity and structural stability. A retrospective comparative quantitative design was used with secondary data from the Philippine Statistics Authority, producing 702 subnational-unit-by-interval observations. Sectoral growth trends were summarized using descriptive statistics, analyzed for structural variations via MANOVA, and classified into economic typologies using PCA-informed hierarchical clustering. Results showed significant differences in six-sector growth structures across temporal intervals. Services and Construction had the highest temporal sensitivity, while Agriculture, Forestry, and Fishing showed the greatest relative stability. Furthermore, hierarchical cluster analysis classified the 117 subnational units into two distinct resilience typologies: Cluster 1, a diversified and more growth-responsive pathway comprising 93 units, and Cluster 2, a lower-growth and volatility-constrained pathway encompassing 24 units characterized by stronger agricultural stability. These findings demonstrate that subnational economic resilience follows distinct structural pathways shaped by sectoral composition and temporal response patterns, providing a data-driven foundation for targeted regional development planning.

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An undergraduate thesis manuscript submitted to the faculty of the Department of Physical Sciences, College of Arts and Sciences, Cavite State University, Indang, Cavite, in partial fulfillment of the requirements for the degree of Bachelor of Science in Applied Mathematics with Contribution No. T-CAS-2026-AM009. Prepared under the supervision of Jayson C. Savilla.

INTRODUCTION

Economic resilience has become a major development concern, aligning with several of the United Nations Sustainable Development Goals (SDGs), particularly SDG 8 on decent work and economic growth, SDG 9 on industry, innovation, and infrastructure, and SDG 11 on sustainable cities and communities. As noted by Eichengreen, Park, and Shin (2024), resilience refers to the capacity of economies to withstand shocks and recover quickly, a view likewise reflected by the Commonwealth Secretariat (n.d.). This concern is especially relevant in both advanced and emerging economies, where financial disruptions, pandemics, supply chain interruptions, and external demand shifts increasingly affect local development trajectories. In Southeast Asia, institutions such as the Asian Development Bank and the Organisation for Economic Co-operation and Development continue to emphasize diversification and regional cooperation as important mechanisms for sustaining stability amid global uncertainty (OECD, 2023a; OECD, 2023b). Achieving such resilience, however, requires detailed and reliable economic information that captures differences in performance across local economies.

In the Philippine context, the Provincial Product Accounts (PPA) developed by the Philippine Statistics Authority provide a structured basis for measuring economic performance at the subnational level. The PPA estimates the Gross Provincial Domestic Product (GPDP) of the subnational economic units included in the dataset, disaggregating economic activity across major sectors. This framework enables localized analysis by capturing production across the broad sectors of Agriculture, Forestry, and Fishing (AFF), Industry, and Services. These broad sectors may further be examined through consolidated economic activities such as Mining and Quarrying, Manufacturing, Construction, Electricity, Steam, Water and Waste Management (ESWWM), and Services. Such disaggregation is important because growth across Philippine provinces remains highly uneven. PSA data show, for example, that Aklan recorded a growth rate of 22.5 percent in 2022, substantially exceeding the performance of many other provinces (PSA, 2024a). Similarly, sectoral data reveal contrasting patterns within individual provinces. In Capiz, the services sector grew by more than 19.6 percent while agriculture declined by 0.8 percent during the same period (PSA, 2024c). These disparities indicate that aggregate subnational economic growth alone is not sufficient to explain how local economies expand, contract, and recover over time.

Across Southeast Asia, resilience remains a strategic priority as economies respond to external shocks and structural transitions. According to the ASEAN+3 Macroeconomic Research Office (2025), the region's resilience is supported by strong domestic demand, diversified export markets, and available policy space that help absorb disturbances. Academic research has also increasingly focused on regional resilience and recovery dynamics. For example, Wang and Li (2022) found that innovation, human capital, and financial development significantly influence the resilience of Chinese provinces by supporting faster recovery and more stable long-term growth. These studies suggest that resilience is not merely a question of

aggregate output, but also of structural conditions that affect how local economies respond to disruption.

In the Philippines, however, empirical studies applying a resilience framework at the provincial level remain limited. One relevant contribution is the PIDS Research Paper Series RPS 2021-04 by Serafica, Vergara, and Oren (2021), which decomposed regional employment growth into national, industry-mix, and regional competitive effects. Although the study focused on employment competitiveness rather than shock absorption, it identified lagging regions and sectors and demonstrated the value of quantitative methods for assessing subnational economic performance. Such work provides an important foundation for extending resilience-oriented analysis to provincial economic structures using more disaggregated production data.

Despite recent improvements in subnational data availability, reliance on Gross Regional Domestic Product (GRDP) continues to limit the analysis of local economic conditions. Gross Regional Domestic Product (GRDP) aggregates economic activity at the regional level and therefore masks the variation among provinces within each region. PSA data illustrate this clearly. In 2021, Laguna accounted for approximately 33.8 percent of CALABARZON's total economic output due to its strong industrial base (PSA, 2022). In contrast, Quezon province contributed a smaller share and is more strongly associated with agricultural activity, particularly coconut production. These differences demonstrate how regional averages can obscure significant variation in economic structure and performance among constituent provinces. While reports from institutions such as the Bangko Sentral ng Pilipinas and the National Economic and Development Authority (NEDA) primarily emphasize regional indicators, the availability of Provincial Product Accounts (PPA) now makes more granular provincial analysis possible.

A similar limitation can be observed in government performance measures such as the Cities and Municipalities Competitiveness Index (CMCI). The Cities and Municipalities Competitiveness Index (CMCI) ranks localities across dimensions such

as economic dynamism, government efficiency, infrastructure, resilience, and innovation (Department of Trade and Industry, n.d.). Although useful as a comparative index, this ranking approach does not fully capture the structural economic characteristics that distinguish one locality from another. Two provinces or cities may achieve similar scores while relying on very different economic foundations. One locality may be driven primarily by services and commercial activity, while another may depend on manufacturing or industrial production. Treating such localities identically in policy design may lead to weakly targeted interventions. Previous studies suggest that relatively few analyses in the Philippine context employ statistical classification methods to identify groups of provinces with similar growth dynamics (Rayel, 2021; Serafica et al., 2021). This gap points to the need for a more structural, data-driven approach to understanding subnational economic performance.

Aggregate growth indicators also conceal the internal composition of local economies. Similar overall growth rates can emerge from very different sectoral patterns. As documented by the International Monetary Fund (2024), sectoral differences in the Philippine economy are substantial. Public construction and consumption have contributed to national growth, while tourism and business process outsourcing have exhibited distinct recovery patterns following recent disruptions. Consequently, two provinces may display comparable total growth while relying on fundamentally different sectoral drivers. One province may experience temporary expansion through extractive industries, while another may achieve comparable growth through sustained service-sector development. Understanding these structural differences is essential for evaluating the resilience and sustainability of provincial economies.

In response to these issues, this study adopts a structured validation-to-classification analytical framework to examine subnational economic resilience in the Philippines. Using the 2018–2024 Provincial Product Accounts (PPA), the analysis focuses on six consolidated economic sectors: Agriculture, Forestry, and Fishing

(AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services. The study first applies descriptive analysis to examine annual and sectoral growth patterns. It then uses Multivariate Analysis of Variance to test whether significant multivariate differences exist in sectoral growth behavior across the six temporal intervals derived from the 2018–2024 data. After validating the presence of structural differences over time, the study applies hierarchical cluster analysis to classify subnational economic units according to similarities in growth velocity and structural stability. Through this sequential framework, the research combines multivariate statistical validation with typological classification, providing a more rigorous understanding of subnational economic resilience than can be obtained from descriptive summaries or ranking systems alone.

Finally, national development initiatives such as the One Town, One Product (OTOP) program aim to strengthen local industries by promoting micro, small, and medium enterprises across the country (Philippine News Agency, 2023). National urban development policies likewise provide frameworks for managing urban growth, infrastructure development, and governance challenges in major metropolitan areas (Centre for Sustainable Cities, 2018). While these initiatives support local economic development, limited empirical analysis directly links national programs to the sectoral performance of individual provinces and cities. By examining sectoral growth patterns and classifying subnational economies based on their structural characteristics, this study contributes to a more evidence-based understanding of subnational economic resilience and offers insights that may support more targeted regional development strategies.

Statement of the Problem

This study examined evidence-based pathways to resilience of subnational economic units through a comprehensive analysis of the 2018 to 2024 Provincial Product Accounts (PPA).

Specifically, it sought answer to the following questions:

1. What were the annual and sectoral changes in growth rates from 2018 to 2024 across six consolidated economic sectors of subnational economic units: Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services?
2. What significant differences existed in the multivariate growth rates across the six economic sectors over the study period in terms of their structural stability and variability?
3. How were subnational economic units classified into distinct economic typologies based on similarities in growth velocity and structural stability across the six economic sectors?

Objectives of the Study

The general objective of this study was to examine evidence-based pathways to resilience of subnational economic units through a comprehensive analysis of the 2018–2024 Provincial Product Accounts (PPA).

Specifically, the study:

1. Described the annual and sectoral changes in growth rates from 2018 to 2024 across six consolidated economic sectors of subnational economic units: Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services.
2. Compared the multivariate growth rates across the six economic sectors over the study period to determine whether statistically significant differences exist in sectoral structural stability and variability over time.
3. Classified subnational economic units into distinct economic typologies based on similarities in growth velocity and structural stability across the six economic sectors.

Hypothesis of the Study

Based on the specific problem regarding the comparison of sectoral growth rates, the following null hypothesis was tested at the 0.05 level of significance:

H_0 : There are no significant differences in the multivariate mean vectors of sectoral growth rates across the six temporal intervals from 2018 to 2019 to 2023 to 2024.

Conceptual Framework of the Study

The conceptual framework of the study as shown in Figure 1 follows a validation-to-classification analytical structure. It begins with data extraction from the Provincial Product Accounts covering 2018 to 2024. The extracted dataset is then prepared by organizing annual sectoral Gross Provincial Domestic Product values for growth rate computation. From these prepared data, real growth rates are computed for the six economic sectors, producing 702 subnational-unit-by-interval observations.

These observations are first subjected to temporal validation through Multivariate Analysis of Variance using Pillai's Trace to determine whether significant multivariate differences exist across the six temporal intervals. After the validation phase, the sectoral indicators are aggregated into resilience profiles for 117 subnational economic units included in the dataset. Two indicators are derived for each unit: growth velocity, measured by the mean sectoral growth rate, and structural stability, measured by the coefficient of variation.

The resulting indicators were subjected to data handling procedures, specifically IQR winsorization and z-score standardization, before Principal Component Analysis is applied to reduce dimensional complexity and retain the most informative latent structure. The reduced component space is then used in the cluster analysis stage through hierarchical clustering using Ward's Minimum Variance Method. The optimal number of clusters is determined using the Silhouette Method. This procedure generates economic typologies based on similarities in growth velocity and structural stability. These typologies provide the basis for identifying subnational

economic resilience pathways and for interpreting structural differences that may inform regional development strategies.

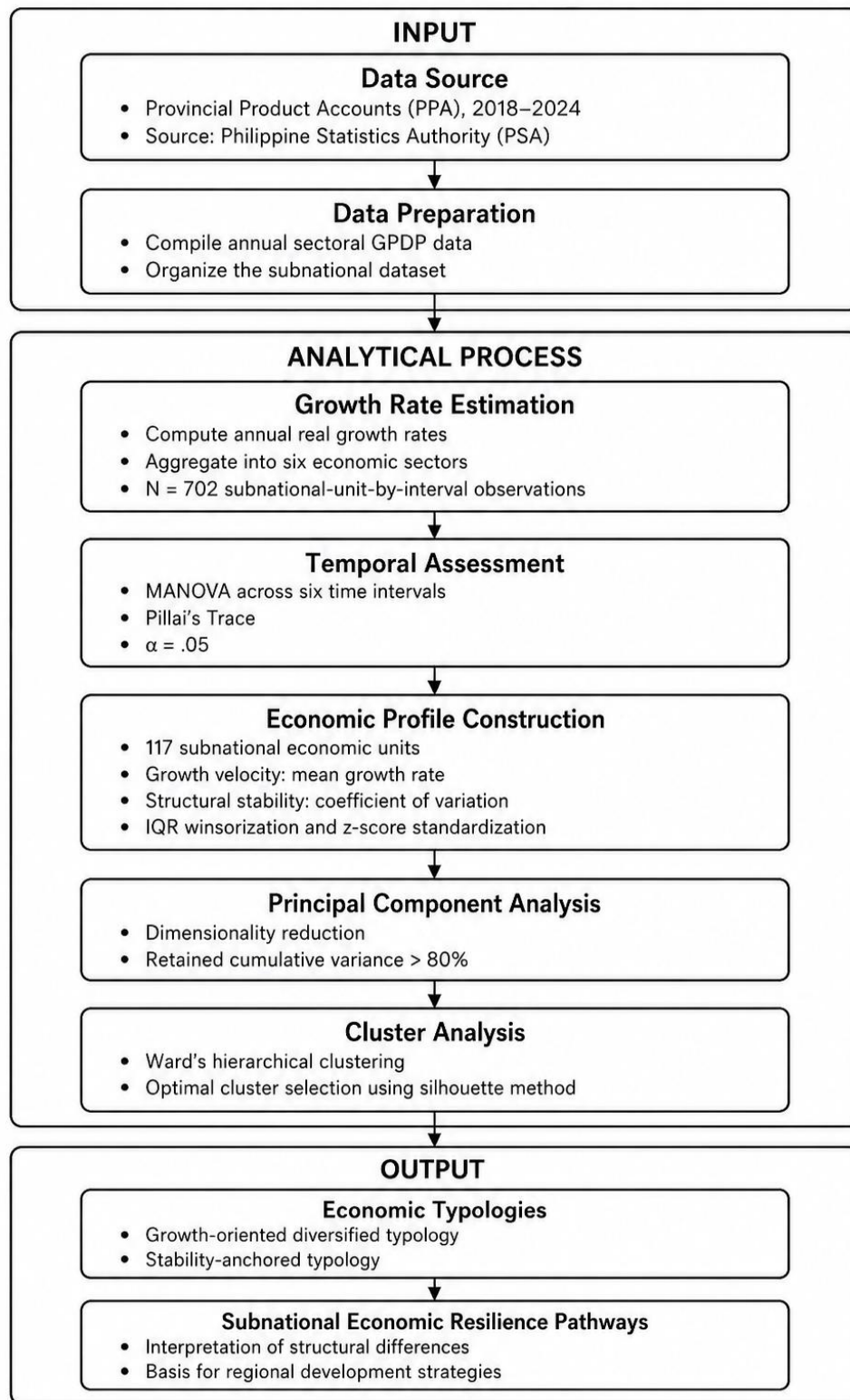


Figure 1. Conceptual Framework of the Study

Significance of the Study

This study provides a subnational-level examination of economic performance in the Philippines using the 2018–2024 Provincial Product Accounts (PPA) compiled by the Philippine Statistics Authority (PSA). By analyzing sectoral growth patterns and structural differences across subnational economic units, the study offers empirical insights that may assist policymakers, planners, investors, and researchers in understanding the dynamics of subnational economic resilience.

The findings of this study demonstrate how subnational-level PPA data can be examined using multivariate statistical techniques. By utilizing detailed PPA data, the research provides evidence on sectoral growth dynamics and structural differences across subnational economic units. These results may support **national agencies** in improving the interpretation of local economic performance and strengthening the analytical use of provincial accounts in development planning.

The study provides empirical evidence at the subnational level by examining sectoral growth patterns and structural stability across **local economies**. The identification of economic typologies may assist **planners** in recognizing sectoral strengths, structural vulnerabilities, and possible development pathways within their jurisdictions.

The study offers an alternative to purely ranking-based evaluation. While competitiveness indices provide comparative scores, the present study applies hierarchical clustering to classify subnational economic units based on similarities in growth velocity and structural stability. The resulting typologies may assist **Department of Trade and Industry (DTI) and policymakers** in designing development strategies that account for structural differences among local economies.

The sectoral growth analysis provides insights into patterns of expansion and variability across the major economic sectors. These findings may help **investors and private-sector planners** better understand local economic conditions and identify sectors that demonstrate relatively stable growth or higher structural sensitivity.

The study contributes to the literature on regional economic analysis by demonstrating how subnational economic data can be analyzed using Multivariate Analysis of Variance and Hierarchical Clustering techniques. The analytical framework developed in this research may serve as a methodological reference for **future studies** examining subnational economic resilience, sectoral growth dynamics, and regional development patterns in the Philippines.

Time and Place of the Study

The geographical scope of the study covered the Philippines, specifically the subnational economic units included in the Provincial Product Accounts (PPA) dataset of the Philippine Statistics Authority (PSA). This allowed the analysis to capture variations in economic structure and sectoral performance across local economies rather than relying solely on national or regional aggregates.

The temporal scope of the study covered the years 2018 to 2024. This period includes the years prior to the major economic disruptions of 2020, the contraction period, and the subsequent recovery phase. Using this timeframe, the study constructed six temporal intervals from 2018 to 2019 to 2023 to 2024 for the multivariate analysis and derives growth velocity and structural stability indicators for the clustering procedure.

Scope and Limitations of the Study

This study examined the economic resilience of subnational economic units in the Philippines using the 2018 to 2024 Provincial Product Accounts (PPA) compiled by the Philippine Statistics Authority (PSA). The analysis covered 117 subnational economic units included in the available dataset. These units include provinces and selected city- or municipality-level units reported in the PPA structure. The study focuses on local economic performance during the period before, during, and after the major economic disruption associated with the Coronavirus Disease 2019 (COVID-19) pandemic.

The research was limited to the structural performance of six consolidated economic sectors: Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services. These six consolidated sectors represent the broad primary, secondary, and tertiary structure of the economy, allowing the study to examine both sector-specific growth behavior and broader structural patterns across subnational economic units. The Services sector was treated as a single aggregated sector because it consists of several service-related subsectors in the PPA. This consolidation allowed the study to maintain a manageable and balanced multivariate structure while still capturing the broad role of service-based activities in subnational economic performance.

The study used sectoral growth rates, mean growth rates, and coefficients of variation to represent two major dimensions of economic resilience: growth velocity and structural stability. Growth velocity refers to the average pace of sectoral expansion or contraction, while structural stability refers to the consistency or volatility of sectoral growth over time. These indicators serve as the basis for the multivariate and clustering procedures.

The empirical analysis was limited to descriptive statistics, Multivariate Analysis of Variance (MANOVA), Principal Component Analysis (PCA), and hierarchical cluster analysis. MANOVA is used to determine whether significant multivariate differences exist in sectoral growth patterns across the six temporal intervals from 2018–2019 to 2023–2024. PCA is used to reduce dimensional complexity among the twelve clustering indicators, while hierarchical clustering is applied to classify subnational economic units into economic typologies based on similarities in growth velocity and structural stability.

Despite its contributions, the study has several limitations. First, the analysis relies entirely on secondary data from official PPA releases. Therefore, the accuracy of the findings depends on the completeness, consistency, and reliability of the

published PSA data. The study does not independently verify the original economic accounts and does not generate primary data from local government units, firms, or households.

Second, some observations for selected local units in the National Capital Region (NCR) were incomplete in the available dataset. To preserve the completeness and balance of the multivariate dataset, corresponding regional growth values were used as proxy values where necessary. While this treatment allowed the inclusion of NCR units in the analysis, it may reduce the extent to which city-level variation within NCR is fully captured. Thus, the results involving NCR units should be interpreted with caution.

Third, the dataset includes Pateros and the City of Isabela, which were retained for dataset completeness and operational consistency with the available PPA structure. However, their inclusion should not be interpreted as a reclassification of their formal administrative status. Pateros is retained as a municipality-level unit, while the City of Isabela is retained as a statistically separate city-level unit in the dataset.

Fourth, the cluster results should be interpreted as structural economic typologies rather than as administrative classifications. The clusters do not represent purely provincial, urban, rural, or HUC-based groupings. Instead, they represent groups of subnational economic units with similar patterns of sectoral growth velocity and structural stability. Therefore, the cluster membership results should not be generalized solely according to administrative type.

Fifth, the dataset includes repeated observations from the same subnational economic units across multiple growth intervals. Although this structure introduces a longitudinal component, the MANOVA treats each subnational-unit-by-interval observation as a distinct economic state for purposes of comparing sectoral growth behavior across time. This simplifying assumption allows temporal comparison but may not fully capture unit-specific dependence across repeated observations.

Sixth, the analysis is limited to economic activities captured within the formal Provincial Product Accounts. It does not include informal economic activity, household-level income variation, employment quality, poverty indicators, fiscal capacity, infrastructure access, disaster exposure, governance quality, or other social and institutional factors that may also influence economic resilience. As a result, the study measures economic resilience only through sectoral growth performance and stability, not through all possible dimensions of local resilience.

Finally, the study is limited to the 2018 to 2024 period. This timeframe includes the pre-disruption period, the 2020 economic contraction, and the subsequent recovery years. The resilience patterns identified in this study may differ under other time periods, alternative macroeconomic conditions, or future economic shocks. Therefore, the findings should be interpreted as evidence-based typologies for the observed period rather than permanent classifications of subnational economies.

Definition of Terms

Cluster analysis is a multivariate statistical technique used to group the 117 subnational economic units into economic typologies based on similarities in sectoral growth velocity and structural stability. In this study, hierarchical clustering using Ward's minimum variance method and Euclidean distance is applied to principal component scores derived from the sectoral indicators.

Economic resilience is the relative capacity of a subnational economy to maintain stable growth despite economic fluctuations. In this study, resilience is operationally represented by the interaction between sectoral growth velocity and structural stability, measured through the mean growth rate and coefficient of variation across six economic sectors.

Economic shocks refer to sudden changes in economic conditions that result in significant fluctuations in sectoral growth rates. In this study, economic shocks refer to the disruptions observed in subnational economic performance during the 2018 to 2024 period.

Economic typologies is the economic classifications produced by the Cluster Analysis are based on similarities in Growth Velocity and Structural Stability across the six economic sectors. In this study, the identified typologies represent distinct patterns of subnational economic structure and performance.

Gross Provincial Domestic Product (GPDP) is the total economic output produced within a province or city. In this study, GPDP values reported in constant 2018 prices are used to compute real growth rates and analyze sectoral economic performance.

Growth velocity is the average growth performance of an economic sector or subnational economic unit over time. In this study, growth velocity is operationally measured using the mean growth rate across the six temporal intervals from 2018 to 2019 to 2023 to 2024.

Multivariate Analysis of Variance (MANOVA) is a statistical technique used to test whether significant differences exist in the combined sectoral growth rates across the six temporal growth intervals from 2018–2019 to 2023–2024. Pillai's Trace is used as the primary multivariate test statistic due to its robustness to violations of statistical assumptions.

Provincial Product Accounts (PPA) is the official dataset compiled by the Philippine Statistics Authority (PSA) that contains provincial-level economic statistics. The 2018 to 2024 PPA dataset serves as the primary data source for the analysis conducted in this research.

Sectoral performance is the quantitative assessment of economic sector behavior within a subnational economic unit, measured using real growth rates and structural stability indicators derived from the Provincial Product Accounts.

Structural stability refers to the degree to which sectoral growth remains consistent over time within a subnational economy. In this study, structural stability is measured using the coefficient of variation of sectoral growth rates.

Subnational economic units are the local economic units in the Philippines included in the Provincial Product Accounts dataset analyzed in this study. These units include provinces and city- or municipal-level units reported in the dataset and serve as the primary units of analysis for examining sectoral growth patterns, growth velocity, structural stability, and economic typologies.

REVIEW OF RELATED LITERATURE

This chapter presents the related literature and studies that provide the theoretical and empirical foundation of the research. The review is organized into five major themes: economic resilience and subnational development, subnational accounting and economic measurement, sectoral structure and economic performance, applications of multivariate statistical analysis in regional studies, and the synthesis of literature leading to the identification of research gaps. The discussion integrates international and local studies to establish the relevance of examining subnational economic resilience in the Philippines through a structured validation-to-classification analytical framework.

Economic Resilience and Subnational Development

Resilience is widely understood as a dynamic and multidimensional capability rather than a fixed condition. Martin and Sunley (2014) identified four major dimensions of resilience: resistance, or the ability to absorb shocks without major declines; recovery, or the speed of return to prior levels of performance; re-orientation, or the adaptive restructuring of the economy; and renewal, or long-term transformation through innovation. This perspective suggests that resilience is not limited to returning to previous conditions, but also includes the capacity of local systems to learn, reorganize, and pursue new developmental paths. Such a view is consistent with evolutionary economic geography, which emphasizes adaptive learning and structural change as core elements of regional development.

A major determinant of resilience is the sectoral structure of the economy. He, Miao, and Qureshi (2022) showed that regions with diversified economic bases tend to recover more rapidly than highly specialized regions. Diversification reduces dependence on a single sector and lowers exposure to sector-specific shocks. Giannakis, Bruggeman, and Mamuneas (2024) likewise emphasize the role of inter-industrial connectivity, arguing that economies with stronger linkages among sectors

are better able to absorb disturbances because shocks are distributed across the value chain. These findings indicate that resilience is closely connected to the internal composition and interaction of productive sectors.

These sectoral characteristics do not operate in isolation. Their effects are shaped by spatial and institutional conditions that influence how local economies respond to disruption. In the Philippine context, Serafica, Vergara, and Oren (2021) note that although services contribute significantly to national growth, their concentration in selected areas reinforces regional inequality. Zhang et al. (2024) further argue that resilience outcomes depend on the fit between industrial upgrading and local capacity. Where upgrading strategies are poorly aligned with local conditions, they may weaken rather than strengthen regional performance. By contrast, adaptive strategies that respond to existing capabilities are more likely to support growth. Relatedly, Li *et al.* (2025) show that structural transformation driven by technological innovation can strengthen long-term adaptive capacity, particularly in agriculture-based regions undergoing modernization.

Spatial configuration also affects resilience. Encarnacion, Magnaye, and Castro (2023) identify spatial clustering in economic dynamism among Philippine local government units, suggesting that proximity to major growth centers generates forms of structural interdependence. However, such advantages are unevenly distributed. Evidence from the World Risk Poll indicates that differences in institutional capacity and resource endowments strongly affect how regions resist and recover from shocks (Lloyd's Register Foundation, 2024). This reinforces the need for localized strategies that reflect the unique economic geography of each area.

Institutional planning likewise plays a role in subnational resilience. Domingo and Manejar (2020) show that disaster risk reduction and management inputs contribute to socioeconomic capacity and reduced vulnerability at both provincial and sub-provincial levels, although efficiency varies considerably across locations. Their findings suggest that targeted allocation and efficient use of public resources can

strengthen resilience outcomes. Overall, the literature indicates that diversification, inter-sectoral integration, spatial positioning, and institutional capacity are important structural determinants of subnational economic resilience.

Subnational Accounting and Economic Measurement

Subnational accounting provides the statistical basis for measuring economic performance below the national level. Traditional regional indicators, particularly Gross Regional Domestic Product (GRDP), are limited in their ability to represent the economic conditions of individual provinces because they aggregate heterogeneous local economies into a single regional value. Such aggregation can obscure substantial differences in growth dynamics and sectoral composition, thereby limiting the usefulness of GRDP for localized planning and policy formulation (PSA, 2025a; OECD, 2023a; OECD, 2023b). This challenge is not unique to the Philippines. Wenz et al. (2023) note that insufficient subnational economic data is a widespread problem internationally, often constraining evidence-based intervention at the local level.

Before the availability of official local product accounts, researchers frequently relied on proxy-based estimation methods to approximate subnational GDP. Bundervoet, Maiyo, and Sanghi (2015) and Puttanapong, Prasertsoong, and Peechapat (2023), for example, used satellite night-light intensity and machine learning techniques to estimate local economic activity in data-scarce settings. Other studies applied mathematical modeling approaches. Aban-Serquina (2024) used input-output data to model provincial GDP, while Curtale and Schiavone (2025) demonstrated the feasibility of disaggregating national GDP into local estimates. Although these methods provide useful approximations, Wenz et al. (2023) argue that they remain second-best alternatives to an official statistical accounting system.

The institutionalization of the Provincial Product Accounts by the Philippine Statistics Authority represents a major improvement in subnational economic measurement. Unlike indirect estimates, the PPA provides official Gross Provincial Domestic Product data using the production approach, which measures economic

output through value added across industries under standardized classification systems and established accounting principles. Initial releases provided a more detailed view of provincial growth, while subsequent updates expanded coverage to provinces and city-level units, thereby strengthening the empirical basis for localized economic analysis (PSA, 2024b; PSA, 2025a). The PPA therefore offers a more robust foundation for examining subnational disparities, sectoral transformation, and resilience patterns across local economies.

This improvement in data availability is central to the present study. Without official provincial accounts, the measurement of structural economic differences would remain largely dependent on proxies and approximations. With the PPA, it becomes possible to examine the six consolidated economic sectors directly and to evaluate subnational performance through a more rigorous statistical framework.

Sectoral Structure and Industry Performance

A resilient economy depends not only on total output but also on the composition of that output across sectors. Structural transformation, commonly defined as the movement of economic activity from traditional agriculture toward industry and services, is widely recognized as a driver of productivity and long-run growth. International evidence indicates that technological innovation plays an important role in this process. Li *et al.* (2025), for instance, find that agricultural modernization raises rural productivity while also facilitating broader structural change.

In the Philippine setting, however, structural transformation remains uneven. Custodio and Sombilla (2025) observed that while some regions have undergone substantial economic restructuring, many provinces continue to depend on lower-productivity sectors. Such variation implies that adaptive capacity differs significantly across local economies. Sectoral composition is therefore not only an indicator of current performance, but also a determinant of vulnerability and recovery.

The literature also suggests that the nature of diversification matters. Lazzeretti, Oliva, and Innocenti (2022) reported that regions diversifying into sectors

related to their existing industrial structure tend to exhibit stronger resistance to shocks, while in some cases diversification into unrelated sectors may support more rapid recovery. These findings imply that resilience is shaped not only by the presence of multiple sectors, but also by the structural relationships among them.

Serafica, Vergara, and Oren (2021) identified a related issue in the Philippines, where the economy has developed a strong service orientation without uniformly broad industrial deepening across all regions. High-value services are concentrated in selected urban locations, whereas many provinces remain reliant on agriculture and other traditional activities. Such uneven structural transformation contributes to divergence in subnational growth trajectories. The Provincial Product Accounts provide an opportunity to analyze these differences more systematically by examining sectoral performance across local economies over time.

In the present study, this sectoral perspective is operationalized through six consolidated economic sectors: Agriculture, Forestry, and Fishing; Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management; and Services. These sectors serve as the mathematical basis for the analysis of growth behavior, temporal variation, and structural typology.

Applications of Multivariate Statistical Analysis in Regional Studies

Multivariate statistical analysis is important in regional studies because economic performance is inherently multidimensional. Subnational economies are characterized by multiple interrelated indicators that cannot be adequately understood through univariate methods alone. By analyzing several variables simultaneously, multivariate techniques provide a more complete picture of structural performance, inter-sectoral variation, and typological similarity.

Cluster analysis has been widely applied in recent literature to classify territories into meaningful groups based on multidimensional economic indicators. Rivai et al. (2024) used Fuzzy C-Means clustering to classify Indonesian provinces according to economic growth-related indicators. Stamenković, Milanović, and

Janković-Milić (2021) combined hierarchical clustering with one-factor MANOVA to validate statistically distinct district-level development groups in Serbia. These studies support the use of clustering, especially when combined with multivariate validation, as a classification tool for identifying structural similarities and disparities among subnational units.

However, statistical classification becomes more defensible when it is anchored to validated structural differences in the data. Multivariate Analysis of Variance (MANOVA) is commonly used to test whether groups differ significantly across multiple variables simultaneously. Stamenković, Milanović, and Janković-Milić (2021), in their study of Serbian districts, combined hierarchical clustering with a one-factor MANOVA to confirm that the resulting clusters were statistically distinct in terms of development indicators. Their work demonstrates the value of combining classification with multivariate validation.

The present study adopted and extended this logic through a validation-to-classification sequence. Rather than applying clustering immediately to subnational economic data, the study first uses MANOVA to test whether significant multivariate differences exist in sectoral growth patterns across time. This step establishes that the data contain meaningful structural variation before typological classification is undertaken. After this validation stage, hierarchical clustering is applied to classify subnational economic units according to similarities in growth velocity and structural stability. In this way, the analysis is not limited to descriptive grouping, but is grounded in statistically verified temporal variation. This sequence reflects the applied mathematical character of the study by integrating multivariate inference and classification into a single analytical framework.

Identification of Research Gaps

Despite advances in regional economic theory and subnational accounting, several important gaps remain. First, empirical analysis at the provincial level in the Philippines remains limited, as many studies continue to rely on aggregated regional

indicators such as GRDP or proxy-based local estimates. This leaves the finer structural dynamics of subnational economies insufficiently examined. Second, much of the literature considers sectoral composition, spatial structure, and growth variation separately rather than within an integrated analytical framework. Third, relatively few studies use longitudinal Provincial Product Accounts data to examine how sectoral growth behavior changes across multiple intervals over time. Fourth, many policy-oriented assessments rely on ranking systems or composite indices that may obscure structural heterogeneity by assigning similar scores to localities with different economic foundations. Finally, there remains limited use of a sequential framework that first validates multivariate sectoral differences and then classifies local economies into structural typologies.

Addressing these gaps, the present study uses the Provincial Product Accounts from 2018 to 2024 to examine subnational economic resilience through a structured validation-to-classification approach. The study first applied Multivariate Analysis of Variance to test whether significant multivariate differences exist in sectoral growth patterns across time. It then applied hierarchical cluster analysis to classify subnational economic units according to similarities in growth velocity and structural stability. By combining longitudinal PPA-based subnational data with multivariate validation and classification, the study provides a more integrated and statistically rigorous framework for understanding how sectoral transformation and growth dynamics shape subnational economic resilience in the Philippines.

Synthesis

The reviewed literature indicates that economic resilience is a multidimensional capability shaped by diversification, structural balance, inter-sectoral connectivity, spatial context, and institutional capacity. Theoretical studies emphasize that resilience includes resistance, recovery, re-orientation, and renewal rather than simple return to pre-shock conditions (Martin & Sunley, 2014). Empirical studies likewise show that diversified and interconnected economies tend to exhibit stronger stability under

economic stress (He, Miao, & Qureshi, 2022; Giannakis, Bruggeman, & Mamuneas, 2024).

Within the Philippine setting, the literature suggests that sectoral composition strongly influences local development trajectories. Service concentration, industrial imbalance, and persistent dependence on lower-productivity sectors contribute to uneven subnational performance (Serafica, Vergara, & Oren, 2021; Custodio & Sombilla, 2025). At the same time, the Provincial Product Accounts now provide the first official statistical basis for analyzing these structural differences at the subnational level (PSA, 2025b). This convergence of resilience theory, sectoral development literature, and improved PPA-based subnational economic data creates a strong basis for a quantitative study of subnational economic resilience.

The methodological literature further supports the use of multivariate approaches for such an analysis. Cluster analysis has been shown to be effective for classifying subnational units with similar economic characteristics, while MANOVA provides a formal statistical basis for validating multivariate differences. Taken together, these methods support a structured analytical framework that is well suited to examining subnational economic performance in a multidimensional and longitudinal manner.

The reviewed literature also suggests that subnational economic resilience should not be interpreted only through administrative categories such as province, city, or region. While urban economies are often assumed to be more diversified and rural economies are often associated with agriculture-based structures, actual resilience depends on the interaction between sectoral growth, volatility, and structural balance. This is important for the present study because the Provincial Product Accounts include both provinces and city-level economic units, making it possible to examine whether economic similarities follow administrative boundaries or cut across them. By applying cluster analysis after multivariate validation, the study extends the literature by showing how provinces and city-level economies may share common resilience

pathways despite differences in formal classification. In this way, the study contributes not only by identifying whether subnational economies differ over time, but also by classifying how these economies differ structurally in terms of growth velocity and stability.

METHODOLOGY

This chapter presents the methodological procedures used to accomplish the objectives of the study. It describes the research design, source of data, population and unit of analysis, data gathering procedure, variable construction, data preparation, and the statistical techniques used to examine subnational economic resilience. The methodological framework follows a validation-to-classification sequence. First, the study determined whether significant multivariate differences existed in sectoral growth behavior across time through Multivariate Analysis of Variance (MANOVA). Second, it applied Hierarchical Clustering to classify subnational economic units according to similarities in growth velocity and structural stability.

Research Design

This study employed a retrospective, comparative quantitative research design based on secondary economic data. As a non-experimental observational study, it analyzed historical Provincial Product Accounts (PPA) data compiled and published by the Philippine Statistics Authority (PSA), without manipulating any variables. The design is appropriate because the study aimed to describe sectoral growth behavior, test whether statistically significant multivariate differences existed across temporal intervals, and classify subnational economic units into economic typologies.

The analytical framework integrated three levels of quantitative analysis. First, descriptive statistics were used to summarize annual and sectoral growth patterns based on the available PPA sectoral structure. Second, for multivariate and classification purposes, the sectoral variables were consolidated into six economic sectors in order to produce a more stable and analytically manageable structure. Multivariate Analysis of Variance (MANOVA) was then used as a validation procedure to determine whether the combined growth behavior of the six economic sectors differed significantly across the six temporal intervals derived from the 2018–2024

dataset. Third, hierarchical cluster analysis was applied to classify subnational economic units based on similarities in growth velocity and structural stability.

Source of Data

The study used secondary data drawn from the Provincial Product Accounts (PPA) compiled by the Philippine Statistics Authority (PSA). The dataset covered the period from 2018 to 2024 and provided annual Gross Provincial Domestic Product (GPDP) values in constant 2018 prices. The use of constant prices allowed the measurement of real economic growth by minimizing inflation effects and ensuring comparability across time.

The analysis covered 117 subnational economic units included in the available dataset. These units served as the empirical basis for examining subnational economic performance in the Philippines. The data were organized according to the production approach and disaggregated into sectoral categories, which were subsequently consolidated into six economic sectors: Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services.

Units of Analysis and Observation Structure

The study included all subnational economic units available in the 2018–2024 Provincial Product Accounts (PPA) dataset that satisfied the requirements of the analysis. The primary units of analysis were the subnational economic units themselves. The temporal structure of the dataset was represented through six annual growth intervals derived from the seven-year panel covering 2018 to 2024.

For the descriptive and cluster analysis components, each subnational economic unit was represented by sector-based indicators of growth velocity and structural stability. For the Multivariate Analysis of Variance (MANOVA) component, the analytical observations consisted of subnational-unit-by-interval records, which were used to compare multivariate sectoral growth patterns across the six temporal intervals.

Data Gathered

The study utilized secondary data from the official open-data releases of the Philippine Statistics Authority (PSA), specifically the Provincial Product Accounts (PPA) tables accessed via PSA OpenStat and related PPA dashboard sources (PSA, n.d.-a; PSA, n.d.-b). The dataset consists of annual Gross Provincial Domestic Product (GPDP) values at constant 2018 prices for all available subnational economic units. The variables retrieved include the name of each subnational unit, the year of observation (2018–2024), total GPDP, and corresponding sectoral values for each industry included in the study.

Data Preparation

The data preparation phase involved a series of structural adjustments to transform the raw PSA datasets into a format suitable for multivariate analysis. The process began with the transposition of initial Excel files to correctly orient economic indicators and temporal intervals, ensuring the data was properly structured for computational modeling. To maintain a consistent subnational focus, economic units were mapped specifically to the provincial and Highly Urbanized City (HUC) levels, while regional-level aggregates were removed to prevent data redundancy and ensure the scope remained strictly subnational.

A critical component of this phase was the transformation of annual Gross Provincial Domestic Product (GPDP) figures into six distinct growth intervals spanning from 2018 to 2024. These growth rates, derived directly from the GPDP data, served as the primary metrics for analyzing economic shifts. Furthermore, the study addressed data gaps through targeted imputation for the National Capital Region (NCR); specifically, where OpenStat records showed missing or zero values for Agriculture and Mining, values were derived from the broader NCR economic accounts and strategically imputed into these sectoral fields. For other incomplete observations within the NCR, the corresponding regional growth rate was utilized as a proxy to preserve analytical continuity.

Throughout this process, Pateros and Isabela City were retained to maintain the integrity of the available PPA data structure. These refined procedures resulted in a complete and balanced panel dataset, allowing for the accurate computation of principal components across all six consolidated sectors.

Variables of the Study

The original sectoral classifications in the Provincial Product Accounts (PPA) were consolidated into six economic sectors to improve analytical stability and reduce dimensional complexity. The final sectors used in the analysis were Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services.

The Services sector was treated as a single aggregated sector formed from the service-related subsectors in the Provincial Product Accounts (PPA). This consolidation was adopted to reduce redundancy among highly related service activities and to maintain a manageable number of variables for multivariate analysis.

Three principal quantitative indicators were constructed in the study. First, the real growth rate was computed for each sector and subnational economic unit using the formula:

$$r_{it} = \frac{Y_{it} - Y_{i,t-1}}{Y_{i,t-1}} \times 100$$

where Y_{it} represents the sectoral value in the current year and $Y_{i,t-1}$ represents the sectoral value in the immediately preceding year.

Second, growth velocity was measured using the arithmetic mean of sectoral growth rates across the six temporal intervals:

$$\text{Mean Growth Rate} = \frac{\sum r_{it}}{n}$$

where $n = 6$ intervals.

Third, structural stability was measured using the coefficient of variation:

$$CV_i = \frac{s_{r_i}}{\bar{r}_i} \times 100$$

where $s_{r,i}$ is the standard deviation of the growth rates of subnational economic unit i , and $\bar{r}_i$ is its mean growth rate.

The mean growth rate captured the average pace of sectoral expansion or contraction, while the coefficient of variation measured the relative dispersion of growth rates around that average. Together, these indicators formed the basis of the clustering procedure.

Descriptive Statistics

Descriptive statistics was used to summarize the annual and sectoral growth behavior of the subnational economic units. The analysis focused on the growth rates of the economic sectors across the study period in order to describe patterns of expansion, contraction, and recovery. These descriptive results provided the empirical baseline for the subsequent multivariate and typological analyses.

To complement the numerical summaries, bar graphs were used to present the annual growth rates of the major sectors and their constituent industries across the six temporal intervals from 2018 to 2019 to 2023 to 2024. The use of bar graphs was appropriate because the study dealt with discrete temporal categories and sector-specific growth values, making visual comparison across intervals more direct and interpretable. The bar graphs also helped illustrate the direction and magnitude of sectoral growth changes prior to the conduct of Multivariate Analysis of Variance (MANOVA) and hierarchical cluster analysis.

Multivariate Analysis of Variance (MANOVA)

Model Specification. A one-way Multivariate Analysis of Variance (MANOVA) was employed to determine whether the joint distribution of sectoral growth rates differed significantly across temporal growth intervals derived from the 2018–2024 dataset. The dependent variables consisted of the growth rates of the six economic sectors: Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services. The independent grouping variable was the temporal growth

interval, represented by six categories: 2018–2019, 2019–2020, 2020–2021, 2021–2022, 2022–2023, and 2023–2024.

Outlier Detection. Prior to the MANOVA, multivariate outliers were identified using Mahalanobis Distance computed from the six-sector growth matrix. This procedure was used to detect observations with unusual joint sectoral growth profiles relative to the multivariate centroid.

The Mahalanobis Distance for observation i is given by:

$$D_i^2 = (x_i - \underline{x})^T S^{-1} (x_i - \underline{x})$$

where,

x_i = vector of sectoral growth rates for observation i

$\underline{x}$ = vector of sample means

S^{-1} = inverse of the sample covariance matrix

Observations exceeding the critical chi-square threshold at $p < 0.01$ with degrees of freedom equal to the number of dependent variables were flagged as potential multivariate outliers. Rather than automatically excluding these cases, the study subjected them to qualitative verification against official releases of the Philippine Statistics Authority (PSA) and contextual economic explanations. Observations supported by official narratives or plausible structural events were retained, since extreme values may reflect genuine economic shocks rather than measurement errors.

Diagnostic Assessment: Multivariate Normality. Prior to conducting the multivariate analysis, the distributional properties of the sectoral growth variables were evaluated to determine whether the assumption of multivariate normality was satisfied. This assumption was assessed using Mardia's test of multivariate skewness and kurtosis, which examines whether the joint distribution of the dependent variables approximates a multivariate normal distribution. Because macroeconomic growth data frequently contain structural shifts and extreme observations, deviations from strict multivariate normality are common.

When the assumption of multivariate normality is not satisfied, robust multivariate test statistics are recommended. For this reason, Pillai's Trace was adopted as the primary MANOVA test statistic. It is widely regarded as the most robust multivariate statistic when the assumptions of multivariate normality and homogeneity of covariance matrices are not fully met.

Assumptions and Statistical Protocols. Before performing the MANOVA procedure, several statistical assumptions were evaluated to ensure the validity of the multivariate model.

Homogeneity of Variance–Covariance Matrices. The equality of covariance matrices across the temporal groups was assessed using Box's M test. This diagnostic evaluates whether the variance-covariance structure of the dependent variables remains statistically equivalent across the six temporal intervals. Because Box's M is highly sensitive to sample size, statistically significant results may occur even when practical differences are modest. When violations occur, Pillai's Trace is preferred because of its robustness to departures from covariance homogeneity.

The Box's M statistic is expressed as:

$$M = (N - g) \ln \ln |S_p| - \sum_{j=1}^g (n_j - 1) \ln \ln |S_j|$$

where:

N = total sample size

g = number of groups

S_p = pooled covariance matrix

S_j = covariance matrix of group j

n_j = sample size of group j

Independence of observations. Each subnational-unit-by-interval observation was treated as an independent statistical unit representing the economic performance of a subnational economic unit during a specific interval. Although the same units

appeared repeatedly across intervals, each observation represented a distinct economic state corresponding to a particular temporal transition. This assumption may be expressed as $Cov(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$ where the covariance between observations i and j is assumed to be zero.

Linearity among dependent variables. To verify that the linear model captures the underlying economic relationships, Scatterplot matrices of the sectoral growth rates were visually inspected for elliptical patterns.

Absence of multicollinearity. The study evaluates the Determinant of the Variance-Covariance Matrix. A non-zero determinant ($|S| > 0$) confirms that the matrix is non-singular and that no sector is a perfect linear combination of the others.

Sample Size and Statistical Power. Adequate sample size was supported by the availability of sufficient error degrees of freedom (df_e). The MANOVA dataset contained $N = 702$ subnational-unit-by-interval observations derived from 117 subnational economic units and $p = 6$ sectoral growth variables, which provided adequate degrees of freedom ($N > p$) for estimating the variance-covariance matrix.

MANOVA Procedure. A one-way MANOVA was performed to test whether the joint distribution of sectoral growth rates differed significantly across the six temporal growth intervals from 2018–2019 to 2023–2024. Pillai's Trace was used as the primary test statistic.

$$V = \sum_{i=1}^s \frac{\lambda_i}{1 + \lambda_i}$$

where λ_i represents the eigenvalues of the test matrix.

Significance level and decision rule. The null hypothesis, that the multivariate mean vectors of sectoral growth rates were identical across the six temporal intervals, was tested at a significance level of $\alpha = 0.05$. Since the calculated p-value associated with Pillai's Trace was less than 0.05, the null hypothesis was rejected, confirming that significant structural shifts occurred during the study period.

Magnitude of Effect. To go beyond simple significance testing, the study reported partial eta squared (η_p^2) to quantify the magnitude of the differences. This metric indicates the proportion of variance in the sectoral growth rates that is attributable to the temporal interval, providing a measure of the practical significance of the economic shifts.

Hierarchical Cluster Analysis

Data Consolidation and Indicator Construction. To capture both the performance and structural stability of subnational economies, the original dataset comprising 16 distinct economic sectors was consolidated into six economic sectors: Agriculture, Forestry, and Fishing (AFF); Mining and Quarrying; Manufacturing; Construction; Electricity, Steam, Water and Waste Management (ESWWM); and Services. For each sector, two indicators were constructed:

Mean Growth Rate. This indicator was used to measure growth velocity, or the average pace of economic expansion or contraction across the observed period.

Coefficient of Variation (CV). This indicator was used to measure structural stability by quantifying the relative dispersion of growth rates across the observed period.

Data Preparation and Methodological Considerations. To improve the validity and stability of the clustering solution, several data preparation procedures and methodological considerations were applied.

Outlier Treatment through Winsorization. Distance-based clustering algorithms are highly sensitive to extreme values. To reduce the influence of mathematically inflated coefficients of variation, Interquartile Range (IQR) winsorization was applied to the structural stability indicators. Values outside the $1.5 \times \text{IQR}$ threshold were capped at the corresponding upper or lower bound prior to distance calculations.

Scale Standardization and Dimensionality Reduction. Because Euclidean distance is sensitive to differences in variable scales and multicollinearity, Principal Component Analysis (PCA) was employed as a preprocessing step. The data were

centered and scaled during PCA extraction. By projecting the variables into orthogonal principal components and retaining only those that explained a cumulative variance of at least 80 percent, the analysis reduced multicollinearity and noise.

Metric Space Validity. The study treated the extracted principal component scores as continuous ratio-scale variables. This satisfied the mathematical requirement for Euclidean distance, ensuring that the distance between any two subnational economic units represented a valid geometric metric in the reduced multidimensional component space.

Hierarchical Structure. The use of Ward's Minimum Variance Method assumes that the underlying relationships among subnational economic units can be represented through a nested hierarchical structure. This method was selected because it minimizes within-cluster variance and tends to produce compact clusters suitable for defining economic typologies.

Clustering Procedure. To generate structural economic typologies, the clustering process was applied to the retained principal component scores rather than to the raw standardized indicators.

Optimal Cluster Selection. Prior to agglomeration, the optimal number of clusters (k) was determined using the Silhouette Method. This procedure evaluated hierarchical cuts using Ward's method to identify the partition that maximized intra-cluster cohesion and inter-cluster separation.

Dissimilarity Measure. The spatial separation between subnational economic units was measured using Euclidean distance on the retained principal components:

$$d(x, y) = \sqrt{\sum_{j=1}^p (z_{xj} - z_{yj})^2}$$

where:

$d(x, y)$ = Euclidean distance between subnational economic units x and y

z_{xj} = score of unit x on component j

z_{yj} = score of unit y on component j

p = number of retained principal components

Clustering Algorithm. Clusters were formed using Ward's Minimum Variance Method (ward.D2), which minimizes the increase in within-cluster variance at each agglomerative step. The linkage criterion may be expressed as

$$D(C_1, C_2) = \frac{n_1 n_2}{n_1 + n_2} \left| \underline{z}_1 - \underline{z}_2 \right|^2$$

where:

$D(C_1, C_2)$ = distance between clusters C_1 and C_2

n_1, n_2 = number of observations in clusters C_1 and C_2

$\underline{z}_1, \underline{z}_2$ = centroid vectors of the two clusters in principal component space.

Cluster Evaluation and Robustness Validation. To assess the structural integrity of the generated typologies, the clustering output was subjected to internal validation and robustness checks.

Typology profiling. The quality of the clustering solution was evaluated using the Average Silhouette Width, Dunn Index, and Calinski-Harabasz Index. These measures were used to assess cluster cohesion, separation, and overall partition quality.

Robustness Check. As an additional check, the hierarchical clustering solution was compared with a k-means solution using the same number of clusters. Consistency in membership patterns was treated as supporting evidence of typological robustness.

Software and Computational Environment

To ensure computational efficiency and statistical rigor, the study used a two-stage software workflow. Initial data preparation, including data cleaning, restructuring, and organization of the Provincial Product Accounts (PPA) dataset, was carried out in

Microsoft Excel. After data preparation, all statistical analyses were performed using the R Statistical Computing Environment.

The analytical workflow in R utilized several packages, including stats, car, psych, cluster, factoextra, fpc, and effectsize. These packages were used for Multivariate Analysis of Variance (MANOVA), assumption diagnostics, descriptive statistics, Principal Component Analysis (PCA), hierarchical clustering, cluster validation metrics, and effect size estimation. Final cluster memberships were exported as CSV files for subsequent geographic mapping and visualization.

RESULTS AND DISCUSSION

Descriptive Statistics

This chapter presents the empirical findings derived from the analysis of the 2018 to 2024 Provincial Product Accounts. The discussion is organized into three major sections: descriptive statistics, Multivariate Analysis of Variance results, and hierarchical cluster analysis results. The descriptive section first examines annual and sectoral growth patterns across the major economic sectors and their constituent industries. This is followed by the MANOVA results, which evaluate whether significant multivariate differences exist across the six temporal intervals. The final section presents the cluster analysis results, which classify subnational economic units according to similarities in growth velocity and structural stability.

The contraction observed during the 2019 to 2020 interval should be interpreted as the 2020 pandemic-related shock because the growth rate compares 2020 sectoral output with 2019 sectoral output. Although the interval is labeled 2019 to 2020, the decline reflects the disruption that occurred in 2020, when lockdowns, mobility restrictions, business closures, weaker demand, and supply-chain interruptions affected the Philippine economy. Reports from GMA Integrated News and the Philippine News Agency documented the sharp contraction of the Philippine economy in 2020, while the World Bank and AMRO linked the downturn to reduced investment, weaker domestic demand, and pandemic-related disruptions. The Technical Education and Skills Development Authority (TESDA) also identified several pandemic-sensitive sectors, including Manufacturing, Construction, Transportation and Storage, Accommodation and Food Service Activities, Education, and Other Services, as among the activities heavily affected during the pandemic period. This context helps explain the contractions observed in mobility-, production-, and demand-sensitive sectors during the 2019–2020 interval (Philippine News Agency, 2021b;

World Bank, 2020; ASEAN+3 Macroeconomic Research Office, 2024; Technical Education and Skills Development Authority, 2020).

Agriculture, Forestry, and Fishing (AFF). During the 2018 to 2019 interval, Agriculture, Forestry, and Fishing recorded a growth rate of 2.10 percent. It showed strong resistance during the 2019–2020 disruption, registering only a slight contraction of -0.10 percent. Growth remained modest at 0.60 percent in 2020–2021, 0.30 percent in 2021–2022, and 1.40 percent in 2022–2023, before declining to -0.60 percent in 2023–2024. The narrow range of movement indicates that AFF functioned as a stabilizing component of the broader economy, helping sustain basic production even when more cyclical sectors weakened. However, the later decline in 2023–2024 suggests that the sector remained vulnerable to agricultural and climate-related pressures. Reports from GMA Integrated News and the Philippine News Agency linked agricultural losses in 2024 to the effects of El Niño, including damage to crops, livestock, and fisheries (GMA Integrated News, 2024; Philippine News Agency, 2024). Economically, this suggests that AFF contributed to resilience through stability, but it did not act as the main driver of rapid economic expansion.

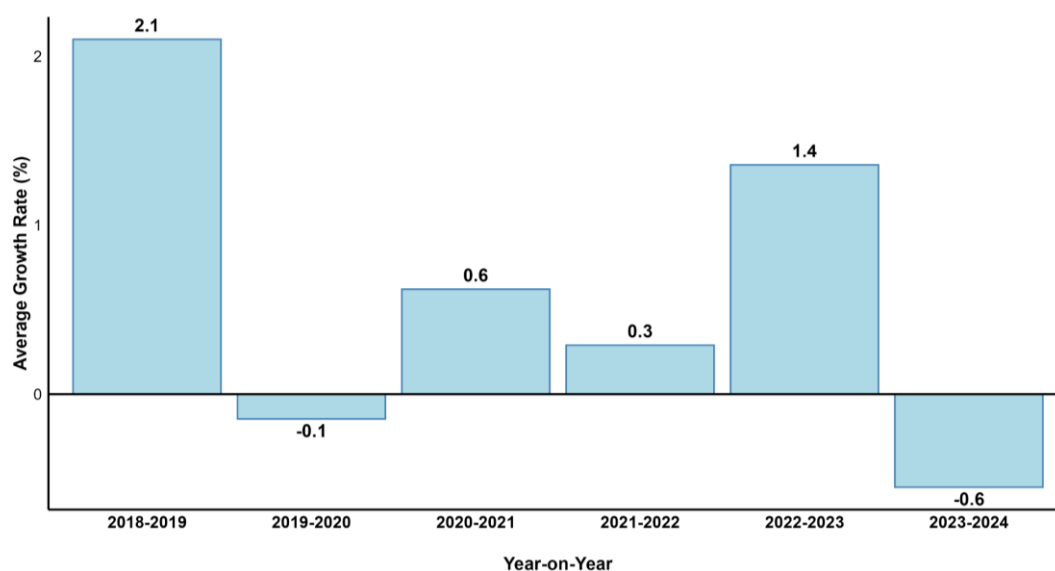


Figure 2. Agriculture, Forestry, and Fishing (AFF)

Industry. This sector includes Mining and Quarrying, Manufacturing, Construction, and Electricity, Steam, Water and Waste Management. Compared with Agriculture, Forestry, and Fishing, these activities generally showed greater sensitivity to economic disruption, but they also displayed stronger rebound patterns during the recovery period. This indicates that the industrial sector played an important role in post-disruption recovery, although the extent of resilience differed across its constituent subsectors.

Mining and Quarrying. This sector recorded 2.30 percent growth in 2018–2019 before contracting sharply to -18.80 percent in 2019–2020. This decline is consistent with the broader disruption experienced by production-oriented and export-linked activities during the early pandemic period, when restrictions affected business operations, logistics, and demand conditions. TESDA also identified Mining and Quarrying as one of the sectors affected during the 2019–2020 pandemic period (World Bank, 2020; Philippine News Agency, 2021b; Technical Education and Skills Development Authority, 2020). The sector then rebounded strongly to 19.00 percent in 2020–2021, remained elevated at 12.00 percent in 2021–2022, and moderated to 6.40 percent and 7.00 percent in the final two intervals. This pattern indicates substantial temporal fluctuation together with strong recovery responsiveness. Economically, the results suggest that Mining and Quarrying was highly exposed to sudden shocks but remained capable of recovering once production and market conditions improved. However, its volatility limits its reliability as a stable foundation for long-term resilience.

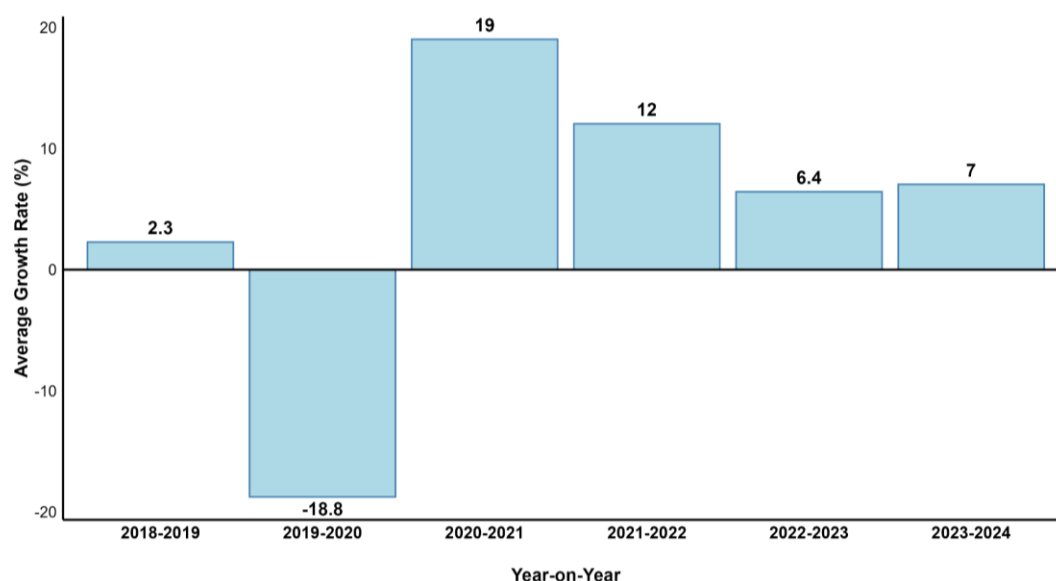


Figure 3. Mining and Quarrying

Manufacturing. This sector recorded 4.50 percent growth in 2018–2019 before declining to -5.50 percent in 2019–2020. The contraction reflects the sector’s vulnerability to disrupted production, weaker demand, and supply-chain interruptions during the early pandemic period, with TESDA also identifying Manufacturing as among the sectors affected by pandemic-related labor market disruption (World Bank, 2020; ASEAN+3 Macroeconomic Research Office, 2024; Technical Education and Skills Development Authority, 2020). It recovered to 7.20 percent in 2020–2021, moderated to 4.20 percent in 2021–2022, slowed further to 1.20 percent in 2022–2023, and rose again to 4.50 percent in 2023–2024. This suggests moderate temporal sensitivity with an eventual return to positive growth, although the recovery path was uneven rather than continuous. Economically, manufacturing retained enough adaptive capacity to resume growth during recovery, but its rebound weakened when broader industrial momentum slowed.

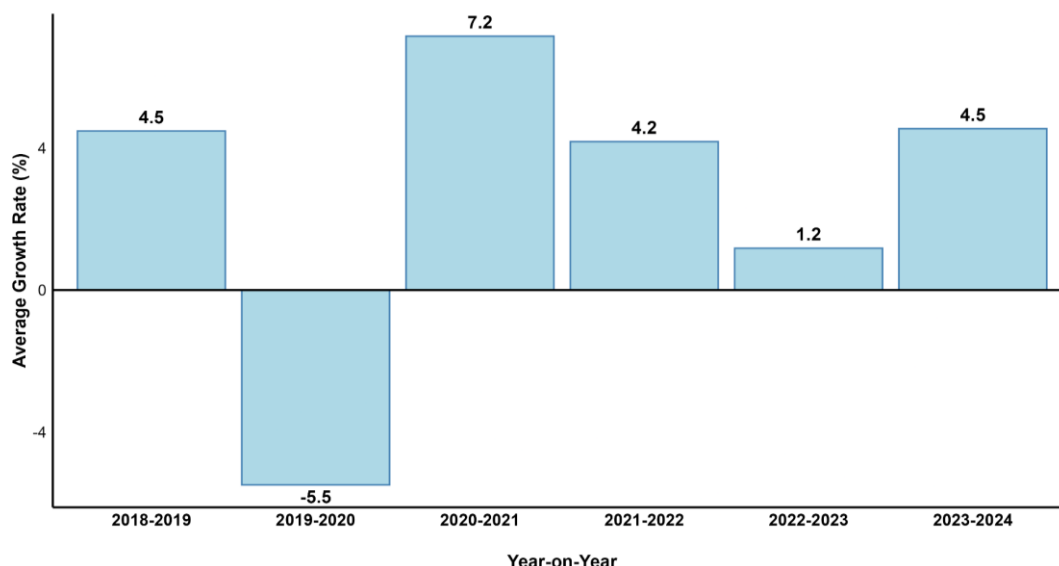


Figure 4. Manufacturing

Electricity, Steam, Water and Waste Management (ESWWM). This sector displayed comparatively stable performance throughout the study period. It recorded 6.30 percent growth in 2018–2019 and only a slight decline of -0.50 percent in 2019–2020. The small contraction suggests that utility-related activities remained relatively resilient because electricity, water, and waste management services continued to be needed by households, firms, and public institutions even during periods of restriction. TESDA’s sectoral discussion and local pandemic-impact assessments support this interpretation, since essential services such as utilities continued to play a necessary role during the quarantine period (Technical Education and Skills Development Authority, 2020; UP Resilience Institute, 2020). Growth remained positive at 4.10 percent in 2020–2021 and 4.90 percent in 2021–2022, before strengthening to 6.50 percent in 2022–2023 and 7.50 percent in 2023–2024. Among the industrial activities, ESWWM showed the greatest overall stability. Economically, this indicates that utility-related infrastructure served as a support sector that helped sustain the continuity of broader economic activity during disruption and recovery.

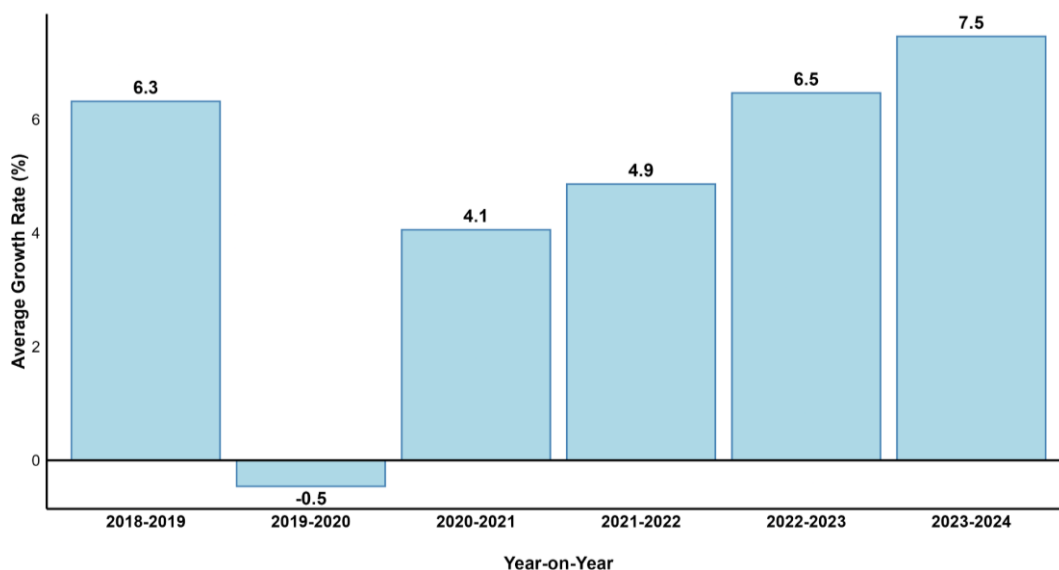


Figure 5. Electricity, Steam, Water and Waste Management (ESWWM)

Construction. This sector recorded 7.10 percent growth in 2018–2019 before contracting sharply to -22.80 percent in 2019–2020. This decline is consistent with the suspension or delay of construction activity during strict quarantine periods, when mobility restrictions, workplace safety rules, and disrupted material flows affected both public and private projects. TESDA identified Construction as one of the sectors heavily affected during the pandemic period, while the Department of Finance emphasized the role of the Build, Build, Build program in supporting recovery. Budget and development reports later showed renewed public spending and investment activity during the recovery period (Technical Education and Skills Development Authority, 2020; Department of Finance, 2020; Department of Budget and Management, 2021; Asian Development Bank, 2022a). The sector rebounded to 11.90 percent in 2020–2021, peaked at 13.20 percent in 2021–2022, and remained high at 9.70 percent and 10.10 percent in the final two intervals. This pattern reflects strong temporal sensitivity and high recovery velocity. Economically, Construction behaved as a cyclical growth sector: highly vulnerable during contraction periods but highly responsive during reopening and renewed investment activity.

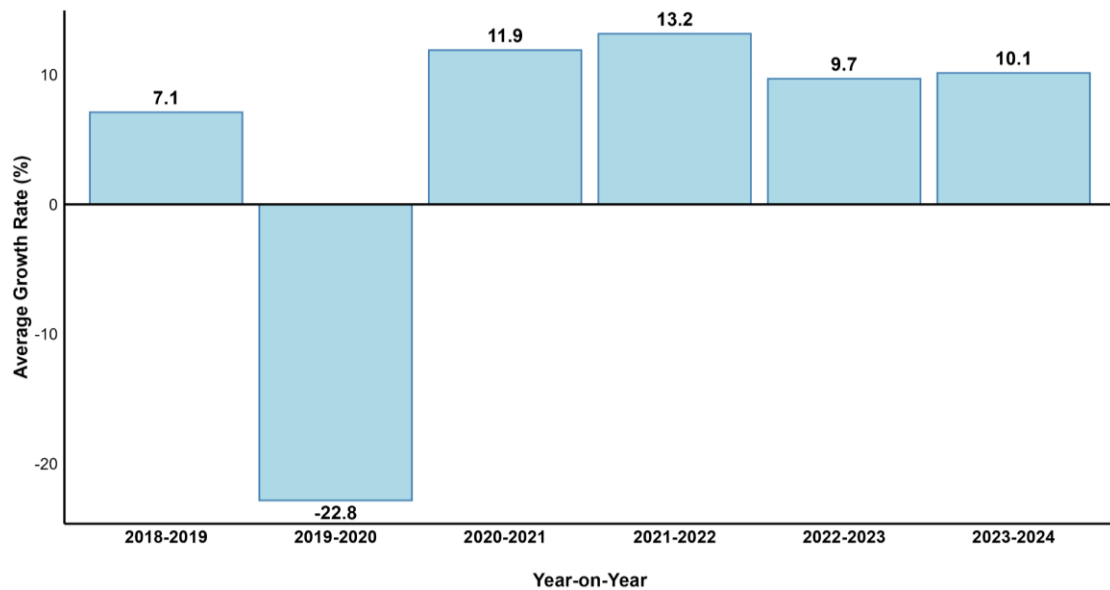


Figure 6. Construction

Services. This sector contains the most diverse set of growth trajectories across the study period. Several service industries experienced sharp declines during 2019–2020, followed by uneven but often substantial rebounds during the later intervals. Some activities, such as Information and Communication and Financial and Insurance Activities, remained positive even during the disruption period, while others, such as Accommodation and Food Service Activities, Transportation and Storage, and Other Services, showed severe contraction followed by multi-stage recovery. This variation indicates that service resilience was uneven, with some activities showing strong continuity and others displaying pronounced shock sensitivity.

For descriptive purposes, the Services sector is discussed through its major PPA service subsectors. However, for MANOVA and cluster analysis, these service subsectors were consolidated into one Services variable to maintain the six-sector analytical framework.

Wholesale and Retail Trade. This sector recorded 7.30 percent growth in 2018–2019 before declining to -4.60 percent in 2019–2020. This contraction is consistent with the effect of quarantine restrictions on malls, non-essential stores, public markets, and face-to-face commercial activity during the early pandemic period. TESDA also identified Wholesale and Retail Trade as one of the sectors affected by pandemic-related disruption (GMA Integrated News, 2020; Philstar, 2021; Technical Education and Skills Development Authority, 2020). Reduced household income and weaker consumer confidence also limited retail activity during the same period. The sector recovered modestly to 4.00 percent in 2020–2021, peaked at 10.70 percent in 2021–2022, and then stabilized at 6.20 percent and 6.40 percent in the final two intervals. This pattern suggests moderate disruption followed by delayed but effective recovery. Economically, trade activity remained closely tied to household consumption, market reopening, and the restoration of consumer demand.

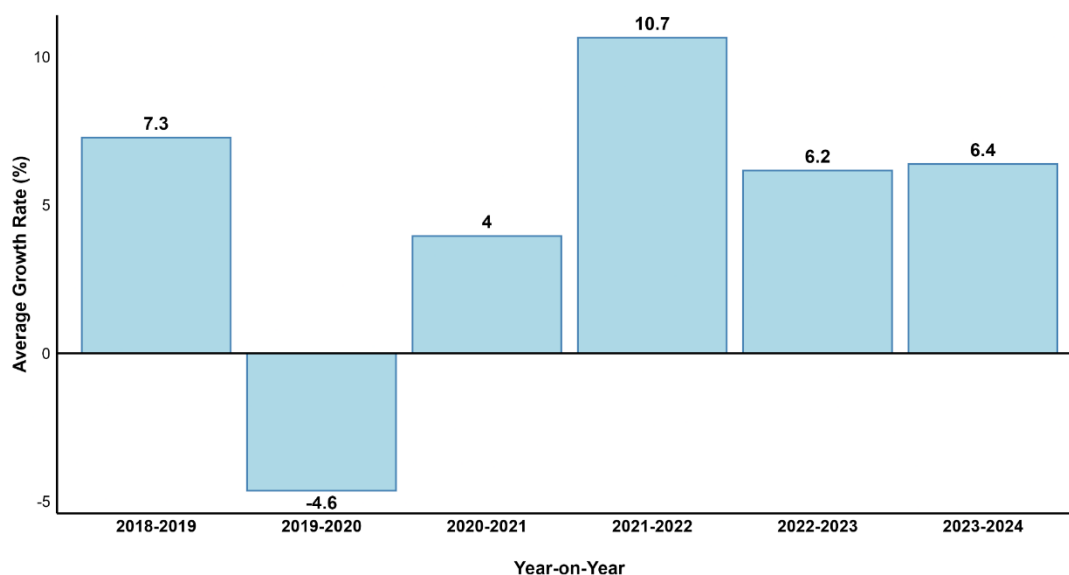


Figure 7. Wholesale and Retail Trade

Transportation and Storage. This Sector recorded 4.80 percent growth in 2018–2019 before experiencing a deep contraction of -28.30 percent in 2019–2020. This reflects the sector's strong dependence on mobility, logistics, travel, and distribution systems, all of which were heavily affected by pandemic restrictions. The sector recovered modestly to 5.40 percent in 2020–2021, then expanded strongly to 24.20 percent in 2021–2022 as mobility restrictions eased and transport activities resumed. TESDA identified Transportation and Storage as one of the sectors strongly affected during the pandemic period, while development and transport-related sources support the later recovery of domestic mobility, economic activity, and logistics movement (Technical Education and Skills Development Authority, 2020; Asian Development Bank, 2022a; World Bank, 2022; De Freitas, 2022). Growth remained elevated at 17.60 percent in 2022–2023 before moderating to 8.20 percent in 2023–2024. Economically, this indicates that Transportation and Storage was highly vulnerable to restrictions but became strongly recovery-responsive once distribution systems and market access were restored.

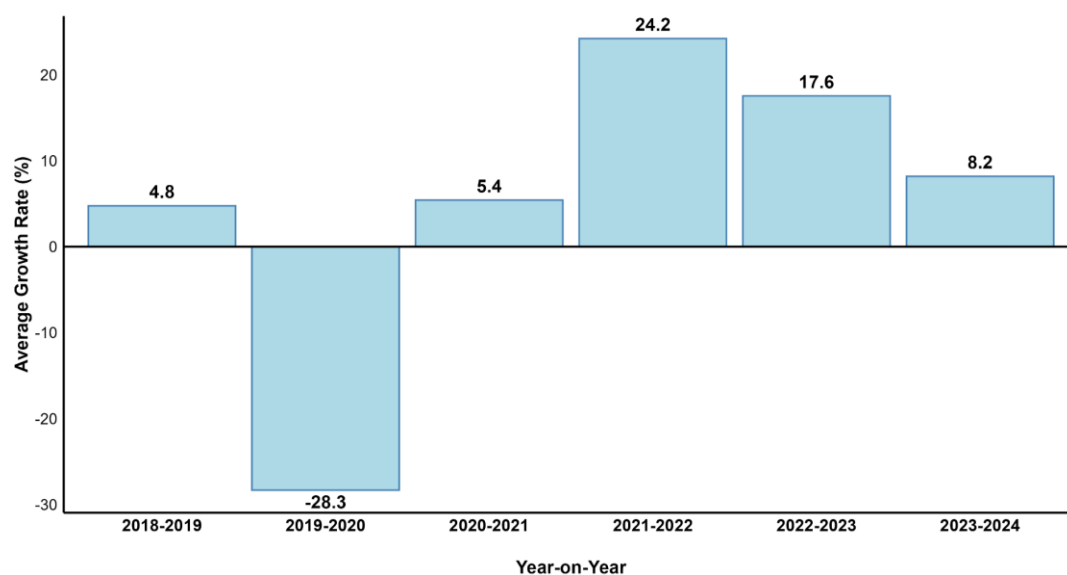


Figure 8. Transportation and Storage

Accommodation and Food Service Activities. This Sector recorded 7.20 percent growth in 2018–2019 before falling sharply to -41.10 percent in 2019–2020. This was one of the deepest contractions among the service subsectors. The decline is consistent with the effect of travel restrictions, reduced tourism activity, hotel losses, and worker displacement in the hospitality sector during the early pandemic period. TESDA also identified Accommodation and Food Service Activities as one of the sectors severely affected by pandemic restrictions (Technical Education and Skills Development Authority, 2020; Cabuenas, 2020a; Cordero, 2020c). The sector recovered to 7.40 percent in 2020–2021, surged to 31.30 percent in 2021–2022, remained high at 19.90 percent in 2022–2023, and moderated to 10.90 percent in 2023–2024. This pattern reflects extreme temporal sensitivity combined with a strong but staged rebound. Economically, the sector was among the most vulnerable to disruptions in face-to-face consumption and tourism, but it also showed strong recovery once mobility, consumer confidence, and hospitality activity improved.

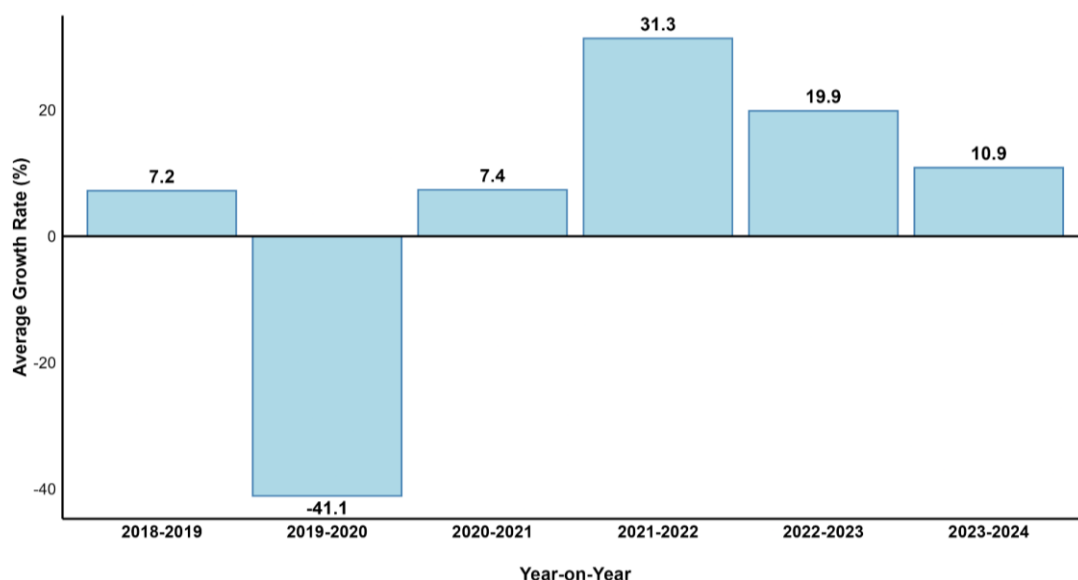


Figure 9. Accommodation and Food Service Activities

Information and Communication. This sector remained comparatively resilient throughout the study period. It recorded 8.70 percent growth in 2018–2019 and stayed positive at 5.60 percent during the 2019–2020 disruption. It then reached 9.40 percent in 2020–2021, followed by 8.20 percent, 4.80 percent, and 5.00 percent in the succeeding intervals. This sustained growth is consistent with the accelerated use of digital platforms, remote work, online learning, e-commerce, and digital communication during and after the pandemic period. TESDA’s sectoral discussion supports the positive performance of Information and Communication during the 2019–2020 interval, while later digital economy reporting shows the continued importance of digital services in the Philippines (Technical Education and Skills Development Authority, 2020; Philippine News Agency, 2021a; International Trade Administration, 2024). Economically, this indicates that digital connectivity became an important stabilizing force for business operations, communication, and service access. The sector’s positive growth across all six intervals suggests that it contributed to adaptive capacity within subnational economies.

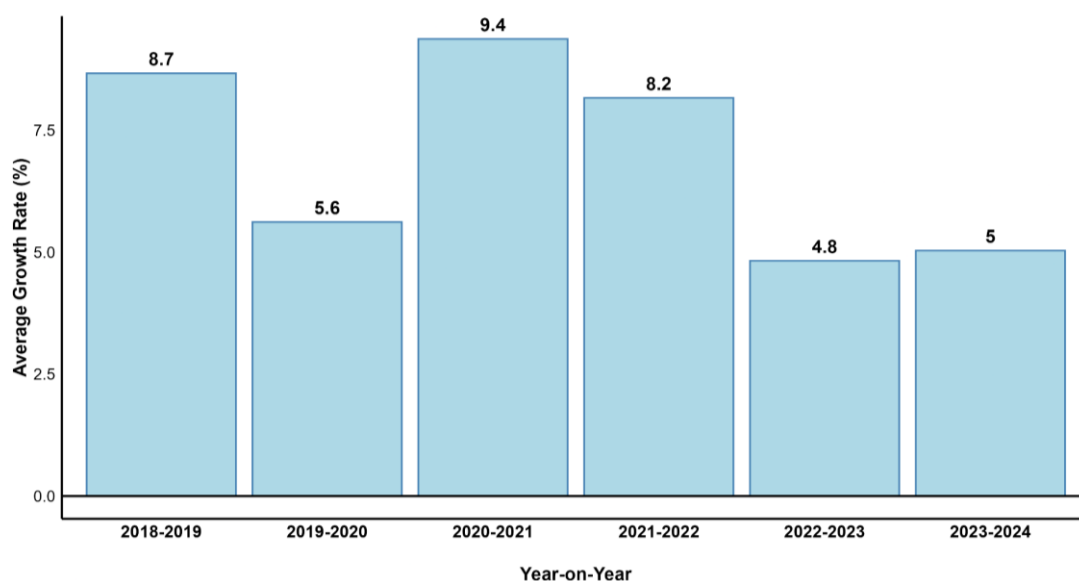


Figure 10. Information and Communication

Financial and Insurance Activities. This sector showed consistently positive growth throughout the study period. It recorded 13.40 percent growth in 2018–2019, moderated to 6.70 percent in 2019–2020, and remained positive at 5.10 percent in 2020–2021. The sector’s resilience is consistent with the continued role of financial intermediation, liquidity support, and regulatory measures during the pandemic period. TESDA’s sectoral discussion supports the positive performance of Financial and Insurance Activities during the 2019–2020 interval. Reports from the Bangko Sentral ng Pilipinas and related news accounts also noted that monetary and regulatory measures helped sustain financial system stability despite weaker credit demand during the disruption period (Technical Education and Skills Development Authority, 2020; Bangko Sentral ng Pilipinas, 2022; Cordero, 2020d; Cordero, 2020e). Growth later increased to 7.50 percent in 2021–2022 and 9.10 percent in both 2022–2023 and 2023–2024. Economically, this indicates that financial and insurance activities functioned as a stable support sector by helping maintain continuity in transactions, credit systems, and broader economic exchange.

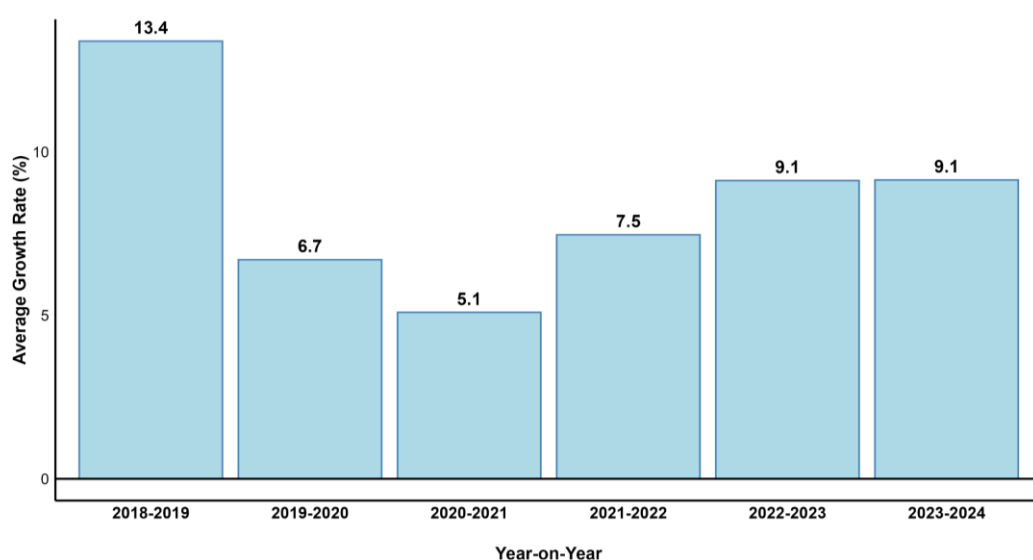


Figure 11. Financial and Insurance Activities

Real Estate Activities. This sector recorded 3.00 percent growth in 2018–2019 before contracting to -13.10 percent in 2019–2020. The decline reflects the effect of restrictions on property viewing, construction-related activity, office transactions, and household investment decisions during the early pandemic period. TESDA identified Real Estate Activities as one of the sectors affected during the 2019–2020 pandemic period. During the same period, the Bangko Sentral ng Pilipinas raised the real estate loan limit for banks to help support property lending and cushion the sector from deeper stress (Technical Education and Skills Development Authority, 2020; Villanueva, 2020). The sector returned to positive growth at 1.60 percent in 2020–2021, improved to 3.80 percent in 2021–2022, recorded 3.10 percent in 2022–2023, and rose to 4.40 percent in 2023–2024. Compared with more flexible service industries, Real Estate recovered more gradually because it depends on longer investment cycles and delayed market adjustment. Economically, the sector followed a slower but steadier recovery path.

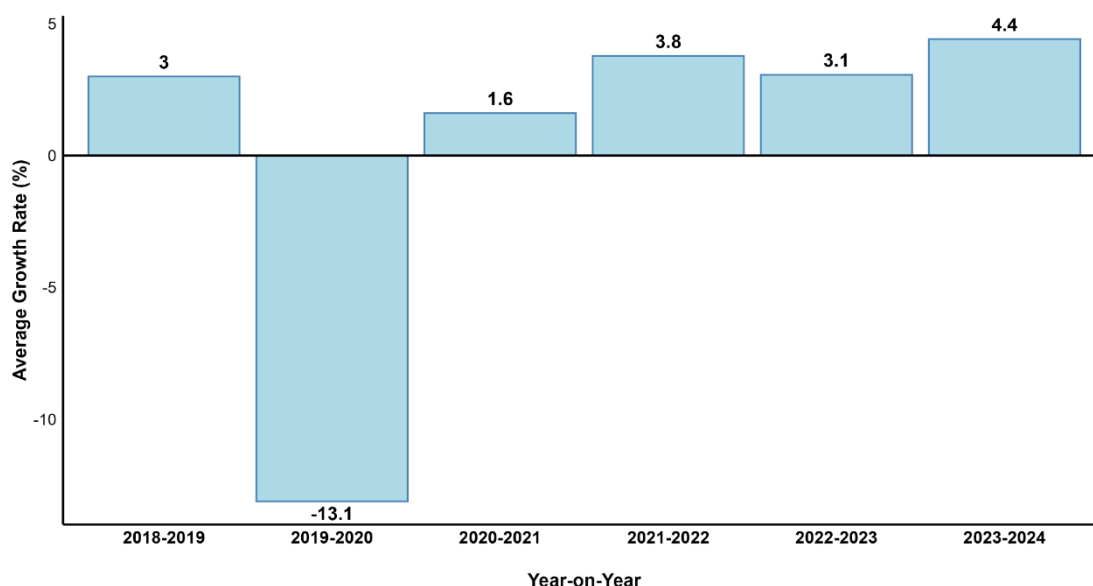


Figure 12. Real Estate Activities

Professional and Business Services. This sector recorded 3.10 percent growth in 2018–2019 before contracting to -6.80 percent in 2019–2020. This decline reflects the effect of disrupted office-based work, reduced business activity, and adjustment costs during the early pandemic period. However, some business support activities were able to adjust through remote work, digital coordination, and the gradual reopening of firms. TESDA identified Professional and Business Services as one of the sectors affected during the 2019–2020 pandemic period. Reports on the pandemic’s effect on regional economic resilience and later GDP revisions also suggest that business-related services gradually recovered as firms adapted to changing operating conditions (Technical Education and Skills Development Authority, 2020; ABS-CBN News, 2020; Cabuenas, 2021). The sector recovered to 5.90 percent in 2020–2021 and then strengthened to 8.30 percent, 8.50 percent, and 10.10 percent in the final three intervals. Economically, this indicates that Professional and Business Services regained strength as firms resumed operations and demand for support, coordination, and specialized services increased.

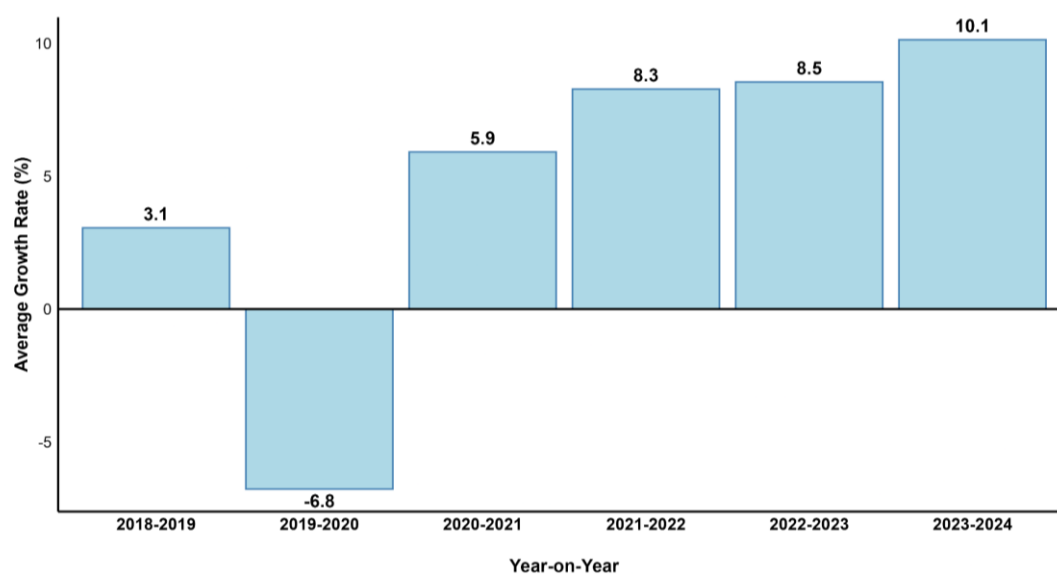


Figure 13. Professional and Business Services

Public Administration and Defense. This sector recorded 13.00 percent growth in 2018–2019 before moderating to 3.50 percent in 2019–2020. Unlike many private-sector activities, this sector remained positive because government services, public administration, and emergency response functions continued during the pandemic period. Budget and disbursement documents show that government spending supported public services, pandemic response, and recovery programs during the disruption and recovery years (Department of Budget and Management, 2021; Department of Budget and Management, 2022; Department of Budget and Management, 2023). The sector rebounded to 6.70 percent in 2020–2021, moderated to 4.40 percent in 2021–2022, and remained positive at 5.50 percent and 5.60 percent in the final two intervals. Economically, this pattern suggests that Public Administration and Defense served as a stabilizing force by sustaining institutional activity and government-supported functions during periods of uncertainty.

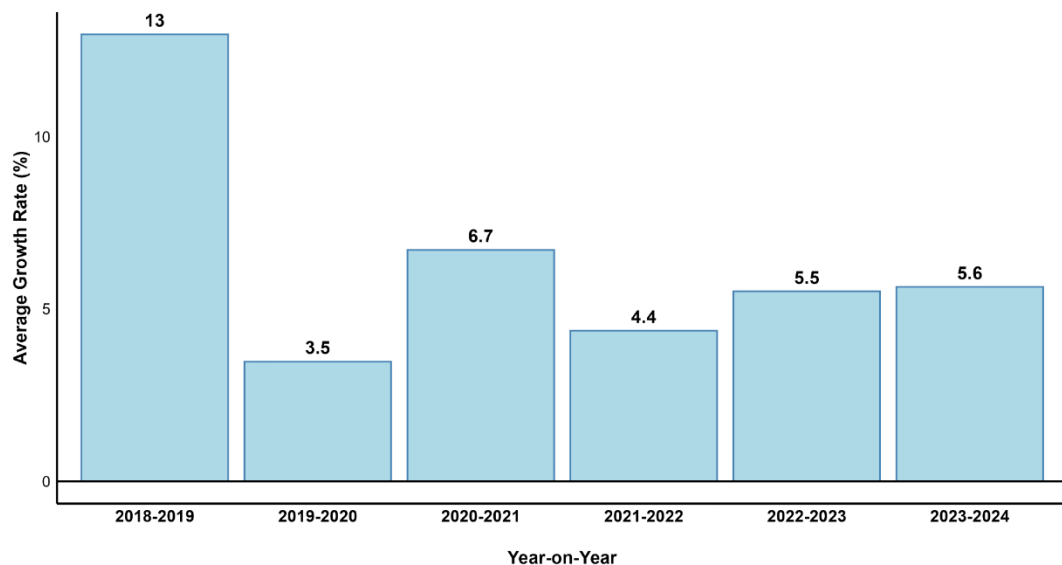


Figure 14. Public Administration and Defense

Education. This sector recorded 6.10 percent growth in 2018–2019 before contracting to -9.10 percent in 2019–2020. This decline reflects the disruption caused by school closures, the suspension of face-to-face classes, and the sudden shift to remote and modular learning during the pandemic. TESDA also identified Education as one of the sectors affected during the pandemic period. Reports during this period showed that many students faced unequal access to learning resources, while many learners transferred from private to public schools because of financial pressure (Magsambol, 2020a; Magsambol, 2020b). Later reports also noted permanent closures among private schools during the pandemic period (Hernando-Malipot, 2022). The sector rebounded to 8.50 percent in 2020–2021 and remained positive at 6.80 percent, 5.60 percent, and 3.90 percent in the succeeding intervals. Economically, this suggests that Education experienced a temporary disruption but gradually adapted as learning systems and institutional operations adjusted.

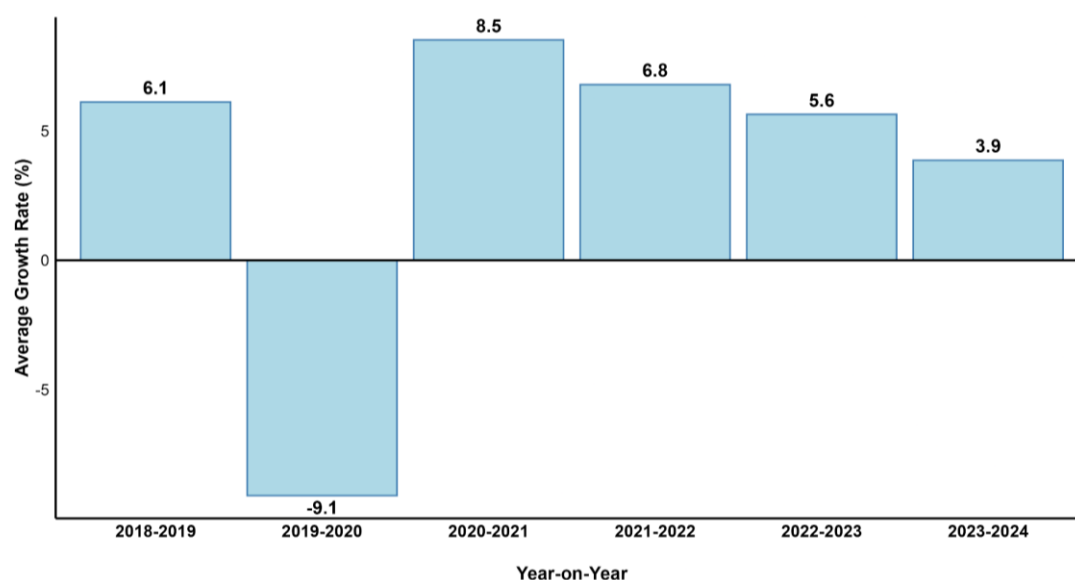


Figure 15. Education

Human Health and Social Work Activities. This sector recorded 5.50 percent growth in 2018–2019 before declining only slightly to -1.90 percent in 2019–2020. The limited contraction suggests that the sector remained relatively necessary during the health crisis, even though routine consultations and non-COVID-19 health services were disrupted. Studies on the health system impact of COVID-19 in the Philippines showed that the pandemic placed strong pressure on health facilities, service delivery, and emergency response systems (Center for Global Development, 2021; UP Resilience Institute, 2020). The sector then surged to 16.90 percent in 2020–2021, reflecting the expansion of health-related services, emergency response, and public health interventions. Growth moderated to 4.90 percent in 2021–2022 and then strengthened again to 7.20 percent and 11.50 percent in the final two intervals. Economically, this indicates that health and social work activities became increasingly important during and after the disruption period, contributing to adaptive and recovery-oriented service provision.

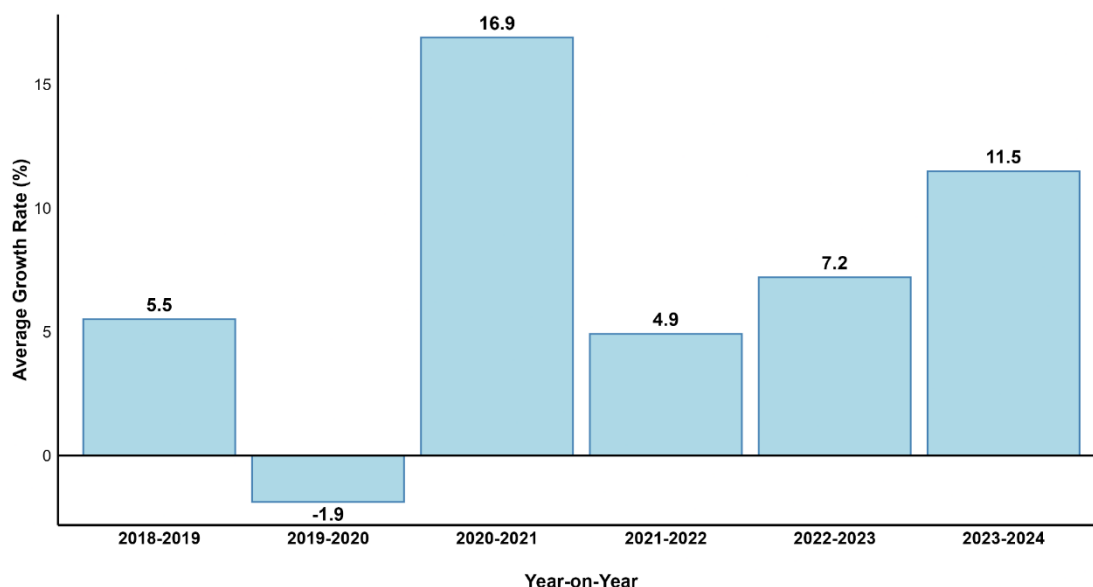


Figure 16. Human Health and Social Work Activities

Other Services. These sectors recorded 11.00 percent growth in 2018–2019 before contracting by -36.70 percent in 2019–2020. The sharp contraction reflects the sector’s dependence on direct personal interaction, local community activity, and small service establishments, many of which were affected by social distancing and mobility restrictions. TESDA identified Other Services as one of the sectors strongly affected during the pandemic period, while local assessments of the pandemic’s impact on micro, small, and medium enterprises reported that small service-oriented businesses faced severe operational disruption during the early pandemic period (Technical Education and Skills Development Authority, 2020; Institute for Climate and Sustainable Cities, 2020). The sector recovered only slightly to 2.50 percent in 2020–2021, then expanded strongly to 24.80 percent in 2021–2022 as routine local economic and social interactions resumed. Growth remained elevated at 20.20 percent in 2022–2023 and moderated to 8.60 percent in 2023–2024. This pattern indicates strong temporal sensitivity alongside a delayed but substantial rebound. Economically, Other Services was heavily affected by disruption, but its recovery depended on the restoration of local mobility, face-to-face activity, and everyday community-based services.

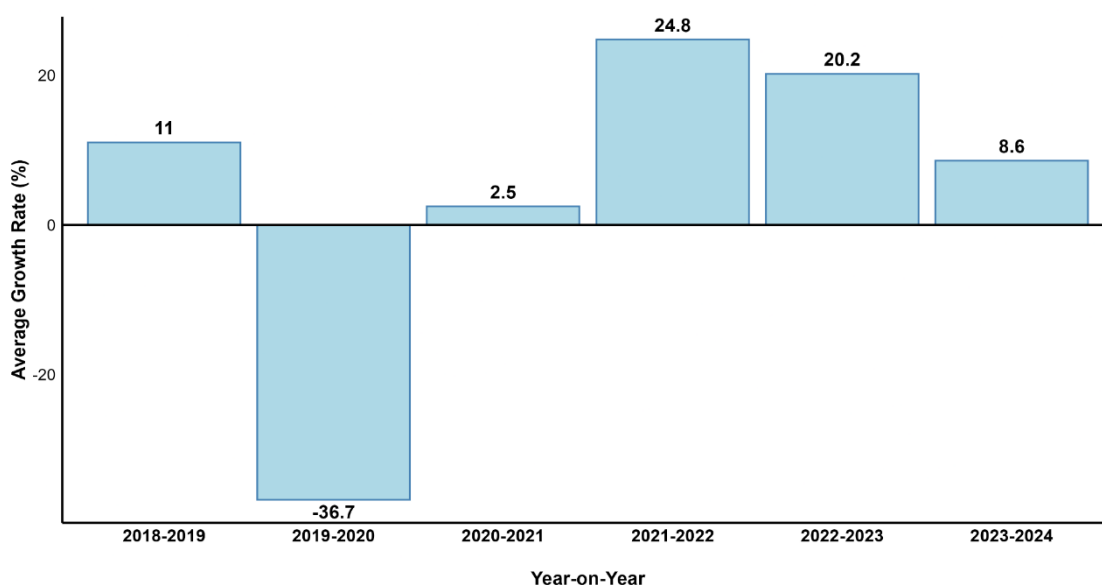


Figure 17. Other Services

The descriptive findings support the view that subnational resilience is closely tied to sectoral composition rather than to aggregate growth alone. The relative stability of Agriculture, Forestry, and Fishing and the stronger volatility of construction- and service-linked sectors suggest that local economies differ not only in their growth levels but also in the structural character of their response to disruption and recovery. This pattern is consistent with the literature emphasizing that resilience is shaped by the internal composition of productive sectors and their differing capacities for resistance, recovery, and adaptation (Martin & Sunley, 2014). It also supports the argument that diversification and inter-sectoral balance matter for resilience, since sectors associated with wider economic linkages, such as utilities, trade, transport, and services, displayed more varied but also more recovery-responsive trajectories (He, Miao, & Qureshi, 2022; Giannakis, Bruggeman, & Mamuneas, 2024). In the Philippine context, the uneven performance of industrial and service subsectors likewise reflects the observation that structural transformation remains uneven across local economies, with some areas more capable of shifting toward higher-productivity sectors than others (Serafica, Vergara, & Oren, 2021; Custodio & Sombilla, 2025).

MANOVA Results

Assessment of Multivariate Outliers. The initial diagnostic procedure involved the computation of Mahalanobis Distance to identify multivariate outliers within the six-sector growth matrix ($N = 702$). This statistic measures the distance of each subnational-unit-by-interval observation from the multivariate centroid while accounting for the covariance structure among the variables. By incorporating the relationships among the six economic sectors, Mahalanobis Distance detects unusual growth profiles based on the combined sectoral behavior rather than isolated deviations.

Using a conservative chi-square threshold at $p < 0.01$ with six degrees of freedom, a total of 18 observations were identified as multivariate outliers.

These observations were retained in the analysis because, within a macroeconomic dataset covering disruption and recovery, extreme values may reflect actual structural shifts rather than measurement errors. Their retention preserves the full range of observed economic variation across the study period.

Table 1. Detection of multivariate outliers

PARAMETER	VALUE
Total Observations	702
Degrees of Freedom	6
Critical Threshold ($p < 0.01$)	22.46
Outliers Detected	18
Outliers Retained	18

Legend:

MD = Mahalanobis Distance; MD > 22.45774 indicates a multivariate outlier at $p < 0.01$.

Assessment of Multivariate Normality. Multivariate normality was assessed using Mardia's test of multivariate skewness and kurtosis. The null hypothesis of multivariate normality was rejected for both skewness and kurtosis because the p-values were less than 0.05. The Mardia skewness coefficient, $b_{1,p}$, was 70.69 with a test statistic of 8270.28, while the Mardia kurtosis coefficient, $b_{2,p}$, was 167.97 with a test statistic of 162.21. These results indicate that the joint distribution of the six sectoral growth variables was not multivariate normal. In practical terms, the data showed asymmetry and heavy-tailed behavior, which is expected in economic growth data affected by shock and recovery periods. Because this assumption was violated, Pillai's Trace was used as the primary MANOVA statistic because it is more robust under non-normal conditions.

Table 2. Test of Multivariate Normality

METRIC	(B)	TEST STATS	P	DECISION	INTERPRETATION
Skewness	70.69	8270.28	<0.01**	Reject H_0	Highly Significant
Kurtosis	167.97	162.21	<0.01**	Reject H_0	Significant

Legend:

$p < 0.05$ indicates significant departure from multivariate normality.

NOTE: Mardia's skewness and kurtosis tests were used.

Assessment of Homogeneity of Covariance Matrices. The equality of covariance matrices across the six temporal intervals was assessed using Box's M

test. The results yielded an approximate chi-square value of 1090.29 with 105 degrees of freedom and $p < 0.01$. Therefore, the null hypothesis of equal covariance matrices was rejected.

This indicates that the covariance structure among the six economic sectors changed across the six temporal intervals. Such variation is consistent with the large-scale structural shifts associated with disruption and recovery. Although the assumption of covariance homogeneity was violated, Pillai's Trace remained the appropriate omnibus statistic because of its robustness to this condition.

Table 3. Test for Equality of Covariance Matrices

TEST	X ²	DF	P	DECISION	INTERPRETATION
Box's M	1090.29	105	<0.01**	Reject H ₀	Highly Significant

Legend:

$p < 0.05$ indicates unequal covariance matrices across temporal intervals.

NOTE: H₀ = the covariance matrices are equal across groups.

Assessment of Multivariate Linearity. Multivariate linearity was assessed using a scatterplot matrix of the six economic sector growth variables. This assumption is important in Multivariate Analysis of Variance (MANOVA) because the procedure relies on the variance-covariance structure of the dependent variables. When the relationships among dependent variables are approximately linear, the covariance matrix can more accurately represent the joint behavior of the variables. By contrast, strong nonlinear relationships may distort that structure and weaken the validity of the multivariate test results.

Visual inspection of the scatterplot matrix showed that most pairwise relationships followed approximately elliptical or linear patterns rather than systematic curvilinear forms. No strong U-shaped or S-shaped relationships were observed among the six economic sectors. Although the strength of association varied across sector pairs, the overall distribution of the plotted points suggested that the assumption of approximate linearity was reasonably met.

This result indicates that the sectoral growth variables maintained an appropriate multivariate structure for MANOVA. Since no strong nonlinear patterns were evident, the variance-covariance relationships among the six economic sectors were considered suitable for inclusion in the multivariate model. Therefore, the assumption of multivariate linearity was considered satisfied.

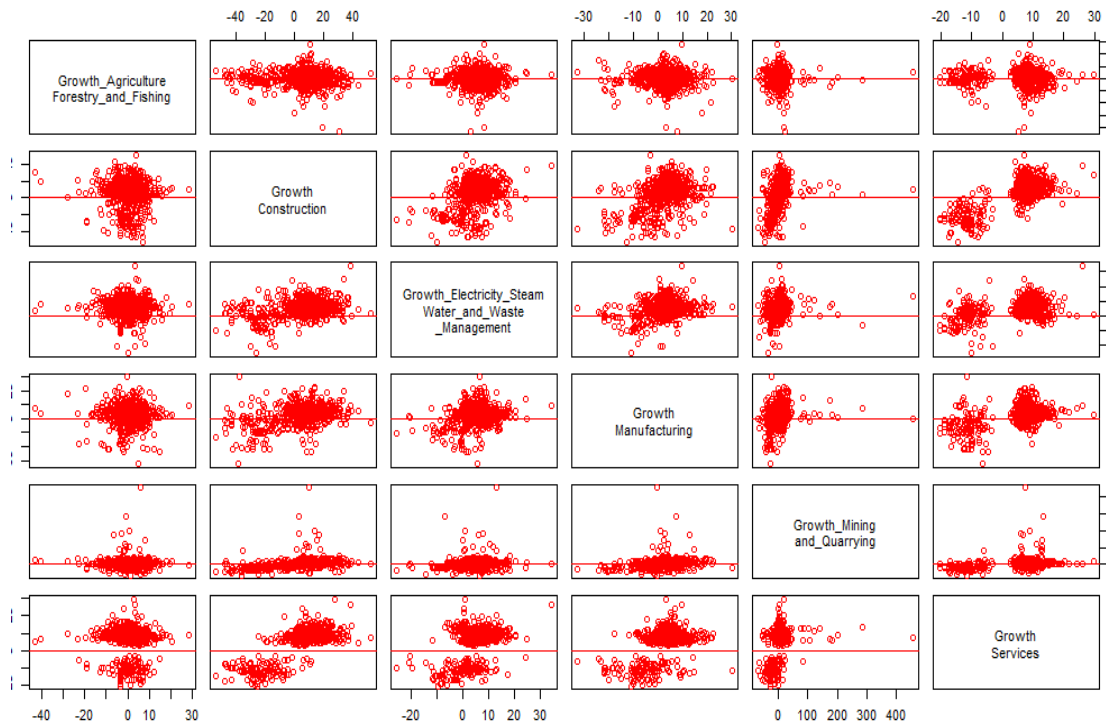


Figure 18. Assessment of Multivariate Linearity

Assessment of Multicollinearity. Multicollinearity was assessed using a Pearson correlation matrix of the six economic sector growth variables. This diagnostic was used to determine whether any pair of dependent variables was too highly correlated to provide distinct information in the MANOVA model. In this study, the correlation matrix did not show problematic multicollinearity because all pairwise correlations remained below the conventional threshold of 0.80.

The highest observed correlation was between Services and Construction at $r = 0.75$. This value indicates a strong positive relationship, meaning that the two sectors tended to move in similar directions across the temporal intervals. However, it did not

exceed the 0.80 threshold, so it was not considered problematic. The remaining sectoral correlations were lower, indicating that the six sectors were related but not redundant.

This result implies that each sector retained sufficient distinctiveness to justify its inclusion as a separate dependent variable in the MANOVA model. In other words, the six economic sectors shared some common movement, which is expected in a macroeconomic system, but not to the extent that one sector could be treated as a near-duplicate of another. Therefore, the correlation structure was considered appropriate for MANOVA, and the assumption of the absence of problematic multicollinearity was regarded as satisfied.

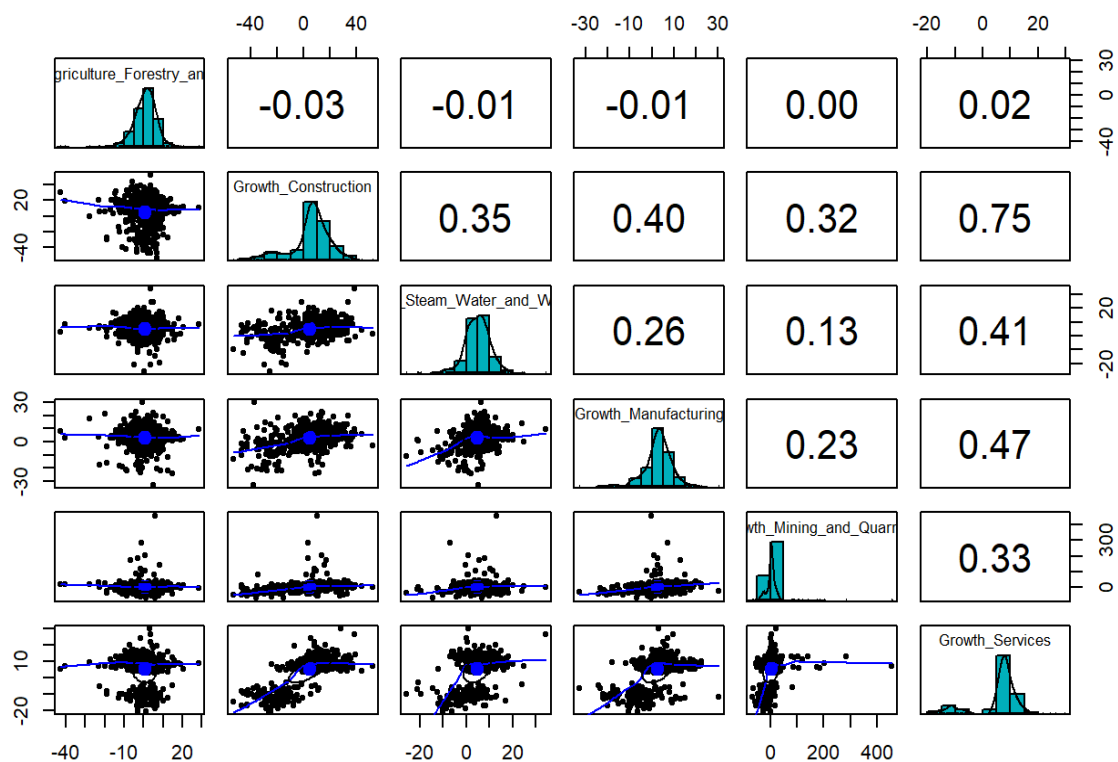


Figure 19. Assessment of Multicollinearity

Assessment of Independent Observations. A residual audit was performed by fitting a multivariate linear model and plotting the residuals against observation order. The residuals appeared randomly distributed around zero, with no obvious trend

or clustering. This pattern provided limited visual support that no strong ordering pattern was present in the residuals. However, because the same subnational economic units appeared across multiple intervals, independence was treated as a simplifying analytical assumption. This assumption was also supported by the research design, where each subnational-unit-by-interval record was treated as a separate analytical observation. Thus, the sectoral growth rates of one subnational economic unit were not assumed to systematically influence those of another.

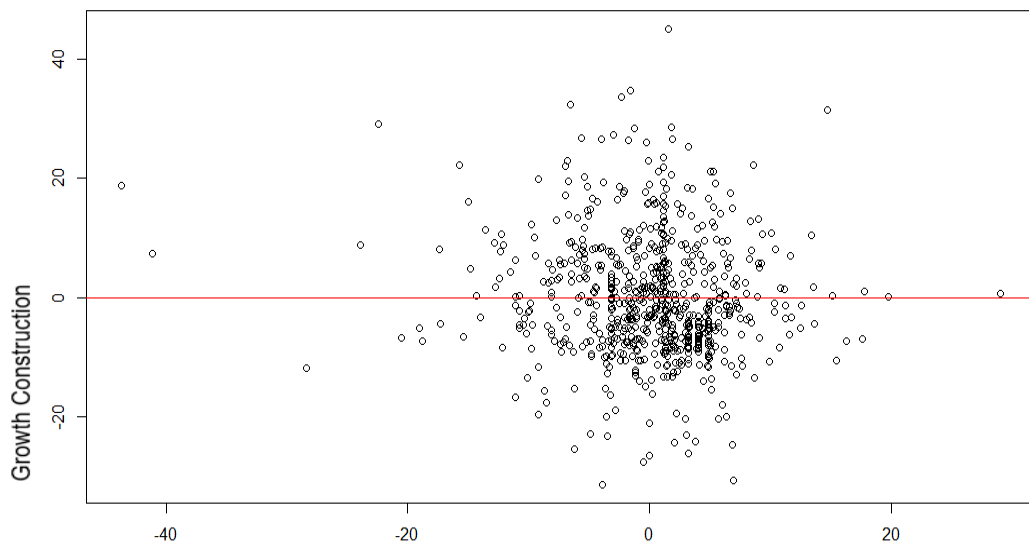


Figure 20. Assessment of Independent Observations

Multivariate Omnibus Test Results. Following assumption checking, a one-way MANOVA was conducted using Pillai's Trace as the omnibus statistic. The results showed a significant multivariate temporal effect, Pillai's Trace = 1.19, $F(30, 3475) = 36.09$, $p < 0.01$. Therefore, the null hypothesis that the multivariate mean vectors of sectoral growth rates were equal across the six temporal intervals was rejected. The partial eta squared value was $\eta^2 = 0.24$, which means that approximately 24 percent of the multivariate variation in sectoral growth structure was associated with temporal differences. This indicates that sectoral growth behavior changed meaningfully across the 2018–2024 period rather than remaining constant over time.

The MANOVA results indicate that the six-sector growth structure changed significantly across the six temporal intervals, confirming that subnational economic performance during 2018–2024 cannot be understood as temporally uniform. This finding supports the study’s validation-to-classification framework by establishing that meaningful multivariate variation existed before cluster formation. In this respect, the results are consistent with the methodological literature arguing that multidimensional economic performance should first be validated as structurally distinct before it is classified into typologies (Stamenković, Milanović, & Janković-Milić, 2021). More broadly, the presence of significant temporal differences also reinforces the resilience literature’s view that economic systems respond to disruption through uneven patterns of resistance, recovery, and re-orientation rather than through a single shared trajectory (Martin & Sunley, 2014). The strong temporal sensitivity observed in Services and Construction, alongside the relatively stable behavior of Agriculture, Forestry, and Fishing, further suggests that sectoral composition influences not only levels of growth but also the pattern of adjustment across time, which is consistent with the sectoral resilience arguments advanced in the literature (He, Miao, & Qureshi, 2022; Giannakis, Bruggeman, & Mamuneas, 2024).

Table 4. Summary of Multivariate Results

VALUE	F	DF	P	η^2	DECISION	INTERPRETATION
1.19	36.09	(30,3475)	<0.01**	0.24	Reject H_0	Highly Significant

Legend:

$p < 0.05$ indicates a significant multivariate temporal effect; η^2 = partial eta squared.

Note: H_0 = the multivariate mean vectors of sectoral growth rates are equal across the six temporal intervals.

Univariate Follow-Up Analysis and Sectoral Sensitivity Hierarchy.

Because the omnibus MANOVA result was significant, univariate follow-up ANOVAs were conducted for each of the six economic sectors. The null hypothesis was rejected for all six sectors because all p-values were below 0.05. Services, Construction, Manufacturing, Electricity, Steam, Water and Waste Management, and Mining and Quarrying were highly significant at $p < 0.01$, while Agriculture, Forestry, and Fishing

was significant at $p = 0.02$. This means that the temporal interval had a statistically detectable effect on the growth behavior of every sector.

The effect sizes showed that the strength of temporal sensitivity differed across sectors. Services had the largest effect size, $\eta^2 = 0.89$, followed by Construction, $\eta^2 = 0.61$. This means that Services and Construction changed most strongly across the six temporal intervals. Manufacturing, $\eta^2 = 0.33$, Electricity, Steam, Water and Waste Management, $\eta^2 = 0.19$, and Mining and Quarrying, $\eta^2 = 0.15$, showed moderate temporal sensitivity. Agriculture, Forestry, and Fishing had the smallest effect size, $\eta^2 = 0.02$, indicating that it remained comparatively stable across the study period.

Economically, these findings suggest that the multivariate temporal effect was driven mainly by service-based and construction-related activity. These sectors were more responsive to changes in mobility, demand, reopening conditions, and investment activity during the disruption and recovery period. Meanwhile, Agriculture, Forestry, and Fishing functioned more as a stabilizing sector because its growth pattern changed less sharply over time. This shows that subnational economic resilience was shaped not only by whether sectors grew or declined, but also by how strongly each sector responded to changing economic conditions.

These findings are consistent with the literature on uneven sectoral transformation. The high temporal sensitivity of Services and Construction supports the view that service concentration and industrial restructuring influence differences in local economic performance (Serafica, Vergara, & Oren, 2021; Custodio & Sombilla, 2025). On the other hand, the lower temporal sensitivity of Agriculture, Forestry, and Fishing supports the idea that agriculture-based sectors may provide continuity during periods of disruption, although they may not generate the same pace of expansion as more diversified sectoral systems (Li et al., 2025; Martin & Sunley, 2014).

Table 5. Technical Summary of Univariate Findings

INDUSTRY SECTOR	F-STAT.	η^2	P	DECISION	INTERPRETATION
Services	1135.80	0.89	<0.01**	Reject H_0	Highly Significant
Construction	213.24	0.61	<0.01**	Reject H_0	Highly Significant
Manufacturing	68.74	0.33	<0.01**	Reject H_0	Highly Significant
Electricity, Steam, Water, & Waste Management	33.47	0.19	<0.01**	Reject H_0	Highly Significant
Mining & Quarrying	23.82	0.15	<0.01**	Reject H_0	Highly Significant
Agriculture, Forestry, and Fishing	2.80	0.02	0.02*	Reject H_0	Significant

Legend:

$p < 0.01$ = highly significant; $0.01 \leq p < 0.05$ = significant; $p \geq 0.05$ = not significant.

NOTE: H_0 = no significant sectoral growth difference across temporal intervals. Higher η^2 values indicate stronger temporal sensitivity.

Cluster Analysis Result

Cluster analysis was conducted to identify structural patterns in subnational economic performance. Because clustering is sensitive to extreme values, the structural stability indicators were first examined using the coefficient of variation. Several coefficients of variation variables showed extreme positive skewness, with large maximum values resulting from near-zero mean growth rates.

To reduce the influence of these mathematically inflated values, an Interquartile Range winsorization procedure was applied. The results show substantial reduction in the maximum coefficient of variation values across sectors while preserving the full set of observations used in clustering.

Variable Standardization and Dimensionality Reduction. Following outlier treatment, all variables were standardized to ensure comparability across sectors. The feature space consisted of 12 indicators: mean growth rate and coefficient of variation for each of the six economic sectors.

Explained Variance of the Principal Components. Because the twelve cluster indicators may be intercorrelated, Principal Component Analysis (PCA) was applied before clustering in order to reduce dimensional complexity and minimize redundancy among the variables. PCA transforms the original indicators into a smaller set of uncorrelated components that retain most of the information contained in the dataset. This step was important because the clustering procedure relied on Euclidean

distance, which can be affected by multicollinearity and overlapping information across variables.

Table 6 presents the eigenvalues, percentage of variance explained, and cumulative variance explained by each principal component. The first principal component (PC1) explained 26.41 percent of the total variance, followed by PC2 with 14.94 percent, PC3 with 12.34 percent, PC4 with 10.74 percent, PC5 with 9.18 percent, and PC6 with 7.63 percent. Collectively, the first six principal components explained 81.24 percent of the total variance in the dataset.

The retention of six components was based on the cumulative variance criterion adopted in the study. Although the eigenvalue of PC6 was below 1.00, retaining the first six components allowed the analysis to exceed the 80 percent cumulative variance threshold established for dimensional reduction. This means that the retained component space preserved most of the structural information in the original twelve indicators while still reducing the complexity of the clustering input.

In practical terms, the PCA results indicate that the multidimensional structure of subnational economic performance can be represented efficiently by a reduced set of principal components without substantial loss of information. Economically, this implies that the variation in growth velocity and structural stability across subnational economic units was driven by several overlapping but compressible dimensions rather than by twelve completely independent indicators. Thus, the retained principal components provided a compact but information-rich basis for the subsequent hierarchical clustering procedure.

Table 6. Explained Variance of the Principal Components

COMPONENT	EIGENVALUE	VAR (%)	CUMULATIVE VAR (%)
PC1	3.17	26.41	26.41
PC2	1.79	14.94	41.36
PC3	1.48	12.34	53.70
PC4	1.29	10.74	64.43
PC5	1.10	9.18	73.61
PC6	0.92	7.63	81.24
PC7	0.71	5.91	87.15

Legend:

PCs with cumulative variance reaching at least 80% were retained.

Table 6. Continued.

COMPONENT	EIGENVALUE	VAR (%)	CUMULATIVE VAR (%)
PC8	0.58	4.80	91.96
PC9	0.41	3.44	95.40
PC10	0.28	2.34	97.74
PC11	0.16	1.36	99.10
PC12	0.11	0.90	100.00

Legend:

PCs with cumulative variance reaching at least 80% were retained.

Scree Plot of Explained Variance by Principal Components. The scree plot shows an elbow around the second principal component, indicating that the largest drop in explained variance occurs early. However, the study retained the first six principal components because these exceeded the 80 percent cumulative variance criterion. This preserved most of the structural information from the twelve indicators while reducing the complexity of the clustering input.

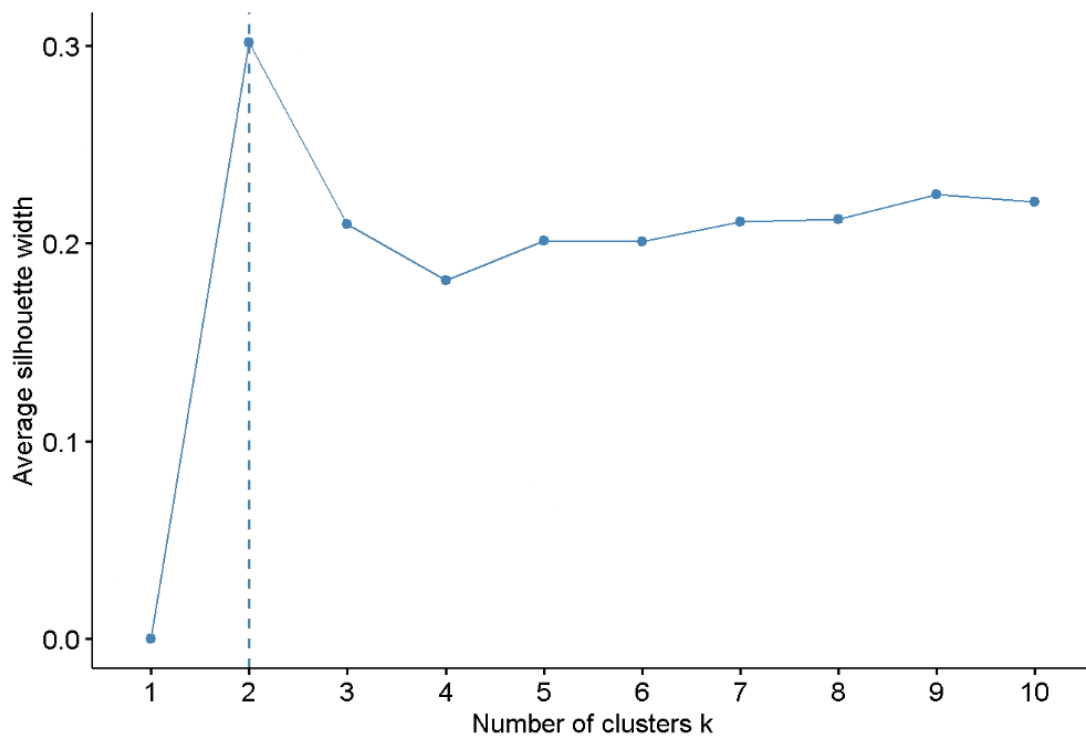


Figure 21. Scree Plot of Explained Variance by Principal Components

Determination of the Optimal Number of Clusters Using Silhouette Plot. The optimal number of clusters was evaluated using the Silhouette Method. The silhouette value measures how well each subnational economic unit fits within its assigned

cluster compared with other clusters. A higher average silhouette width indicates better within-cluster similarity and stronger separation between clusters. The highest average silhouette width was obtained for a two-cluster solution, which means that the data were most interpretable when divided into two broad economic typologies. However, because the later validation result showed an average silhouette width of about 0.29, the cluster separation should be interpreted as weak but meaningful for exploratory classification. Economically, this suggests that the 117 subnational economic units do not form sharply separated groups, but they still show enough structural difference to support a useful two-pathway typology for resilience analysis.

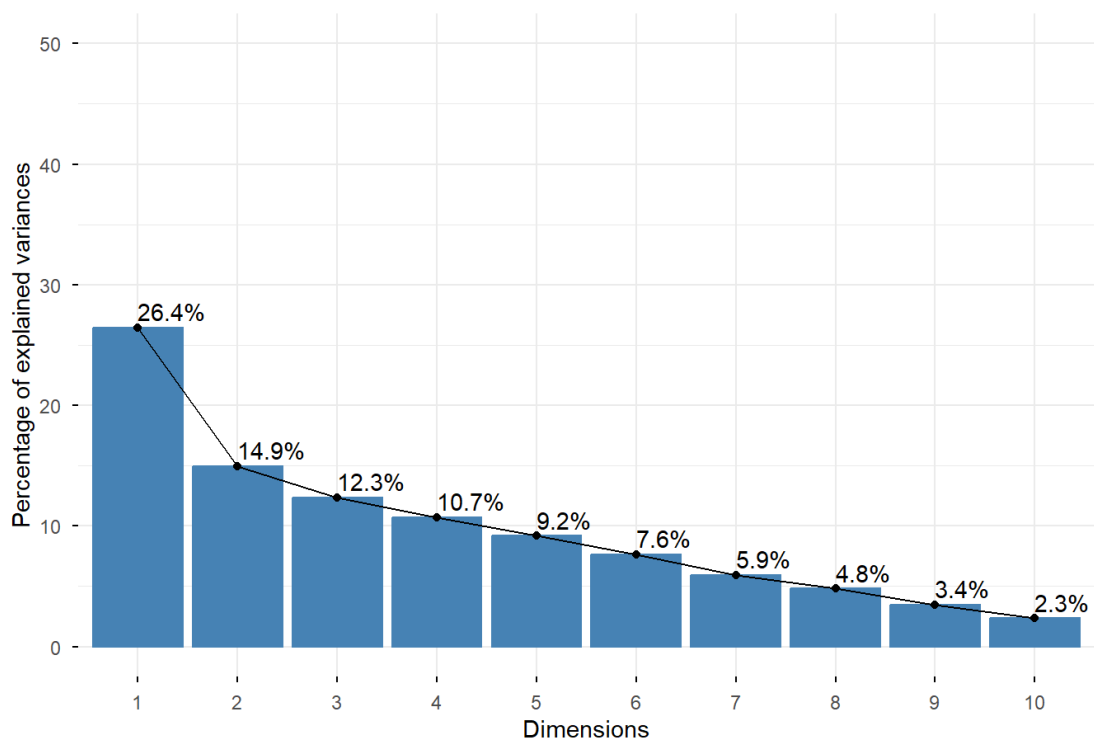


Figure 22. Determination of the Optimal Number of Clusters Using Silhouette Plot

Hierarchical Clustering Procedure

Hierarchical clustering was then applied to the retained principal component scores using Ward's Minimum Variance Method. The resulting dendrogram was cut at the silhouette-optimized level of two clusters. The cluster membership distribution was as follows: Cluster 1: 93 subnational economic units and Cluster 2: 24 subnational economic units. This indicates a dominant cluster containing most of the subnational

economic units in the dataset and a smaller cluster representing a more distinct structural group.

Dendrogram of Subnational Economic Units. The cluster membership results showed that the two clusters were not composed exclusively of either provinces or Highly Urbanized Cities (HUCs). Instead, both clusters contained a mixture of administrative types, indicating that the clustering solution was driven primarily by similarities in sectoral growth velocity and structural stability rather than by administrative classification alone. Cluster 1 consisted of 93 subnational economic units, comprising 74 provinces, 18 Highly Urbanized Cities, and the City of Isabela. Cluster 2 consisted of 24 subnational economic units, comprising 8 provinces, 15 Highly Urbanized Cities, and the Municipality of Pateros. Appendix E presents the complete list of subnational economic units under each cluster.

This composition suggests that the resulting typologies should not be interpreted as a simple rural-versus-urban classification. Although Cluster 2 contained a strong concentration of city-level units from the National Capital Region, both clusters still included a combination of provinces and urbanized economies. The clustering structure therefore reflected similarities in multivariate economic behavior across sectors rather than formal administrative status.

Cluster 2 included the provinces of Albay, Bohol, Camarines Norte, Catanduanes, Masbate, Negros Oriental, Siquijor, and Sorsogon, together with several city-level units from the National Capital Region. Economically, Cluster 2 was characterized by lower mean growth performance in Electricity, Manufacturing, Mining and Quarrying, and Services, as well as higher instability in several sectoral indicators. Although the cluster recorded a slightly higher mean in Agriculture, Forestry, and Fishing, the difference was not statistically significant. Thus, Cluster 2 is more appropriately interpreted as a structurally constrained and lower-growth pathway rather than a purely agriculture-based economy group.

In contrast, Cluster 1 demonstrated stronger mean growth performance across several key sectors, particularly Services, Electricity, Manufacturing, and Mining and Quarrying. The cluster also exhibited relatively lower instability in several sectoral indicators, particularly in comparison with Cluster 2. These characteristics suggest a more diversified and growth-responsive economic pathway. The presence of both provinces and Highly Urbanized Cities within Cluster 1 further indicates that stronger economic resilience was associated not simply with urban status, but with broader sectoral diversification and stronger industrial-service expansion.

Overall, the cluster membership profile confirms that subnational economic resilience cannot be generalized solely according to administrative classification. Provinces and city-level economies may follow similar resilience pathways when they exhibit comparable patterns of sectoral growth and structural stability. The two clusters therefore represent broader structural resilience typologies rather than simple province-city groupings.

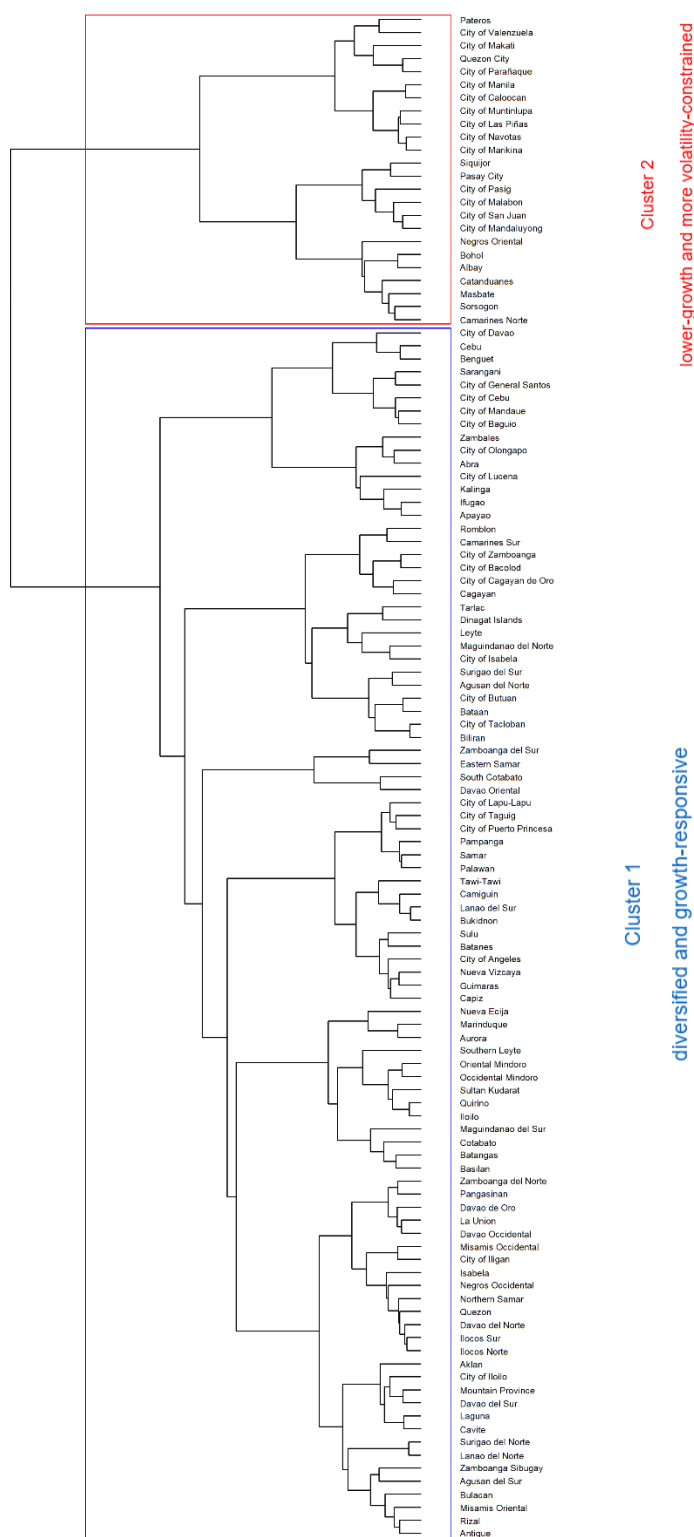


Figure 23. Dendrogram of Subnational Economic Units

Table 7. Cluster Membership Distribution

CLUSTER	NO. OF ECONOMIC UNITS
Cluster 1	93
Cluster 2	24

Legend:

Cluster 1 = diversified and growth-responsive;

Cluster 2 = lower-growth and more volatility-constrained.

NOTE: Clusters represent economic typologies, not administrative categories.

Cluster Validation Metrics. To evaluate the reliability of the clustering solution, several internal validation metrics were calculated. The optimal number of clusters was 2, with an Average Silhouette Width of 0.29, a Dunn Index of 0.20, and a Calinski-Harabasz Index of 34.54.

An average silhouette width of approximately 0.29 indicates a weak but interpretable clustering structure. Given the high heterogeneity typically observed in regional economic data, this level of separation remains acceptable for exploratory structural classification.

Table 8. Cluster Validation Metrics

METRIC	VALUE
Optimal k	2.00
Average Silhouette Width	0.29
Dunn Index	0.20
Calinski-Harabasz (CH) Index	34.54

Legend:

Higher silhouette, Dunn, and Calinski-Harabasz values indicate stronger cluster quality.

NOTE: The optimal cluster solution was selected using the silhouette method.

Cluster Profiling and Inferential Significance. To determine which indicators differentiated the two clusters, Welch's unequal-variance t-tests were performed on the indicators of growth velocity and structural stability. The results showed that not all indicators differed significantly between the clusters. Based on the p-value results, Mean Agriculture, Forestry, and Fishing, Mean Construction, and CV Construction were not significant, indicating that the observed differences for these variables were not statistically strong enough at the 0.05 level. In contrast, several indicators showed either significant or highly significant differences, demonstrating that the two clusters were separated more clearly in specific dimensions of growth and stability.

Highly significant differences ($p < 0.01$) were observed in CV Agriculture, Forestry, and Fishing, Mean and CV Electricity, Mean Manufacturing, Steam, Water, and Waste Management, CV Mining and Quarrying, and Mean Services. Significant differences ($p < 0.05$) were likewise observed in CV Manufacturing, Mean Mining and Quarrying, and CV Services. These findings indicate that the clusters differed most clearly in the growth velocity of key industrial and service sectors and in the structural stability of nearly all sectors, with the exception of Construction.

Economically, the results suggest that the distinction between the two clusters was driven less by uniform performance across all sectors and more by specific structural differences in sectoral growth and volatility. Cluster 1 showed substantially stronger growth in Electricity, Steam, Water, and Waste Management, Manufacturing, Mining and Quarrying, and Services, while Cluster 2 exhibited significantly weaker growth in these areas. Conversely, Cluster 2 tended to display higher coefficients of variation in most sectors, most notably in Mining and Quarrying, which indicates greater instability. This means the two clusters represent substantively distinct economic structures. Cluster 1 reflects a more robust, higher-growth profile, whereas Cluster 2 reflects a more constrained and volatile structure.

The non-significant results are also important. The absence of significant differences in Mean Agriculture, Forestry, and Fishing and both Construction indicators suggests that these variables did not strongly distinguish the two clusters. In practical terms, this means that the clustering solution was not defined by the primary sector's average growth or the construction industry. Instead, the most meaningful separation emerged from differences in industrial and service-sector momentum and from variation in structural stability. Accordingly, the two clusters should be interpreted as economically distinct typologies rather than arbitrary statistical partitions.

The clustering stage of the analysis is consistent with recent studies that use classification methods to identify meaningful subnational development patterns from multidimensional indicators. Prior studies have shown that clustering can distinguish

territories with stronger growth momentum from those with more constrained or structurally different economic profiles (Rivai et al., 2024; Stamenković et al., 2021). In the present study, the two-cluster solution similarly reveals that subnational economic units in the Philippines do not follow a single development pathway. Instead, they form structurally distinct groupings based on growth velocity and structural stability. This supports the use of hierarchical clustering not merely as a descriptive grouping tool, but as a statistically grounded typology-building method following prior validation through MANOVA.

Table 9. Cluster Profiling Significance Using Welch's t-Test

INDICATOR	C1 MEAN	C2 MEAN	T- STATS	P	DECISION	INTERPRETATION
Mean AFF	0.42	1.34	-1.69	0.10 ^{ns}	Fail to Reject H ₀	Not Significant
CV AFF	5.40	2.26	5.24	< 0.01**	Reject H ₀	Highly Significant
Mean Construction	5.04	4.21	1.05	0.30 ^{ns}	Fail to Reject H ₀	Not Significant
CV Construction	4.55	4.80	-0.33	0.74 ^{ns}	Fail to Reject H ₀	Not Significant
Mean ESWWM	5.77	0.95	15.33	< 0.01**	Reject H ₀	Highly Significant
CV ESWWM	0.99	3.38	-20.47	< 0.01**	Reject H ₀	Highly Significant
Mean Manufacturing	2.95	1.62	3.35	< 0.01**	Reject H ₀	Highly Significant
CV Manufacturing	3.02	4.25	-2.25	0.03*	Reject H ₀	Significant
Mean Mining & Quarrying	5.67	0.74	3.28	< 0.01**	Reject H ₀	Highly Significant
CV Mining & Quarrying	6.64	18.22	-5.92	< 0.01**	Reject H ₀	Highly Significant
Mean Services	5.78	3.94	5.09	< 0.01**	Reject H ₀	Highly Significant
CV Services	4.74	6.39	-2.26	0.03*	Reject H ₀	Significant

Legend:

p < 0.01 = highly significant; 0.01 ≤ p < 0.05 = significant; p ≥ 0.05 = not significant.

NOTE: H₀ = no significant difference between Cluster 1 and Cluster 2. Mean = growth velocity;
CV = structural stability.

Comparison of Mean Sectoral Growth Rates of Cluster 1 and Cluster 2.

This compares the mean sectoral growth rates of Cluster 1 and Cluster 2 across the

six economic sectors. The figure does not indicate that a sector belongs exclusively to one cluster; rather, it shows the average growth performance of the same sector within each of the two clusters. Thus, both clusters have corresponding values for Agriculture, Construction, Electricity, Manufacturing, Mining and Quarrying, and Services. The purpose of the figure is to compare how the two clusters differ in their sectoral growth profiles.

The results show that Cluster 1 recorded higher average growth rates than Cluster 2 in most sectors, particularly in Electricity, Steam, Water and Waste Management, Manufacturing, Mining and Quarrying, and Services. Cluster 2 showed a slightly higher mean only in Agriculture, Forestry, and Fishing. Construction also showed a higher mean in Cluster 1, although the difference was less pronounced. These patterns indicate that Cluster 1 had a stronger growth profile across most industrial and service-oriented sectors, while Cluster 2 exhibited a comparatively weaker growth structure.

This visual pattern is consistent with the inferential results of the Welch's t-test. Statistically significant differences in mean growth were found for Electricity, Steam, Water and Waste Management, Manufacturing, Mining and Quarrying, and Services, whereas the mean differences for Agriculture, Forestry, and Fishing and Construction were not statistically significant. Economically, this implies that Cluster 1 was supported by more dynamic industrial and service-sector activity, which gave it a broader expansion profile across the study period. In contrast, Cluster 2 showed weaker average growth in most sectors, suggesting a more constrained economic structure with less momentum outside agriculture. This means that the separation between the two clusters was not arbitrary, but reflected a meaningful divide between a more growth-responsive group of subnational economic units and a comparatively slower-growing group.

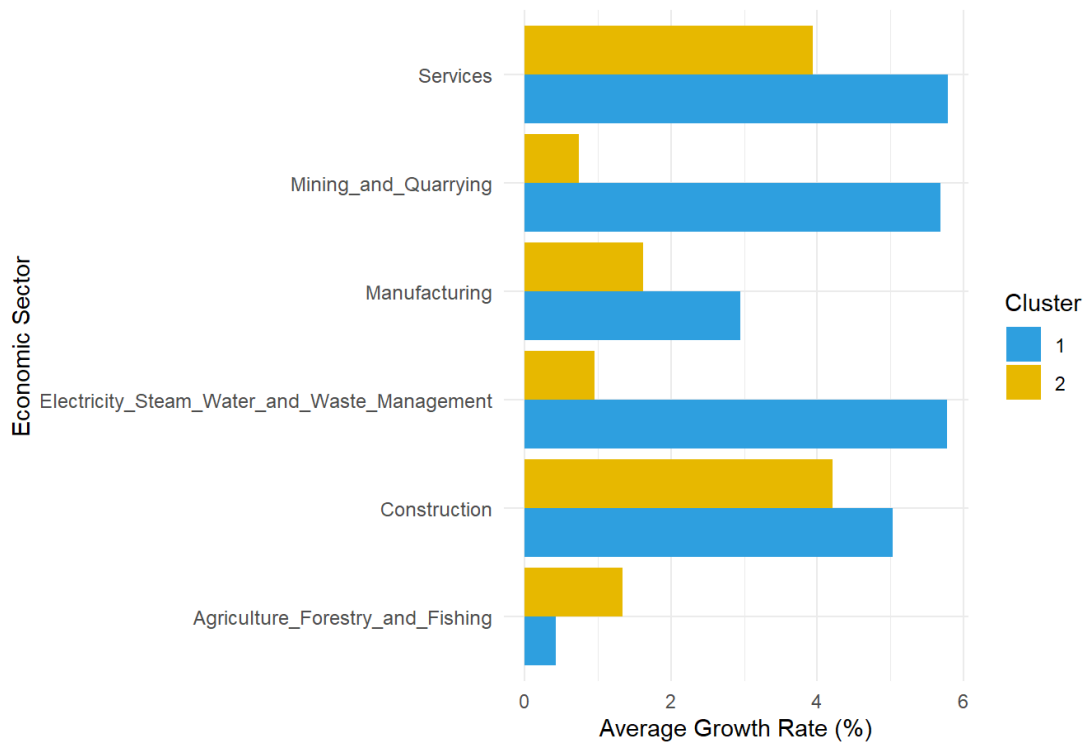


Figure 24. Comparison of Mean Sectoral Growth Rates Between Cluster 1 and Cluster 2

Cluster Typology Visualization. The clustering results reveal two broad economic typologies derived from the growth velocity and structural stability profiles of the 117 subnational economic units. These typologies do not refer to single sectors, but to groups of subnational economic units that exhibit similar overall sectoral growth patterns across the six economic sectors. Thus, the two clusters represent different structural configurations of subnational economic resilience rather than arbitrary statistical partitions.

Cluster 1 represents the larger group, consisting of 93 subnational economic units. This cluster exhibited higher mean growth in Services, Electricity, Steam, Water and Waste Management, Manufacturing, and Mining and Quarrying, indicating stronger expansion in both service-oriented and selected industrial activities. The Welch's t-test results further showed that these sectors significantly differentiated the

two clusters, particularly in Electricity, Manufacturing, Mining and Quarrying, and Services. This means that Cluster 1 was distinguished by a broader and more dynamic growth structure rather than by dependence on only one sector. Economically, this suggests that Cluster 1 reflects a more diversified and growth-responsive typology, with stronger post-shock momentum in the sectors most closely associated with production linkages, utilities, and service-sector expansion.

Cluster 2 represents the smaller group, consisting of 24 subnational economic units. Compared with Cluster 1, this group showed slower average growth in most industrial and service sectors, while the mean for Agriculture, Forestry, and Fishing was slightly higher, although that difference was not statistically significant. More importantly, the broader cluster profile suggests that this group was less driven by service- and industry-based expansion and showed comparatively stronger agricultural stability. Economically, this implies a more conservative growth profile: Cluster 2 may exhibit stronger stability in agriculture-related activity, but it also appears to have more limited momentum in the sectors that typically drive faster post-shock recovery and structural transformation.

Taken together, the two clusters indicate that subnational economic resilience during the 2018–2024 period followed two broad pathways. The first pathway is a higher-growth diversified pathway, represented by subnational economic units with stronger service and industrial growth capacity. The second pathway is a structurally constrained, stability-oriented pathway, represented by units with more limited industrial and service expansion and comparatively stronger agricultural stability. This distinction suggests that resilience at the subnational level is not uniform. Rather, it depends on how local economies combine growth momentum, sectoral diversification, and structural stability across the major sectors.

Consistent with the literature, this pattern supports the view that diversified and interconnected economies tend to display stronger adaptive and recovery capacity, while more structurally constrained economies may show greater baseline stability but

slower structural expansion. The typology therefore reinforces the argument that subnational resilience should be interpreted in relation to sectoral composition and structural differences, not only in terms of aggregate growth performance.

The distinction between the two typologies is consistent with the literature emphasizing the role of diversification, sectoral balance, and structural connectivity in economic resilience. Cluster 1, which exhibited stronger growth in services and several industrial sectors, reflects the type of more diversified and growth-responsive structure that the literature associates with stronger adaptive capacity and faster recovery from shocks (He, Miao, & Qureshi, 2022; Giannakis, Bruggeman, & Mamuneas, 2024). Cluster 2, by contrast, reflects a more structurally constrained and stability-oriented pathway, with comparatively stronger agricultural stability but weaker industrial and service momentum. This pattern is also consistent with Philippine studies showing that uneven structural transformation contributes to divergence in local development trajectories, particularly when high-value services and deeper industrial activity remain concentrated in selected areas (Serafica, Vergara, & Oren, 2021; Custodio & Sombilla, 2025). Thus, the two typologies identified in the present study do not merely describe different levels of growth. Rather, they represent different structural pathways through which subnational economies respond to volatility, sustain activity, and pursue recovery.

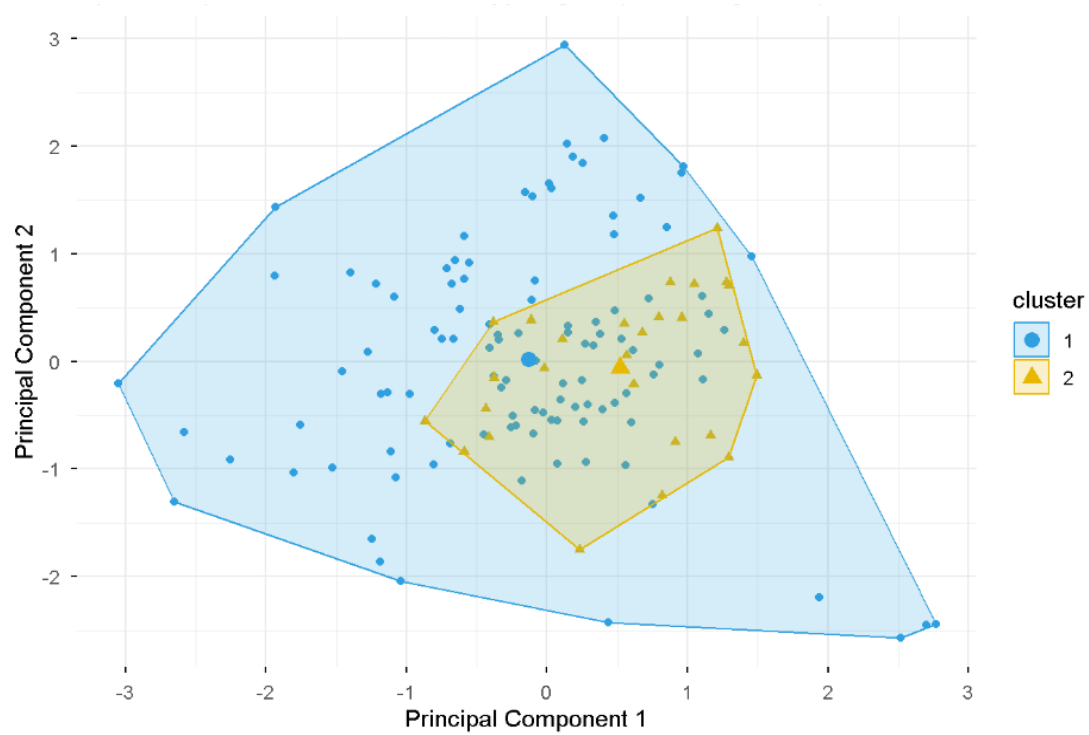


Figure 25. Cluster Typology Visualization

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter summarizes the major findings of the study, presents the conclusions based on the statistical results, and gives practical recommendations for national agencies, local planners, and future researchers. The discussion is written in simple and direct form so that the meaning and use of the findings can be understood by both technical and non-technical readers.

Summary

This study examined evidence-based pathways to resilience of subnational economic units in the Philippines using 2018–2024 Provincial Product Accounts data. The analysis covered 117 subnational economic units and produced 702 subnational-unit-by-interval observations. The study used descriptive statistics, Multivariate Analysis of Variance, Principal Component Analysis, and hierarchical cluster analysis to understand how sectoral growth patterns changed over time and how subnational economies could be grouped according to growth velocity and structural stability.

The descriptive results showed that the 2018–2024 period included clear shock and recovery patterns. Services and Construction showed strong contractions during the 2019–2020 interval but also showed strong recovery in the following intervals. This indicates that these sectors were highly sensitive to changes in mobility, demand, reopening conditions, and investment activity. In contrast, Agriculture, Forestry, and Fishing showed smaller changes across the six temporal intervals. This means that AFF served more as a stabilizing sector than as a main driver of fast recovery.

The MANOVA results confirmed that the six-sector growth structure changed significantly across the six temporal intervals. Pillai's Trace was significant, $p < 0.01$, and the null hypothesis was rejected. The partial eta squared value, $\eta^2 = 0.24$, showed that temporal differences explained a meaningful portion of the multivariate variation in sectoral growth. The univariate follow-up results further showed that all six sectors were significantly affected by time, but the strength of effect differed across sectors.

Services and Construction had the largest effect sizes, while Agriculture, Forestry, and Fishing had the smallest effect size.

The cluster analysis classified the 117 subnational economic units into two broad economic typologies. Cluster 1 included 93 subnational economic units and represented a generally diversified and growth-responsive pathway. This group showed stronger growth in Services, Electricity, Manufacturing, and Mining and Quarrying. Cluster 2 included 24 subnational economic units and represented a structurally constrained and lower-growth pathway. This group showed weaker average growth in most industrial and service sectors and higher instability in several indicators. Although Cluster 2 had a slightly higher mean in Agriculture, Forestry, and Fishing, the difference was not statistically significant, so it should not be interpreted as a purely agriculture-based group.

Conclusions

Based on the findings, the study concludes that subnational economic resilience in the Philippines was shaped by both sectoral growth behavior and structural differences across local economies. The rejection of the null hypothesis shows that sectoral growth patterns were not the same across the six temporal intervals. This means that the 2018–2024 period involved meaningful changes in the way subnational economies responded to disruption and recovery.

The findings also show that aggregate economic growth alone is not enough to explain resilience. A subnational economy may appear to grow overall, but its internal sectors may behave very differently. Some sectors may recover quickly, while others may remain slow or unstable. This is important for planning because local governments and national agencies need to look beyond total output and examine the sectoral structure behind growth.

The study further concludes that resilience does not follow only one pathway. Cluster 1 economies showed stronger growth responsiveness because they had broader service and industrial expansion. Cluster 2 economies showed more

constrained growth and greater instability in several indicators. This means that policies should not treat all subnational economic units in the same way. Development strategies should be based on the structural needs of each economic typology.

The results support the goals of sustainable and inclusive development, especially SDG 8 on decent work and economic growth, SDG 9 on industry, innovation, and infrastructure, and SDG 11 on sustainable cities and communities. By identifying which sectors and subnational units are more sensitive or more stable, the study provides evidence that may help improve local economic planning, strengthen recovery strategies, and support more targeted regional development.

Recommendations

Based on the findings, the following recommendations are proposed:

1. National agencies such as NEDA, DTI, and PSA may use cluster-based analysis together with existing ranking systems. This can help policymakers design programs based on the actual structure of local economies rather than relying only on broad regional or competitiveness scores.
2. Cluster 1 subnational economic units may benefit from policies that strengthen service productivity, digital infrastructure, supply-chain efficiency, and industrial support systems. Since these units are more growth-responsive, they may also need stronger risk-management strategies during future disruptions.
3. Cluster 2 subnational economic units may benefit from programs that support agricultural modernization, agro-industrial linkages, food processing, logistics, and value-added production. These strategies may help improve growth velocity while maintaining the stability associated with agriculture-related activity.
4. Local government units and planners may regularly examine their PPA-based sectoral composition to identify which sectors drive growth and which sectors

create vulnerability. This can help them prepare more targeted development plans and recovery programs.

5. Future researchers may expand the analysis by including variables not covered in this study, such as employment quality, poverty, fiscal capacity, informal economic activity, infrastructure access, disaster exposure, and governance indicators. These additional variables may provide a broader view of local resilience beyond sectoral growth alone.
6. Future studies may also apply predictive models, such as Random Forest or Gradient Boosting, to forecast possible vulnerabilities to future economic shocks. This would allow the current classification framework to be extended from retrospective analysis to forward-looking planning

REFERENCES

- Aban-Serquina, M. (2024). Prediction model for gross domestic product: A production approach. *International Journal of Research Publication and Reviews*, 5(7), 1089–1097. <https://ijrpr.com/uploads/V5ISSUE7/IJRPR31344.pdf>
- ABS-CBN News. (2019, March 5). 28 'Build, Build, Build' projects finished by 2022: Official. <https://www.abs-cbn.com/business/03/05/19/28-build-build-build-projects-finished-by-2022-official>
- ABS-CBN News. (2020, October 23). What 2019 regional GDP tells us about the pandemic, PH economic resilience. <https://news.abs-cbn.com/business/10/23/20/dissecting-data-what-2019-regional-gdp-tells-us-about-the-pandemic-ph-economic-resilience>
- ASEAN+3 Macroeconomic Research Office. (2024, December). Scarring effects of COVID-19 and post-pandemic Philippine economic recovery. https://amro-asia.org/wp-content/uploads/2024/12/Selected-Issue-2_Scarring-Effects-of-COVID-19-and-Post-pandemic-Philippine-Economic-Recovery-3.pdf
- ASEAN+3 Macroeconomic Research Office. (2025). ASEAN+3 regional economic outlook 2025 (AREO). https://amro-asia.org/wp-content/uploads/2025/04/AMRO-AREO-2025-Full-Report_final.pdf
- Asian Development Bank. (2022a, December 14). ADB forecasts 7.4% growth for the Philippines in 2022. <https://www.adb.org/news/adb-forecasts-7-4-growth-philippines-2022>
- Asian Development Bank. (2022b). Supporting post-COVID-19 economic recovery in Southeast Asia (TA 9964). <https://www.adb.org/sites/default/files/publication/793526/post-covid19-economic-recovery-southeast-asia.pdf>
- Bangko Sentral ng Pilipinas. (2022). Report on regional economic developments in the Philippines 2022. https://www.bsp.gov.ph/Media_And_Research/Report%20on%20Regional%20Economic%20Developments/REDP_2022.pdf
- Bangko Sentral ng Pilipinas. (2024). Annual report 2024: Deploying digital solutions and fostering financial inclusion. https://www.bsp.gov.ph/Media_And_Research/Annual%20Report/AnnRep_2024.pdf
- Brooks, G. P., Adjanin, N., Oppong, F., & Liu, Y. (2024, April 11–14). The homogeneity of covariances assumption in MANOVA: Differential impact of heterogenous variances and covariances [Paper presentation]. 2024 American Educational Research Association (AERA) Annual Meeting, Philadelphia, PA, United States. https://sites.ohio.edu/brooksg/AERA24_MANOVA_Final_240411.pdf
- Bundervoet, T., Maiyo, L., & Sanghi, A. (2015). Bright lights, big cities: Measuring national and sub-national economic growth in Africa from outer space, with an application to Kenya and Rwanda (Policy Research Working Paper No.

7461). The World Bank.
<https://documents1.worldbank.org/curated/en/912151468188369841/pdf/WPS7461.pdf>

BusinessMirror. (2020, October 23). Digital transformation in PH accelerated by five years due to COVID-19 pandemic.

<https://businessmirror.com.ph/2020/10/23/digital-transformation-in-ph-accelerated-by-five-years-due-to-covid-19-pandemic/>

Cabuenas, J. V. D. (2020a, March 3). Hotels to cut rates as COVID-19 outbreak brings losses to P22B. GMA News Online.

<https://www.gmanetwork.com/news/money/companies/728177/hotels-to-cut-rates-as-covid-19-outbreak-brings-losses-to-p22b/story/>

Cabuenas, J. V. D. (2020b, October 14). Philippine economy to lag behind ASEAN peers in 2020 - IMF. GMA News Online.

<https://www.gmanetwork.com/news/money/economy/759860/philippine-economy-to-lag-behind-asean-peers-in-2020-imf/story/>

Cabuenas, J. V. D. (2021, August 9). Philippine first-quarter GDP growth revised upward to -3.9%. GMA News Online.

<https://www.gmanetwork.com/news/money/economy/798591/philippine-first-quarter-gdp-growth-revised-upward-to-3-9/story/>

Catilogo, M. I. U. (2024, August 14). PSA: PH health spending rose to 5.9% of GDP in 2023. Philippine Daily Inquirer. <https://business.inquirer.net/474666/psa-ph-health-spending-rose-to-5-9-of-gdp-in-23>

Center for Global Development. (2021). Health systems impact of COVID-19 in the Philippines.

https://www.cgdev.org/sites/default/files/WP569_Health_systems_impact_of_COVID_in_the_Philippines.pdf

Centre for Sustainable Cities. (2018). The Philippines: National urban policies and city profiles for Manila and Batangas.

<https://centreforsustainablecities.ac.uk/wp-content/uploads/2018/10/Full-Report-The-Philippines-National-Urban-Policies-and-City-Profiles-for-Manila-and-Batangas.pdf>

Commonwealth Secretariat. (n.d.). Economic resilience: Concepts and measurements (7th ed.). Commonwealth iLibrary.

<https://www.thecommonwealth-ilibrary.org/index.php/comsec/catalog/download/92/89/375?inline=1>

Cordero, T. (2020a, April 3). ADB: Philippine GDP to slide at 2% in 2020 due to COVID-19; rebound to 6.5% in 2021. GMA News Online.

<https://www.gmanetwork.com/news/money/economy/732505/adb-philippine-gdp-to-slide-at-2-in-2020-due-to-covid-19-rebound-to-6-5-in-2021/story/>

Cordero, T. (2020b, May 12). Moody's sees PHL economy to contract 2.5% in 2020. GMA News Online.

<https://www.gmanetwork.com/news/money/economy/737848/moody-s-sees-phl-economy-to-contract-2-5-in-2020/story/>

- Cordero, T. (2020c, May 27). Around 75k hotel workers will be displaced due to COVID-19 — industry group. GMA News Online.
<https://www.gmanetwork.com/news/money/economy/739997/around-75k-hotel-workers-will-be-displaced-due-to-covid-19-industry-group/story/>
- Cordero, T. (2020d, September 4). Bank lending growth slowed in July amid COVID-19 pandemic — BSP. GMA News Online.
<https://www.gmanetwork.com/news/money/economy/754218/bank-lending-growth-slowed-in-july-amid-covid-19-pandemic-bsp/story/>
- Cordero, T. (2020e, October 2). Bank lending growth slowed anew in August — BSP. GMA News Online.
<https://www.gmanetwork.com/news/money/economy/758178/bank-lending-growth-slowed-anew-in-august-bsp/story/>
- Cordero, T. (2020f, December 8). World Bank expects Philippine economy to shrink deeper by 8.1% in 2020. GMA News Online.
<https://www.gmanetwork.com/news/money/economy/767186/world-bank-expects-philippine-economy-to-shrink-deeper-by-8-1-in-2020/story/>
- Cornish, R. (2007). Statistics: 3.1 Cluster analysis. Mathematics Learning Support Centre. <https://www.statstutor.ac.uk/resources/uploaded/clusteranalysis.pdf>
- Crawley, A., Daigneault, A., & Bowen, K. M. (2025). Metrics for analyzing regional resilience: A bibliographic and cluster analysis approach. *Agricultural and Resource Economics Review*, First View, 1–16.
<https://doi.org/10.1017/age.2025.10010>
- Curtale, R., & Schiavone, M. (2025). A novel approach to regionalize country-level GDP projections. *Journal of Forecasting*. Advance online publication.
<https://doi.org/10.1002/for.70052>
- Custodio, A., & Sombilla, M. (2025). Structural transformation and rural development in the Philippines: Sectoral shifts and adaptive capacity. *Land*, 14(2), 253.
<https://doi.org/10.3390/land14020253>
- De Freitas, G. (2022, July 20). Philippines' maritime logistics sector set for growth despite Covid setbacks. HKTDC Research.
<https://research.hktdc.com/en/article/MTA3NjcwNTEwNA>
- Department of Budget and Management. (2021, July 30). National Government disbursement performance (December 2020). https://www.dbm.gov.ph/wp-content/uploads/DBCC/2020/NG-Disbursements_December-2020_final_updated-as-of-July-30-2021.pdf
- Department of Budget and Management. (2022). 2022 People's Enacted Budget: Building resilience amidst the pandemic, continuing the legacy.
<https://www.dbm.gov.ph/images/pdf/files/2022-Peoples-Enacted-Budget-final.pdf>
- Department of Budget and Management. (2023). 2023 People's Enacted Budget: Agenda for prosperity. <https://www.dbm.gov.ph/images/pdf/files/2023-Peoples-Enacted-Budget.pdf>

- Department of Finance. (2020, June). 'Build, Build, Build' to break vicious circle of weak supply & demand. <https://www.dof.gov.ph/build-build-build-to-break-vicious-circle-of-weak-supply-demand/>
- Department of Finance. (2022, January 9). Dominguez thanks President Duterte, Congress for amended Retail Trade Liberalization Act. <https://www.dof.gov.ph/dominguez-thanks-president-duterte-congress-for-amended-retail-trade-liberalization-act/>
- Department of Trade and Industry. (n.d.). Cities and Municipalities Competitiveness Index (CMCI) manual (Edition X). <https://cmci.dti.gov.ph/downloads/forms/CMCI%20Manual.pdf>
- Domingo, S. N., & Manejar, A. J. A. (2020). Looking at local government resilience through network data envelopment analysis (PIDS Discussion Paper No. DP 2020-19). Philippine Institute for Development Studies. <https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidsdps2019.pdf>
- Eichengreen, B., Park, D., & Shin, K. (2024). Economic resilience: Why some countries recover more robustly than others from shocks. *Economic Modelling*, 136, Article 106748. <https://doi.org/10.1016/j.econmod.2024.106748>
- Encarnacion, R. H., Magnaye, D. C., & Castro, A. G. M. (2023). Spatial analysis of local competitiveness: Relationship of economic dynamism of cities and municipalities in major regional metropolitan areas in the Philippines. *Sustainability*, 15(2), 950. <https://doi.org/10.3390/su15020950>
- Giannakis, E., Bruggeman, A., & Mamuneas, T. P. (2024). Regional economic resilience, productivity growth and sectoral interconnectedness. *Papers in Regional Science*, 103(2), Article 100010. <https://doi.org/10.1016/j.pirs.2024.100010>
- GMA Integrated News. (2020, August 6). Philippine economy falls into recession as GDP shrinks by 16.5% in Q2 amid lockdown. GMA News Online. <https://www.gmanetwork.com/news/money/economy/750131/philippine-economy-falls-into-recession-as-gdp-shrinks-by-16-5-in-q2-amid-lockdown/story/>
- GMA Integrated News. (2024, May 21). Agriculture damage due to El Niño hits P9.5B. GMA News Online. <https://www.gmanetwork.com/news/money/economy/907517/agriculture-damage-due-to-el-nino-hits-p9-5b/story/>
- Greenlaw, S. A., Shapiro, D., & MacDonald, D. (2022). Principles of economics for AP courses 2e. OpenStax. <https://openstax.org/books/principles-economics-ap-courses-2e/pages/1-introduction>
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2010). *Multivariate data analysis* (7th ed.). Pearson Prentice Hall.
- He, D., Miao, P., & Qureshi, N. A. (2022). Can industrial diversification help strengthen regional economic resilience? *Frontiers in Environmental Science*, 10, 987396. <https://doi.org/10.3389/fenvs.2022.987396>

- Hernando-Malipot, M. (2022, August 19). 425 private schools have 'permanently closed' amid the Covid-19 pandemic: DepEd. Manila Bulletin. <https://mb.com.ph/2022/8/19/425-private-schools-have-permanently-closed-amid-the-covid-19-pandemic-deped>
- Institute for Climate and Sustainable Cities. (2020, November). Rapid situation analysis of the impacts of COVID-19 on MSMEs in Malolos and Paombong. http://www.icsc.ngo/governance-for-resilience/wp-content/uploads/2020/12/BulacanTechReport_Nov2020_FINAL_DIGITA.pdf
- International Monetary Fund. (2024). Philippines: 2024 Article IV Consultation — Staff Report [Country report]. <https://www.imf.org/-/media/Files/Publications/CR/2024/English/1phlea2024001-print-pdf.ashx>
- International Trade Administration. (2024, December 18). Philippines - Digital economy. U.S. Department of Commerce. <https://www.trade.gov/country-commercial-guides/philippines-digital-economy>
- Jin, T., & Li, Z. (2025). Sectoral linkages and their influence on structural change: The case of China. PLOS ONE, 20(9), e0330908. <https://doi.org/10.1371/journal.pone.0330908>
- Johnson, R. A., & Wichern, D. W. (2007). Applied multivariate statistical analysis (6th ed.). Pearson Prentice Hall.
- Laerd Statistics. (n.d.). One-way MANOVA in SPSS Statistics. Retrieved December 2, 2025, from <https://statistics.laerd.com/spss-tutorials/one-way-manova-using-spss-statistics.php>
- Lazzeretti, L., Oliva, S., & Innocenti, N. (2022). Unfolding smart specialisation for regional economic resilience: The role of industrial structure. *Investigaciones Regionales - Journal of Regional Research*, 54, 5–25. <https://www.redalyc.org/journal/289/28973325001/>
- Li, C., Luo, C., Jia, H., Yue, M., & Wu, L. (2025). Advancing agricultural economic growth through technology innovation and structural transformation: A multilevel analysis. *Journal of Innovation & Knowledge*, 10(4), Article 100743. <https://www.sciencedirect.com/science/article/pii/S2444569X25000885>
- Liu, H., Fang, Y., Liu, J., & Chen, Y. (2023). The interaction of cultural and creative industries clusters and regional economic resilience from the perspective of spatial analysis. *Sustainability*, 15(6), 5542. <https://www.mdpi.com/2071-1050/15/6/5542>
- Lloyd's Register Foundation. (2024). World risk poll 2024 report: Resilience in a changing world. <https://doi.org/10.60743/c0rm-h862>
- Magsambol, B. (2020a, June 24). No student left behind? During pandemic, education 'only for those who can afford'. Rappler. <https://rappler.com/newsbreak/in-depth/education-only-for-people-who-can-afford-coronavirus-pandemic>
- Magsambol, B. (2020b, July 18). Over 200,000 students transfer from private to public schools amid pandemic. Rappler. <https://rappler.com/nation/students-private-schools-transfer-to-public-coronavirus-pandemic>

- Martin, R., & Sunley, P. (2014). On the notion of regional economic resilience: Conceptualisation and explanation (Revised working paper). Cambridge University, Department of Geography.
<https://www.geog.cam.ac.uk/files/research/projects/cger/Onthenotion.pdf>
- MathWorks. (n.d.). What is Multivariate Analysis of Variance (MANOVA)?
<https://www.mathworks.com/discovery/manova.html>
- Montrimas, A., Bruneckienė, J., Navickas, V., & Martinkienė, J. (2024). Measuring national economic resilience through industrial portfolios. *Journal of International Studies*, 17(1), Article 8.
https://www.jois.eu/files/8_1421_Montrimas%20et%20al.pdf
- National Economic and Development Authority. (2021). Updated Philippine development plan 2017–2022. https://pdp.depdev.gov.ph/wp-content/uploads/2021/08/8312021_Updated-PDP-2017-2022.pdf
- Office for National Statistics. (2022). Subnational economic activity statistics: User guide.
<https://www.ons.gov.uk/economy/grossdomesticproductgdp/methodologies/subnationaleconomicactivitystatisticsuserguide>
- Organisation for Economic Co-operation and Development. (2023a). Economic outlook for Southeast Asia, China and India 2023: Reviving tourism post-pandemic. OECD Publishing. <https://doi.org/10.1787/f677c529-en>
- Organisation for Economic Co-operation and Development. (2023b). OECD regional outlook 2023: The longstanding geography of inequalities. OECD Publishing. <https://doi.org/10.1787/92cd40a0-en>
- Oxford Business Group. (2021). Philippines country profile 2021: Industry & agri-business. Oxford Business Group.
<https://oxfordbusinessgroup.com/reports/philippines/2021-report/country-profile>
- Philippine News Agency. (2021a, January 28). PH digital economy holds vast potential for pandemic recovery. <https://www.pna.gov.ph/articles/1128854>
- Philippine News Agency. (2021b, January 28). PH economy contracts 9.5% in 2020; strong rebound seen. <https://www.pna.gov.ph/articles/1128813>
- Philippine News Agency. (2023, September 3). One town, one product law ‘perfect’ for recovering MSMEs. <https://www.pna.gov.ph/articles/1209083>
- Philippine News Agency. (2024, August 2). DA records P15.3-B agri damage due to El Niño phenomenon. <https://www.pna.gov.ph/articles/1230394>
- Philippine Statistics Authority. (n.d.-a). PPA dashboard — Region XI.
<https://rso11.psa.gov.ph/statistical/ppa/dashboard>
- Philippine Statistics Authority. (n.d.-b). Provincial Product Accounts (PPA) dashboard – Region II. <https://rso02.psa.gov.ph/statistics/ppa/dashboard>
- Philippine Statistics Authority. (2022, December 19). Laguna is the biggest economy in CALABARZON; Cavite and Batangas lead growth in the region in 2021

[Press release]. <https://psa.gov.ph/content/laguna-biggest-economy-calabarzon-cavite-and-batangas-lead-growth-region-2021>

Philippine Statistics Authority. (2024a, February 19). 2022 economic performance of the provinces in the Philippines (Special Release No. 2024-SSO-013). <https://psa.gov.ph/content/2022-economic-performance-provinces-philippines>

Philippine Statistics Authority. (2024b, October 30). PSA release: Results of the Provincial Product Accounts — 82 provinces and 33 highly urbanized cities (Reference No. 2024-SSO-154). <https://www.psa.gov.ph/content/psa-release-results-provincial-product-accounts-82-provinces-and-33-highly-urbanized-cities>

Philippine Statistics Authority. (2024c, November 26). Capiz's economy records a 6.8 percent increase in 2023 [Press release]. https://rso06.psa.gov.ph/sites/default/files/article/PR_CAPIZ.pdf

Philippine Statistics Authority. (2025a, January 22). Provincial Product Accounts: 2023 economic performance of the Province of Capiz (Report No. SR-2025-004). <https://rso06.psa.gov.ph/sites/default/files/article/SR-2025-004%20PPA.pdf>

Philippine Statistics Authority. (2025b, September 25). Institutionalizing the Provincial Product Accounts: PSA releases the 2024 economic performance of 82 provinces and 33 highly urbanized cities (Reference No. 2025-357). <https://psa.gov.ph/statistics/ppa/node/1684080852>

Philstar. (2021, January 29). Philippine economy shrank 9.5 percent in 2020. The Philippine Star. <https://www.philstar.com/headlines/2021/01/29/2073878/philippine-economy-shrank-95-percent-2020>

PortCalls Asia. (2025, April 14). PH transport and storage sectors hit trillion-peso mark in 2023. <https://portcalls.com/ph-transport-and-storage-sectors-hit-trillion-peso-mark-in-2023/>

Puttanapong, N., Prasertsoong, N., & Peechapat, W. (2023). Predicting provincial gross domestic product using satellite data and machine learning methods: A case study of Thailand. *Asian Development Review*, 40(2), 39–85. <https://doi.org/10.1142/S0116110523400024>

Rayel, R. R. (2021). Conceptualization & correlates of competitiveness (Working paper). National Competitiveness Council / Department of Trade and Industry. <https://www.ioer-imrj.com/wp-content/uploads/2023/05/Examining-the-Conceptualization-and-Correlates-of-Competitiveness-of-Cities-and-Municipalities-in-the-Philippines.pdf>

Reed, E. (2024, October 31). Economic shock. SmartAsset. <https://smartasset.com/financial-advisor/economic-shock>

Rivai, M., Fraskisa, N., & Rosni. (2024). Analysis of fuzzy C-means clustering method on grouping provinces in Indonesia based on economic growth in 2023. *International Journal of Scientific Research in Science, Engineering and Technology*, 11(6), 380–388. <https://doi.org/10.32628/IJSRSET2411466>

- Santos Knight Frank. (2024, January 8). Philippine real estate outlook 2024: A new path to growth. <https://santosknightfrank.com/real-estate-outlook-2024/>
- Serafica, R. B., Vergara, J. M., & Oren, Q. C. A. (2021). Regional analysis of the Philippine services sector (PIDS Research Paper Series No. RPS 2021-04). Philippine Institute for Development Studies. <https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidsrp2104.pdf>
- Stamenković, M., Milanović, M., & Janković-Milić, V. (2021). Multivariate statistical analysis of regional economic disparities at district level in Serbia. *TEME*, 45(2), 681–698. <https://doi.org/10.22190/TEME200715039S>
- Statistics How To. (n.d.). Homogeneity of covariance. Retrieved December 2, 2025, from <https://www.statisticshowto.com/homogeneity-of-covariance/>
- Statistics Solutions. (n.d.). Understanding MANOVA: A comprehensive analysis of multiple dependent variables. Retrieved December 2, 2025, from <https://www.statisticsolutions.com/free-resources/directory-of-statistical-analyses/manova/>
- Sustainable Prosperity. (n.d.). Economic resilience. <https://sustainable-prosperity.eu/sustainable-prosperity/economic-resilience/>
- Swiss Federal Department of Foreign Affairs. (2022). Economic report 2022: Philippines. https://www.eda.admin.ch/content/dam/countries/countries-content/philippines/en/Economic_Report_2022_Philippines.pdf
- Tan, J., Hu, X., Hassink, R., & Ni, J. (2020). Industrial structure or agency: What affects regional economic resilience? Evidence from resource-based cities in China. *Cities*, 106, 102906. <https://www.sciencedirect.com/science/article/abs/pii/S0264275120312543>
- Technical Education and Skills Development Authority. (2020, December). Looking into the skills for the post-pandemic labor market. TESDA Planning Office. https://www.tesda.gov.ph/Uploads/File/LMIR%202020/20.12.01_LMIR-no.-5_Skills-for-the-Post-Pandemic-Labor-Market_Draft-3.pdf
- United Nations. (n.d.). Sustainable Development Goals. <https://sdgs.un.org/goals>
- United Nations Human Settlements Programme. (2016). Structural transformation in developing countries: Cross regional analysis. UN-Habitat. <https://unhabitat.org/structural-transformation-in-developing-countries-cross-regional-analysis>
- UP Resilience Institute. (2020). Assessment of the socio-economic impact of the COVID-19 outbreak in selected communities. <https://resilience.up.edu.ph/projects/assessment-of-the-socio-economic-impact-of-the-covid-19-outbreak-in-selected-communities/>
- USDA Foreign Agricultural Service. (2023, December 14). Philippines: Food service - hotel restaurant institutional annual (Report No. RP2023-0062). U.S. Department of Agriculture. <https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Food+Service+->

+Hotel+Restaurant+Institutional+Annual_Manila_Philippines_RP2023-0062.pdf

- Villanueva, J. (2020, August 20). BSP hikes real estate loan limit to 25%. Philippine News Agency. <https://www.pna.gov.ph/articles/1112934>
- Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2016). *Probability & statistics for engineers & scientists* (9th ed.). Pearson.
- Wang, X., & Li, M. (2022). Determinants of regional economic resilience to economic crisis: Evidence from Chinese economies. *Sustainability*, 14(2), 809. <https://doi.org/10.3390/su14020809>
- Wenz, L., Carr, R. D., Kögel, N., Kotz, M., & Kalkuhl, M. (2023). DOSE: Global data set of reported sub-national economic output. *Scientific Data*, 10(1), 425. <https://doi.org/10.1038/s41597-023-02323-8>
- Wisdomlib. (2025, November 2). Provincial GDP: Significance and symbolism. <https://www.wisdomlib.org/concept/provincial-gdp>
- World Bank. (2020, May). Philippines monthly economic developments. World Bank Group. <https://documents1.worldbank.org/curated/en/402921590507349355/txt/Philippines-Monthly-Economic-Developments-May-2020.txt>
- World Bank. (2022, June). Philippines economic update June 2022 edition. World Bank Group. <https://documents1.worldbank.org/curated/en/099325006072264961/pdf/P177408091735101c0b25405cf39f312c86.pdf>
- Zhang, L., Lin, G., Lyu, X., & Su, W. (2024). Suppression or promotion: Research on the impact of industrial structure upgrading on urban economic resilience. *Humanities and Social Sciences Communications*, 11, Article 843. <https://doi.org/10.1057/s41599-024-03329-2>
- Zhong, F. (2024). Assessing regional resilience in China using a sustainable livelihood framework. *Habitat International*, 147, 102622. [https://doi.org/10.1016/S1470-160X\(24\)00045-1](https://doi.org/10.1016/S1470-160X(24)00045-1)

APPENDICES

Appendix 1. Cleaned Excel Data Preparation Workflow

Gross Domestic Product, by Industry, by Province and HUC, Growth Rates

G		At Current Prices		At Constant 2018 Prices	
		2022-2023	2023-2024	2022-2023	2023-2024
Agriculture, Forestry, and Fishing	National Capital Region (NCR)	11.0	8.1	5.4	0.8
	..City of Caloocan	/s	/s	/s	/s
	..City of Las Piñas	/s	/s	/s	/s
	..City of Makati	/s	/s	/s	/s
	..City of Malabon	/s	/s	/s	/s
	..City of Mandaluyong	/s	/s	/s	/s
	..City of Manila	/s	/s	/s	/s
	..City of Marikina	/s	/s	/s	/s
	..City of Muntinlupa	/s	/s	/s	/s
	..City of Navotas	/s	/s	/s	/s
	..City of Parañaque	/s	/s	/s	/s
	..Pasay City	/s	/s	/s	/s
	..City of Pasig	/s	/s	/s	/s
	..Quezon City	/s	/s	/s	/s
	..City of San Juan	/s	/s	/s	/s
	..City of Taguig	/s	/s	/s	/s
	..City of Valenzuela	/s	/s	/s	/s
	..Pateros	/s	/s	/s	/s
	Cordillera Administrative Region (CAR)	13.9	5.7	-1.2	1.1
	..Abra	1.1	3.6	-1.8	-9.1
	..Apayao	1.3	3.5	-2.8	-9.2
	..Benguet	26.4	7.2	-0.4	8.1
	..Ifugao	-1.5	4.1	-3.9	1.4
	..Kalinga	4.6	9.8	-0.1	-6.8
	..Mountain Province	16.5	-0.5	-0.2	-1.3
	..City of Baguio	23.9	2.5	-2.5	3.6
	Region I (Ilocos Region)	7.2	9.4	3.3	-0.6
	..Ilocos Norte	8.7	13.1	3.8	2.5
	..Ilocos Sur	7.8	7.4	3.5	-2.2
	..La Union	2.6	5.6	1.9	-2.2
	..Pangasinan	7.6	9.6	3.4	-0.8
	Region II (Cagayan Valley)	9.5	9.4	2.3	-0.8
	..Batanes	37.6	8.7	14.9	4.0
	..Cagayan	11.7	5.8	4.9	-3.4
	..Isabela	7.4	11.0	0.3	0.7
	..Nueva Vizcaya	10.7	18.3	4.1	2.7
	..Quirino	8.9	2.8	1.2	-4.6
	Region III (Central Luzon)	11.2	4.0	2.2	-0.3
	..Aurora	3.5	-1.9	-4.4	-11.6
	..Bataan	-3.2	21.0	-9.4	7.9
	..Bulacan	4.6	9.9	0.3	-0.3
	..Nueva Ecija	12.4	-0.3	2.9	0.0
	..Pampanga	20.1	-0.1	7.5	-2.9
	..Tarlac	10.9	13.2	3.3	5.3
	..Zambales	-5.2	-2.1	-12.6	-8.6

Presentation on screens limited to 1,000 rows and 50 columns

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Appendix 2. R Code for Descriptive

LIBRARIES AND DATA LOADING

```
library(tidyverse)
data_raw <- read.csv("file_path_of_data")
```

DATA TRANSFORMATION (LONG FORMAT + YEAR RANGE)

```
data_long <- data_raw %>%
  pivot_longer(
    cols = starts_with("Growth_"),
    names_to = "Sector",
    values_to = "Growth_Rate"
  ) %>%
  mutate(
    Sector = str_replace(Sector, "Growth_", ""),
    Sector = str_replace_all(Sector, "_", " "),
    Year_Range = paste0(as.numeric(Year) - 1, "-", Year)
  )
```

AGGREGATION (MEAN GROWTH PER SECTOR)

```
data_summary <- data_long %>%
  group_by(Year_Range, Sector) %>%
  summarise(
    Mean_Growth = mean(Growth_Rate, na.rm = TRUE),
    .groups = "drop"
  )
```

VISUALIZATION (DISPLAY PLOTS)

```
sectors <- unique(data_summary$Sector)

for (s in sectors) {

  sector_data <- data_summary %>%
    filter(Sector == s)

  p <- ggplot(sector_data, aes(x = Year_Range, y = Mean_Growth)) +
```

```
geom_hline(
  yintercept = 0,
  color = "black",
  linewidth = 0.8,
  alpha = 0.3
) +
```

```
geom_col(
  fill = "lightblue",
  color = "steelblue",
  linewidth = 0.6
) +
```

```
geom_text(
  aes(
    label = round(Mean_Growth, 1),
    vjust = ifelse(Mean_Growth >= 0, -0.6, 1.4)
  ),
  size = 5,
  fontface = "bold"
) +
```

```
theme_minimal() +
theme(
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  axis.line = element_line(color = "black", linewidth = 0.8),

  axis.text.x = element_text(size = 13, face = "bold", color = "black"),
  axis.title.x = element_text(size = 14, face = "bold"),
```

```

axis.text.y = element_text(size = 12, color = "black"),
axis.title.y = element_text(size = 14, face = "bold"),

plot.subtitle = element_text(
  size = 16,
  face = "bold",
  hjust = 0.5,
  margin = margin(b = 15)
),

plot.margin = margin(20, 20, 20, 20)
) +

labs(
  title = NULL,
  subtitle = paste("Sector:", s),
  x = "Year-on-Year",
  y = "Average Growth Rate (%)"
)

print(p)
}
# OUTPUT GENERATION AND EXPORT
export_folder <- "file_path_of_desired_location"

if (!dir.exists(export_folder)) {
  dir.create(export_folder, recursive = TRUE)
}

for (s in names(plot_list)) {

  file_name <- gsub("[[:punct:]]", "", s)

```

```
ggsave(  
  filename = paste0(export_folder, file_name, ".png"),  
  plot = plot_list[[s]],  
  width = 12,  
  height = 7,  
  dpi = 300  
)  
}
```

Appendix 3. R Code for MANOVA

LIBRARIES AND DATA LOADING

```
pacman::p_load(
  tidyverse,
  heplots,
  psych,
  effectsize,
  car
)
```

```
file_path <- "file_path_of_data"
```

```
data <- read.csv(file_path)
```

PHASE 2: DATA PREPARATION (SERVICES CONSOLIDATION)

```
data <- data %>%
```

```
  mutate(
```

```
    Growth_Services = (
```

```
      Growth_Accommodation_and_Food_Service_Activities +
```

```
      Growth_Education +
```

```
      Growth_Financial_and_Insurance_Activities +
```

```
      Growth_Human_Health_and_Social_Work_Activities +
```

```
      Growth_Information_and_Communication +
```

```
      Growth_Other_Services +
```

```
      Growth_Professional_and_Business_Services +
```

```
      Growth_Public_Administration_and_Defense_Compulsory_Social_Activities +
```

```
      Growth_Real_Estate_and_Ownership_of_Dwellings +
```

```
      Growth_Transportation_and_Storage +
```

```
      Growth_Wholesale_and_Retail_Trade_Repair_of_Motor_Vehicles_and_Motorcycles
    ) / 11
```

```
)
```

```
cols_growth <- c(
```

```
  "Growth_Agriculture_Forestry_and_Fishing",
```

```
  "Growth_Construction",
```

```
"Growth_Electricity_Steam_Water_and_Waste_Management",
"Growth_Manufacturing",
"Growth_Mining_and_Quarrying",
"Growth_Services"
)
```

```
dv_matrix <- as.matrix(data[, cols_growth])
data$Year_Factor <- as.factor(data$Year)
```

MULTIVARIATE ASSUMPTION TESTING

A. Multivariate Outliers (Mahalanobis Distance)

```
m_dist <- mahalanobis(dv_matrix, colMeans(dv_matrix), cov(dv_matrix))
cutoff <- qchisq(0.999, df = ncol(dv_matrix))
outliers <- which(m_dist > cutoff)
```

```
cat("Outliers detected (p < 0.01):", length(outliers), "\n")
```

B. Multivariate Normality (Mardia's Test)

```
psych::mardia(dv_matrix, plot = FALSE)
```

C. Homogeneity of Covariance Matrices (Box's M Test)

```
box_results <- heplots::boxM(dv_matrix, data$Year_Factor)
print(box_results)
```

D. Multivariate Linearity

```
pairs(
  dv_matrix,
  main = "Sectoral Linearity Matrix",
  pch = 19,
  col = rgb(0, 0, 1, 0.2)
)
```

E. Correlation Structure

```
cor_matrix <- cor(dv_matrix)
print(round(cor_matrix, 3))

psych::pairs.panels(
  dv_matrix,
  method = "pearson",
  hist.col = "#00AFBB",
  density = TRUE,
  ellipses = TRUE,
  main = "Correlation Matrix: Sectoral Interdependence"
)
```

F. Independence of Observations

```
mlm_model <- lm(dv_matrix ~ data$Year_Factor)

plot(
  residuals(mlm_model),
  main = "Residuals vs Observation Order"
)

abline(h = 0, col = "red")
```

MANOVA ANALYSIS

```
manova_result <- manova(dv_matrix ~ Year_Factor, data = data)

summary(manova_result, test = "Pillai")

effectsize::eta_squared(manova_result)
```

UNIVARIATE FOLLOW-UP ANALYSIS

```
univariate_results <- summary.aov(manova_result)
print(univariate_results)
```

```
sectors <- colnames(dv_matrix)
sector_eff_sizes <- list()

for (s in sectors) {
  formula_s <- as.formula(paste(s, "~ Year_Factor"))
  model_s <- lm(formula_s, data = data)

  sector_eff_sizes[[s]] <- effectsize::eta_squared(model_s, partial = TRUE)
}

cat("\n--- SECTORAL EFFECT SIZES ---\n")
print(sector_eff_sizes)
```

Appendix 4. R Code for Hierarchical Cluster Analysis

LIBRARIES AND DATA LOADING

```
library(tidyverse)
```

```
library(cluster)
```

```
library(factoextra)
```

```
library(fpc)
```

```
library(broom)
```

```
data_raw <- read.csv("file_path_of_data")
```

CONSOLIDATION (6-SECTOR MUTATION)

```
data_consolidated <- data_raw %>%
```

```
  mutate(
```

```
    Mean_Services = (
```

```
      Mean_Accommodation_and_Food_Service_Activities +
```

```
      Mean_Education +
```

```
      Mean_Financial_and_Insurance_Activities +
```

```
      Mean_Human_Health_and_Social_Work_Activities +
```

```
      Mean_Information_and_Communication +
```

```
      Mean_Other_Services +
```

```
      Mean_Professional_and_Business_Services +
```

```
      Mean_Public_Administration_and_Defense_Compulsory_Social_Activities +
```

```
      Mean_Real_Estate_and_Ownership_of_Dwellings +
```

```
      Mean_Transportation_and_Storage +
```

```
      Mean_Wholesale_and_Retail_Trade_Repair_of_Motor_Vehicles_and_Motorcycles
```

```
    ) / 11,
```

```
    CV_Services = (
```

```
      CV_Accommodation_and_Food_Service_Activities +
```

```
      CV_Education +
```

```
      CV_Financial_and_Insurance_Activities +
```

```
      CV_Human_Health_and_Social_Work_Activities +
```

```
      CV_Information_and_Communication +
```

```

CV_Other_Services +
CV_Professional_and_Business_Services +
CV_Public_Administration_and_Defense_Compulsory_Social_Activities +
CV_Real_Estate_and_Ownership_of_Dwellings +
CV_Transportation_and_Storage +
CV_Wholesale_and_Retail_Trade_Repair_of_Motor_Vehicles_and_Motorcycle
s
) / 11
) %>%
select(
  Name,
  Mean_Agriculture_Forestry_and_Fishing, CV_Agriculture_Forestry_and_Fishing,
  Mean_Construction, CV_Construction,
  Mean_Electricity_Steam_Water_and_Waste_Management,
  CV_Electricity_Steam_Water_and_Waste_Management,
  Mean_Manufacturing, CV_Manufacturing,
  Mean_Mining_and_Quarrying, CV_Mining_and_Quarrying,
  Mean_Services, CV_Services
)

```

IQR WINSORIZATION

```

winsorize_iqr <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1
  lower <- Q1 - 1.5 * IQR_val
  upper <- Q3 + 1.5 * IQR_val
  pmin(pmax(x, lower), upper)
}

df_winsorized <- data_consolidated %>%
  mutate(across(starts_with("CV_"), winsorize_iqr))

```

```

df_final <- df_winsorized %>%
  column_to_rownames("Name")

cv_vars <- grep("^CV_", names(data_consolidated), value = TRUE)

outlier_report <- data.frame(
  Variable = cv_vars,
  Max_Before = sapply(data_consolidated[cv_vars], max, na.rm = TRUE),
  Max_After = sapply(df_winsorized[cv_vars], max, na.rm = TRUE)
)

print(outlier_report)

# PRINCIPAL COMPONENT ANALYSIS

pca_model <- prcomp(df_final, center = TRUE, scale. = TRUE)

pca_variance_table <- data.frame(
  Component = paste0("PC", 1:length(pca_model$sdev)),
  Eigenvalue = pca_model$sdev^2,
  Variance_Explained = (pca_model$sdev^2 / sum(pca_model$sdev^2)) * 100,
  Cumulative_Variance = cumsum((pca_model$sdev^2 / sum(pca_model$sdev^2)) *
100)
)

print(pca_variance_table)

fviz_eig(pca_model, addlabels = TRUE)

fviz_pca_var(
  pca_model,
  col.var = "contrib",
  gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
  repel = TRUE

```

```

) +
  labs(title = "PCA Variable Distribution")

cum_var <- cumsum(pca_model$sdev^2 / sum(pca_model$sdev^2))
num_pcs <- which(cum_var >= 0.80)[1]

pca_scores <- pca_model$x[, 1:num_pcs]

cat("Selected PCs:", num_pcs, "\n")

# SILHOUETTE OPTIMAL CLUSTERS
opt_plot <- fviz_nbclust(
  pca_scores,
  FUN = hcut,
  method = "silhouette",
  hc_method = "ward.D2"
)

print(opt_plot)

sil_data <- opt_plot$data
optimal_k <- sil_data$clusters[which.max(sil_data$y)]

cat("Optimal clusters:", optimal_k, "\n")

# HIERARCHICAL CLUSTERING
dist_matrix <- dist(pca_scores)
hc_model <- hclust(dist_matrix, method = "ward.D2")
clusters <- cutree(hc_model, k = optimal_k)

df_final$Cluster <- as.factor(clusters)

```

EVALUATION METRICS

```

cluster_stats <- cluster.stats(dist_matrix, clusters)
sil <- silhouette(clusters, dist_matrix)

evaluation_table <- data.frame(
  Metric = c("Optimal k", "Avg Silhouette", "Dunn Index", "CH Index"),
  Value = c(optimal_k, mean(sil[,3]), cluster_stats$dunn, cluster_stats$ch)
)

print(evaluation_table)

cluster_distribution <- as.data.frame(table(clusters))
colnames(cluster_distribution) <- c("Cluster", "Count")

print(cluster_distribution)

```

VISUALIZATION

```

pca_cluster_plot <- fviz_cluster(
  list(data = pca_scores, cluster = clusters),
  geom = "point",
  palette = c("#2E9FDF", "#E7B800"),
  ellipse.type = "convex",
  show.clust.cent = TRUE,
  ggtheme = theme_minimal(),
  xlab = "Principal Component 1",
  ylab = "Principal Component 2"
) +
  labs(title = "Spatial Separation of Economic Typologies (PCA Projection)")
print(pca_cluster_plot)

png("c:/DOWNLOADS/dendrogram_landscape.png", width = 1200, height = 800)
plot(as.dendrogram(hc_model), horiz = TRUE)

```

```
dev.off()
```

```
profile_data <- df_final %>%
  group_by(Cluster) %>%
  summarise(across(starts_with("Mean_"), mean, na.rm = TRUE)) %>%
  pivot_longer(-Cluster, names_to = "Sector", values_to = "Value") %>%
  mutate(Sector = gsub("Mean_", "", Sector))
```

```
ggplot(profile_data, aes(Sector, Value, fill = Cluster)) +
  geom_col(position = "dodge") +
  coord_flip() +
  theme_minimal()
```

WELCH T-TEST

```
variables <- setdiff(names(df_final), "Cluster")
```

```
profiling_table <- lapply(variables, function(v) {
  test <- t.test(as.formula(paste(v, "~ Cluster")), data = df_final)
  data.frame(
    Variable = v,
    Cluster_1_Mean = unname(test$estimate[1]),
    Cluster_2_Mean = unname(test$estimate[2]),
    T_Statistic = unname(test$statistic),
    P_Value = test$p.value
  )
}) %>% bind_rows()
```

```
print(profiling_table)
```

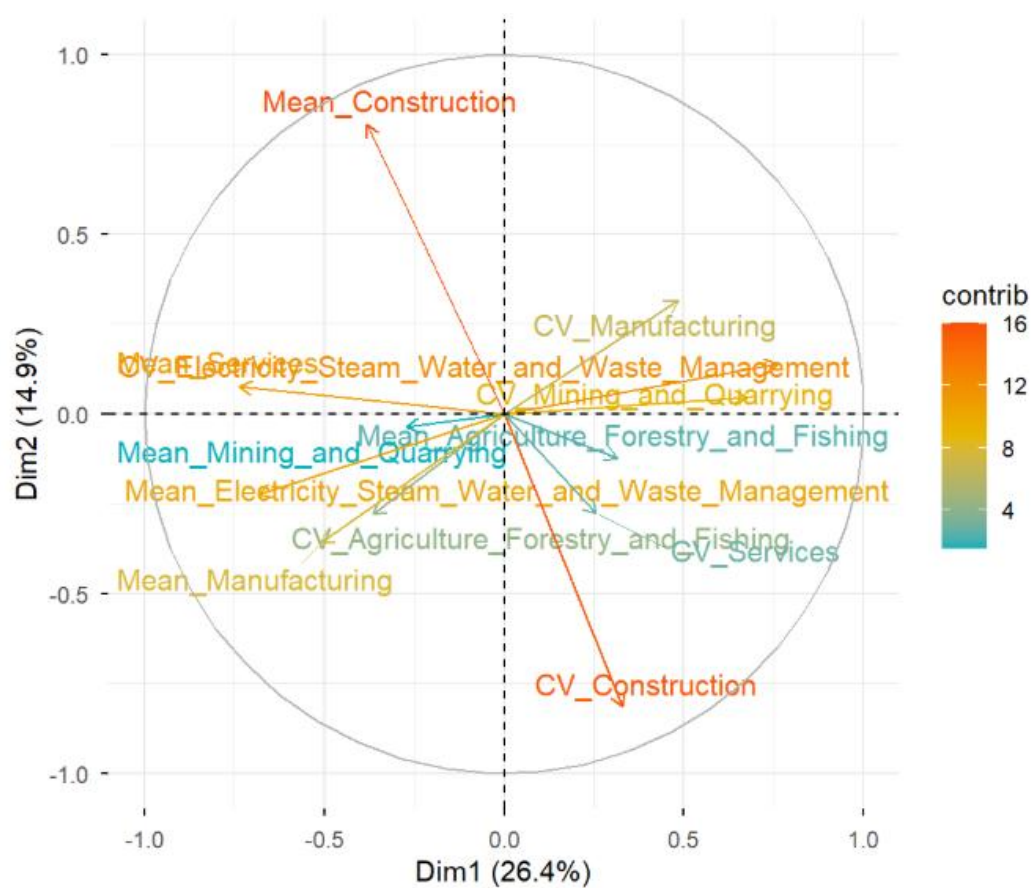
ROBUSTNESS CHECK (K-MEANS)

```
set.seed(123)
```

```
km_res <- kmeans(pca_scores, centers = optimal_k, nstart = 25)
```

```
print(table(km_res$cluster, clusters))
```

Appendix 5. PCA Variable Contributions



Appendix 6. Cluster Membership by Subnational Economic Unit

Cluster 1
Abra
Agusan del Norte
Agusan del Sur
Aklan
Antique
Apayao
Aurora
Basilan
Bataan
Batanes
Batangas
Benguet
Biliran
Bukidnon
Bulacan
Cagayan
Camarines Sur
Camiguin
Capiz
Cavite
Cebu
City of Angeles
City of Bacolod
City of Baguio
City of Butuan
City of Cagayan de Oro
City of Cebu

City of Davao

City of General Santos

City of Iligan

City of Iloilo

City of Isabela

City of Lapu-Lapu

City of Lucena

City of Mandaue

City of Olongapo

City of Puerto Princesa

City of Tacloban

City of Taguig

City of Zamboanga

Cotabato

Davao Occidental

Davao Oriental

Davao de Oro

Davao del Norte

Davao del Sur

Dinagat Islands

Eastern Samar

Guimaras

Ifugao

Ilocos Norte

Ilocos Sur

Iloilo

Isabela

Kalinga

La Union

Laguna

Lanao del Norte

Lanao del Sur

Leyte

Maguindanao del Norte

Maguindanao del Sur

Marinduque

Misamis Occidental

Misamis Oriental

Mountain Province

Negros Occidental

Northern Samar

Nueva Ecija

Nueva Vizcaya

Occidental Mindoro

Oriental Mindoro

Palawan

Pampanga

Pangasinan

Quezon

Quirino

Rizal

Romblon

Samar

Sarangani

South Cotabato

Southern Leyte

Sultan Kudarat

Sulu

Surigao del Norte

Surigao del Sur

Tarlac

Tawi-Tawi

Zambales

Zamboanga Sibugay

Zamboanga del Norte

Zamboanga del Sur

Cluster 2
Albay
Bohol
Camarines Norte
Catanduanes
City of Caloocan
City of Las Piñas
City of Makati
City of Malabon
City of Mandaluyong
City of Manila
City of Marikina
City of Muntinlupa
City of Navotas
City of Parañaque
City of Pasig
City of San Juan
City of Valenzuela
Masbate
Negros Oriental
Pasay City
Pateros
Quezon City
Siquijor
Sorsogon

Appendix 7. Thesis Forms



Republic of the Philippines
CAVITE STATE UNIVERSITY
 (CvSU)
 Don Severino De Las Alas Campus
 Indang, Cavite

COLLEGE OF ARTS AND SCIENCES
 Department of Physical Sciences

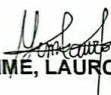
**APPROVAL OF THESIS TITLE AND
 REQUEST FOR ADVISER AND TECHNICAL CRITIC**

Name of Researcher/s: **Lauro Jr. H. Gime; Mark Anthony O. Nene; Renze Mico P. Rabanzo**

Course/Field of Specialization: **BS Applied Mathematics**

Working Title:

**Evidence-Based Pathways to Provincial Resilience: A Data-Driven Industry Classification
 and Performance Analysis Using 2022-2024 Provincial Product Accounts**


GIME, LAURO JR., H.; NENE, MARK ANTHONY, O.; RABANZO, RENZE MICO, P.
 Signature of Researcher

November 18, 2025
 Date:

CONFORME:


JAYSON C. SAVILLA, MAS
 Thesis Adviser
 Date: 11-18-25


RACHEL MAE O. PANGANIBAN
 Technical Critic
 Date: 11-18-25

Recommending Approval:


LANI S. RODIS
 Program Research Coordinator
 Date: 11-19-25


RENE B. BETONIO, PhD
 Department Chairperson
 Date: 11-19-25


ALCONA MAE B. CARAMILLO, MSc.
 College R&D Coordinator
 Date: 11-11-25

Approved:


AMMIE P. FERRER, PhD
 Dean
 Date: 12-11-25



Republic of the Philippines
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 Indang, Cavite

COLLEGE OF ARTS AND SCIENCES
 Department of Physical Sciences

REQUEST FOR PROPOSAL DEFENSE

Date:

ALCONA MAE P. BALTAZAR, MSc.
 College Research Coordinator

Madam:

I wish to request for an oral defense of my thesis proposal titled: **Evidence-Based Pathways to Provincial Resilience: A Data-Driven Industry Classification and Performance Analysis Using 2022–2024 Provincial Product Accounts** on **December 9, 2025, 1:00-2:00 PM** at **PL 105**.

I have herewith attached my thesis proposal. I understand that it is necessary to give copies of the full thesis proposal to the panel three (3) working days before the schedule of oral defense.

I am hoping for your favorable response on this request.

Very truly yours,

  
GIME, LAURO JR. H.; NENE, MARK ANTHONY O.; RABANZO, RENZE MICO P.
 Signature over Printed Name

Recommending Approval:


JAYSON C. SAVILLA, MAS
 Thesis Adviser

11/25/2025
 Date


RACHEL MAE O. PANGANIBAN
 Technical Critic

11/25/2025
 Date

Approved:


LANI S. RODIS
 Dept. Research Coordinator

12/02/2025
 Date


RENE B. BETONIO, PhD
 Department Chairperson

12/2/25
 Date


ALCONA MAE B. CARAMILLO, MSc.
 College R&D Coordinator

12-11-25
 Date



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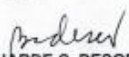
COMMENTS/SUGGESTIONS

Researcher/s: Lauro Jr. H. Gime; Mark Anthony O. Nene; Renze Mico P. Rabanzo

Working Title: Evidence-Based Pathways to the Resilience of Subnational Economic Units

Comments and Suggestions	Action Taken	Remarks
Thesis Title The panel suggested to make the thesis title to be short and concise (10-15 words).	The researchers have addressed the recommendation by changing the title into a short and concise form.	
Introduction The researchers are advised to include the Purpose of the Study, ensure that the Statement of the Problem is aligned with the Objectives, use operational definitions in the Definitions of Terms, and revise the Conceptual Framework.	The researchers have addressed the recommendations by including the Purpose of the Study (page 5), aligning the Statement of the Problem with the Objectives (page 6), revising the Definitions of Terms into operational definitions (pages 10-12), and updating the Conceptual Framework (page 12).	
Review of Related Literature The panel suggested adding more citations to support the Cluster Analysis.	The researchers have added supporting studies on Cluster Analysis (pages 18-21).	
Methodology The panel recommended removing the Ethical Considerations section.	The researchers removed the Ethical Considerations section (page 29).	

Rated by:


BENJARDE C. DESOR 01/20/26
 Panel Date



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 Department of Physical Sciences

COMMENTS/SUGGESTIONS

Researcher/s: Lauro Jr. H. Gime; Mark Anthony O. Nene; Renze Mico P. Rabanzo

Working Title: Evidence-Based Pathways to the Resilience of Subnational Economic Units

Comments and Sugestions	Action Taken	Remarks
Thesis Title No Comments/Suggestions		
Introduction The panel recommended removing Question 4 from the Statement of the Study, deleting the fourth objective in the Objectives of the Study, removing H_a from the Hypothesis of the Study, and including the date of release of the PPA in the Time and Place section.	The researchers have addressed the recommendations by removing Question 4 from the Statement of the Study (page 6), deleting the fourth objective (pages 6–7), and including the date of release of the PPA in the Time and Place section of the study (page 9).	
Review of Related Literature No Comments/Suggestions		
Methodology The panel suggested removing all of the rule of thumbs and replacing them with references from more reliable sources.	The researchers have removed all the rule of thumbs and added more reliable citations (pages 22-29).	

Rated by:

JAPHET LLEVARES

Panel

01/20/26

Date



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 Department of Physical Sciences

COMMENTS/SUGGESTIONS

Researcher/s: Lauro Jr. H. Gime; Mark Anthony O. Nene; Renze Mico P. Rabanzo

Working Title: Evidence-Based Pathways to the Resilience of Subnational Economic Units

Comments and Sugestions	Action Taken	Remarks
Thesis Title No Comments/Suggestions		
Introduction No Comments/Suggestions		
Review of Related Literature The panel recommended adding more RRL that discusses the relationships between variables, providing a justification for the sectoral structure and industry performance, and separating the Synthesis from the Identification of Gaps.	The researchers have added more supporting RRL and discussed the relationships between variables (pages 14–21), provided a justification for the sectoral structure and industry performance (page 23), and separated the Synthesis from the Identification of Gaps (pages 20–21).	
Methodology The panel recommended finding additional citations to support the use of a Comparative Qualitative Design and advised presenting the information in paragraph form rather than using bullet points.	The researchers have added more citations to support the Comparative Qualitative Design (page 22) and removed the use of bullet points (page 22).	

Rated by:


JONEL C. ABELLA

Panel

1/20/2024
 Date

VPAA-QF-17



Republic of the Philippines
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 Indang, Cavite

COLLEGE OF ARTS AND SCIENCES
 Department of Physical Sciences

THESIS OUTLINE APPROVAL FORM

Thesis Title: Evidence-Based Pathways to the Resilience of Subnational Economic Units

Name of Author(s): Lauro Jr. H. Gime; Mark Anthony O. Nene; Renze Mico P. Rabanzo

Degree Program: Bachelor of Science in Applied Mathematics

 JAYSON C. SAVILLA, MAS Adviser	 Date	 RACHEL MAE O. PANGANIBAN Technical Critic	 Date
 RENE B. BETONIO, PhD Department Chair	 Date	 LANI S. RODIS Department R and D Coordinator	 Date
 ALCONA MAE B. CARAMILLO, MSc. College/Campus R and D Coordinator	 Date		
 AMMIE P. FERRER, PhD Dean/Campus Administrator	 Date		

CAS Thesis Form 8



Republic of the Philippines
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 Indang, Cavite
COLLEGE OF ARTS AND SCIENCES
 Department of Biological Sciences
FINAL DEFENSE EVALUATION

Researcher/s: **LAURO JR. H. GIME**
MARK ANTHONY O. NENE
RENZE MICO P. RABANZO

Title: **EVIDENCE-BASED PATHWAYS TO RESILIENCE OF SUBNATIONAL ECONOMIC UNITS**

Comments/Suggestions	Action Taken	Remarks
Title		
No Comment/Suggestions		
Introduction		
No Comment/Suggestions		
Objectives		
No Comment/Suggestions		
Significance of the Study		
No Comment/Suggestions		
Review of Related Literature		
No Comment/Suggestions		
Methodology		
No Comment/Suggestions		
Results and Discussion		
• Align all tables with the paragraph format and change all column headings to uppercase.	• Aligned the results tables with the paragraph format and changed all column headings to uppercase.	o/c
• Change the notation np^2 to pn^2 in Table 1.4.	• Revised np^2 to pn^2 in Table 1.4 and in the related discussion.	o/c
• Remove the additional title "Dendrogram of Subnational Economic Units" in Figure 4.3.	• Removed the additional figure title from Figure 4.3.	o/c

- | | | |
|--|---|------------------|
| <ul style="list-style-type: none"> • Add descriptions for each cluster, including the provinces, HUCs, and other subnational units included in Cluster 1 and Cluster 2. | <ul style="list-style-type: none"> • Added a cluster membership discussion explaining that Cluster 1 consists of 93 subnational economic units and Cluster 2 consists of 24 subnational economic units. The discussion also described the composition of each cluster by provinces, HUCs, City of Isabela, and Paternos. | <p>o k</p> |
| <ul style="list-style-type: none"> • Clarify what Cluster 1 and Cluster 2 are called based on their economic characteristics. | <ul style="list-style-type: none"> • Described Cluster 1 as a generally diversified and growth-responsive typology and Cluster 2 as a structurally constrained, lower-growth, and more volatility-constrained typology. | <p>Very good</p> |

Conclusion

No Comment/Suggestions

Recommendation

No Comment/Suggestions

Rated by:


GAMALIEL B. BELUANG
 Evaluator

9/29/26
 Date

CAS Thesis Form 8



Republic of the Philippines
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COLLEGE OF ARTS AND SCIENCES
 Department of Biological Sciences
FINAL DEFENSE EVALUATION

Researcher/s: **LAURO JR. H. GIME**
MARK ANTHONY O. NENE
RENZE MICO P. RABANZO

Title: **EVIDENCE-BASED PATHWAYS TO RESILIENCE OF SUBNATIONAL ECONOMIC UNITS**

Comments/Suggestions	Action Taken	Remarks
Title		
No Comment/Suggestions		
Introduction		
No Comment/Suggestions		
Objectives		
No Comment/Suggestions		
Scope and Limitation of the Study		
Clarify the scope from provincial-level analysis to subnational economic units.	Revised the scope to focus on 117 subnational economic units.	
Specify the included units, sectors, and data period.	Clarified that the study covers provinces, selected city-level units, Pateros, six consolidated sectors, and 2018–2024 PPA data.	
Strengthen the study limitations.	Added limitations on NCR proxy values, Pateros and City of Isabela inclusion, cluster interpretation, repeated observations, excluded non-economic factors, and the 2018–2024 timeframe.	
Review of Related Literature		
No Comment/Suggestions		
Methodology		
No Comment/Suggestions		

Results and Discussion

- | | |
|--|---|
| ✓ Add legends below the results tables. | ✓ Added legends below the results tables. |
| ✓ Improve the arrangement of data inside the tables. | ✓ Revised the arrangement of data inside the tables for clearer presentation. |
| ✓ Adjust table sizes and align them with the text. | ✓ Adjusted table sizes and aligned them with the text. |
-

Conclusion

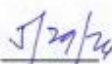
No Comment/Suggestions

Recommendation

No Comment/Suggestions

Rated by:


BENJARDE C. DESOR
Evaluator


Date

CAS Thesis Form 8



Republic of the Philippines
CAVITE STATE UNIVERSITY
 (CvSU)
 Don Severino De Las Alas Campus
 Indang, Cavite
COLLEGE OF ARTS AND SCIENCES
 Department of Biological Sciences
FINAL DEFENSE EVALUATION

Researcher/s: **LAURO JR. H. GIME**
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Title: **EVIDENCE-BASED PATHWAYS TO RESILIENCE OF SUBNATIONAL ECONOMIC UNITS**

Comments/Suggestions	Action Taken	Remarks
Title		
No Comment/Suggestions		
Introduction		
No Comment/Suggestions		
Objectives		
No Comment/Suggestions		
Scope and Limitation of the Study		
No Comment/Suggestions		
Review of Related Literature		
No Comment/Suggestions		
Methodology		
No Comment/Suggestions		
Results and Discussion		
• Add descriptions for each cluster, including the provinces, HUCs, and other subnational units included in Cluster 1 and Cluster 2.	• Added a cluster membership discussion explaining that Cluster 1 consists of 93 subnational economic units and Cluster 2 consists of 24 subnational economic units. The discussion also described the composition of each cluster by provinces, HUCs, City of Isabela, and Pateros.	
• Emphasize more the procedure in Cluster Analysis.	• The researchers revised the Cluster Analysis procedure to a much clearer and emphasized paragraph.	

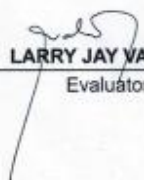
Conclusion

No Comment/Suggestions

Recommendation

No Comment/Suggestions

Rated by:


LARRY JAY VALERO

Evaluator

02/02/2024
Date

CAS Thesis Form 9:



Republic of the Philippines
CAVITE STATE UNIVERSITY
 (CvSU)

Don Severino De Las Alas
 Campus Indang, Cavite

COLLEGE OF ARTS AND SCIENCES
 Department of Physical Sciences

CERTIFICATE OF COMPLETION

May 29, 2026

To whom it may concern,

This is to certify that **Lauro Jr. H. Gime, Mark Anthony O. Nene, and Renze Mico P. Rabanzo**, students of **BS Applied Mathematics** has defended and submitted the revised thesis proposal titled: **Evidence-Based Pathways to Resilience of Subnational Economic Units**


JAYSON C. SAVILLA, MAS
 Thesis Adviser

05-29-2026
 Date


RACHEL MAE O. PANGANIBAN
 Technical Critic

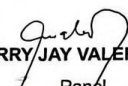
05/29/26
 Date


BENJARDE C. DESOR, LPT, MAT
 Panel

5/29/26
 Date


GAMALIEV B. BELUANG
 Panel

5/29/25
 Date


LARRY JAY VALERO, LPT
 Panel

06/02/26
 Date


LANI S. RODIS
 Dept. Research Coordinator

06/02/26
 Date


RENE B. BETONIO, PHD
 Department Chairperson

6/2/2026
 Date


ALCONA MAE B. CARAMILLO, MSC.
 College Research Coordinator

22 JUN 2026
 Date



Republic of the Philippines
CAVITE STATE UNIVERSITY
 Don Severino delas Alas Campus
 Indang, Cavite

COLLEGE OF ARTS AND SCIENCES
 Department of Physical Sciences

ROUTING SLIP
 (for Undergraduate)

Name of Student(s): Lauro Jr. H. Gime; Mark Anthony O. Nene; Renze Mico P. Rabanzo

Type of Study: ☐ Thesis ☐ Narrative ☐ EDP
☐ Design Project ☐ Case Study ☐ Teaching Portfolio

Title of Study: Evidence-Based Pathways to Resilience of Subnational Economic Units

Type of Manuscript: ☐ Outline (if applicable) ☐ Final Manuscript

	DATE RECEIVED	DATE RELEASED	REMARKS	SIGNATURE
Adviser	05-19-2026	05-21-2026	Edit	<i>[Signature]</i>
	05-23-2026	05-24-2026	Edit	<i>[Signature]</i>
	05-28-2026	06-01-2026	ok	<i>[Signature]</i>
Technical Critic	06-01-2026	06-02-2026	ok	<i>[Signature]</i>
Department Research/OJT Coordinator	06-02-2026	06-02-2026	ok	<i>[Signature]</i>
Department Chairperson	6-2-2026	6-2-2026	ok	<i>[Signature]</i>
English Critic	06-03-2026	06-05-2026	for 2nd reading	<i>[Signature]</i>
	06-09-2026	06-11-2026	for final edit	<i>[Signature]</i>
	06-14-2026	06-19-2026	okay	<i>[Signature]</i>
College Research/OJT Coordinator	22 JUN 2026	22 JUN 2026	ok	<i>[Signature]</i>
Director for Research (for thesis only)	6.26.26	6.26.26	ok	<i>[Signature]</i>
Dean	JUN 22 2026	JUN 22 2026	ok	<i>[Signature]</i>

Appendix 8. English Critic Certification



Republic of the Philippines
CAVITE STATE UNIVERSITY
 (CvSU)
 DON SEVERINO DELAS ALAS CAMPUS
 Indang, Cavite



CENTER FOR LANGUAGE LEARNING AND RESEARCH

CERTIFICATION

June 18, 2026

This is to certify that the thesis manuscript/ student teaching portfolio/ narrative report/ capstone project manuscript/ EDP manuscript/design project manuscript titled EVIDENCE-BASED PATHWAYS TO RESILIENCE OF SUBNATIONAL ECONOMIC UNITS of LAURO JR. H. GIME, MARK ANTHONY O. NENE, and RENZE MICO P. RABANZO has been read and edited by the undersigned English Critic.


MARISOL C. CRIZALDO
 English Critic

OFFICIAL RECEIPT:

Name of the student/s: LAURO JR. H. GIME, MARK ANTHONY O. NENE, and RENZE MICO P. RABANZO

Manuscript Title: EVIDENCE-BASED PATHWAYS TO RESILIENCE OF SUBNATIONAL ECONOMIC UNITS

No. of times read: 3

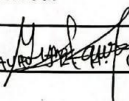
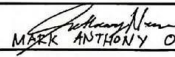
No of pages: 110

Total Amount: **P2200**

(TO BE ACCOMPLISHED BY THE STUDENT(S))

This is to certify that I/WE have paid two thousand and two hundred pesos (in words) (P2200) as payment for **Ms. Marisol C. Crizaldo** who edited my/our thesis manuscript/ student teaching portfolio/ narrative report/ capstone project manuscript/ EDP manuscript/ design project manuscript titled, EVIDENCE-BASED PATHWAYS TO RESILIENCE OF SUBNATIONAL ECONOMIC UNITS

Received:

 LAURO JR. H. GIME  MARK ANTHONY O. NENE  RENZE MICO P. RABANZO

(Name and Signature of the student/s)